Retained in every vertebrate, lost repeatedly elsewhere: a 503-genome census of the IP_3 receptor family

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Abstract

The inositol 1,4,5-trisphosphate receptor (IP_3R, gene family *ITPR*) is the endoplasmic reticulum's ligand-gated calcium-release channel, and in vertebrates it exists as three paralogues whose origin, distribution and retention have never been measured against a declared search space. We enumerated the family across 7,691 reference proteomes (77.6 million proteins) and 503 genomes — 309 vertebrate and 194 not — with the ryanodine receptors, which carry every IP_3R-diagnostic Pfam domain, searched alongside as an internal control at every stage.

Three results follow. First, the family is ancestrally eukaryotic and has been lost repeatedly and independently: it is absent from all 384 land-plant and all 1,353 Dikarya reference proteomes, from Apicomplexa, Microsporidia, Rhodophyta, diatoms and four other fungal phyla, and from all 634 archaeal and 3,537 bacterial proteomes — and every one of 35 clade-level absences holds at assembly level in genomes where a measured positive control recovers a comparably long, deeply conserved gene. Second, the three vertebrate paralogues are not lost at all: across 927 genome × paralogue cells in 309 vertebrate genomes, zero reach the absence state, and in the 189 assemblies contiguous enough to carry the gene the minimum is three gene-equivalents. Dollo parsimony places no loss anywhere on the tree, and we report instead the settings that would manufacture one. Third, the family's public record is much worse than the family: 55.0 % of full-length *ITPR* protein records carry no usable gene symbol, and 940 of 1,232 demonstrated genes (76.3 %) cannot be reached by any protein-database search.

The duplications are dated and their asymmetry explained: *ITPR2* and *ITPR3* are sisters (approximately unbiased test $p = 0.476$, with both alternatives rejected at $p < 2 \times 10^{-5}$), the *ITPR1*/(*ITPR2*, *ITPR3*) split sits on the vertebrate stem and the *ITPR2*/*ITPR3* split on the gnathostome stem at 462–563 Ma, and the surviving two-round paralogon links run through *ITPR1*. *ITPR1* alone was doubled and retained after the teleost genome duplication (97.3 % of 73 teleost genomes), and *ITPR1* alone is held under roughly twice the purifying selection of its sisters ($\omega = 0.024$ against 0.043 and 0.042; RELAX k = 9.36, intensified). The three paralogues share 46 to 49 of their roughly 58 intron positions in every genome that carries them, an 85- to 88-fold enrichment; the ryanodine receptors, which carry every diagnostic domain of the family, share one. Mapping constraint onto the channel identifies the gate and the selectivity filter as the least changeable

elements, places the ten measured IP_3 contacts outside the Pfam domain named after the ligand, and finds the constrained unit at the ligand site to be a 15 Å pocket rather than the contacts themselves. The IP_3-binding core is the *less* constrained of the two functional modules, and that comparison reverses on whether a 50-residue luminal loop — the least conserved sequence in the receptor, lying about fifty residues from the most conserved — is counted as part of the pore.

Contents

Introduction

Inositol 1,4,5-trisphosphate releases calcium from a non-mitochondrial intracellular store [1], and the channel that does it is a ~2,700-residue, four-fold symmetric protein of the endoplasmic reticulum membrane [2, 3]. From its first sequence it was recognised as the smaller relative of the ryanodine receptor [4], and the two families have been read against each other ever since: they share the MIR, RIH and RIH-associated domains, the six-transmembrane pore module, and — as this work repeatedly had to accommodate — every Pfam signature used to diagnose either one. In vertebrates the receptor is encoded by three genes, *ITPR1*, *ITPR2* and *ITPR3* [5], with distinct tissue distributions [6, 7], distinct single-channel behaviour [8] and distinct disease associations: *ITPR1* in spinocerebellar ataxia 15 and 29 and in Gillespie syndrome [9, 10, 11, 12], *ITPR2* in isolated anhidrosis [13], and *ITPR3* in Charcot-Marie-Tooth neuropathy and a multisystem disorder [14, 15, 16].

Almost everything known about the family's distribution outside the vertebrates rests on searches that were never scoped. Surveys of calcium signalling machinery report IP_3 receptors in animals, amoebae and various protists and their absence from land plants and from the yeasts [17, 18, 19, 20], but an absence is a claim about a search space, and the search spaces behind those statements are not declared, not controlled, and not separable from the annotation quality of the databases they were run in. The same is true of the vertebrate paralogues. Their number is known; whether any lineage has lost one is not, because no study has asked the question of genomes rather than of gene sets, and a gene missing from a gene set is more often an annotation than a biological fact.

Three properties of this family make it a hard case, and each of them shapes what follows. It is long, so its locus is longer than the contigs of a large fraction of published assemblies. It is paralogous with a sister family that carries all of its diagnostic domains, so any search sensitive enough to find divergent IP_3 receptors also finds ryanodine receptors. And it is poorly named: much of its record sits under placeholder symbols or under no symbol at all, so name-based retrieval systematically under-counts it.

We therefore built the census as a genomic one, with the sister family searched alongside as an internal control at every stage, and with every absence required to carry a positive control demonstrating that the search reached the assembly. The scope is declared and finite throughout: 763 vertebrate and 6,928 other reference proteomes, 309 vertebrate genomes selected as one per order plus every species the protein record left in doubt, and 194 non-vertebrate genomes sampled most finely where the negative claims are. Every claim below is a claim about that space.

Results

The family is ancestrally eukaryotic and has been lost repeatedly

We swept both family profiles — one built from IP_3 receptor seeds, one from ryanodine receptor seeds — over 763 vertebrate and 6,928 other reference proteomes: 77,559,719 proteins and 31.9 billion residues, with the four non-vertebrate eukaryote groups partitioning Eukaryota against the vertebrate set so that the two denominators add. Combined with an exhaustive InterPro enumeration of the family's Pfam signatures and the gene models recovered from 503 genomes, the census holds 18,065 records, of which 8,990 are called IP_3 receptor across 1,402 taxa (Fig. 1).

Where the family is. 662 of the 6,928 non-vertebrate reference proteomes carry an IP_3 receptor. It is present in every sampled sponge, cnidarian, mollusc, annelid, echinoderm, rotifer and non-vertebrate chordate proteome, in 292 of 311 arthropod and 106 of 110 nematode proteomes, and outside the animals in Amoebozoa, Discoba, ciliates, oomycetes, dinoflagellates, haptophytes and green algae. Two lineages carry a full-length member of *both* families: the choanoflagellate *Salpingoeca rosetta* and the filasterean *Capsaspora owczarzaki*, so the IP_3 receptor / ryanodine receptor duplication predates Metazoa, as comparative surveys of the calcium toolkit have suggested [17, 21].

Where it is not, and what makes that a measurement. The family is absent from all 384 land-plant (Streptophyta) reference proteomes against 15 of 48 in Chlorophyta; from all 1,353 Dikarya proteomes (Ascomycota 0/1,034, Basidiomycota 0/319) against 18 of 34 in Mucoromycota and 6 of 16 in Chytridiomycota; from all 60 apicomplexan, 29 microsporidian, 27 glomeromycote, 18 mortierellomycote and 35 kickxellomycote proteomes; from all 16 diatom and 12 red algal proteomes; and from all 634 archaeal and 3,537 bacterial proteomes. The fungal pattern is not a single basal event but repeated independent loss across five phyla.

Each negative carries a positive control inside the same search. PF08709, the IP_3-binding core that names the family, returns zero substantial matches in land plants against 26 in Chlorophyta, and zero in Dikarya against 16 in Mucoromycota; PF02815 (MIR), which every eukaryote carries on unrelated proteins, returns 633 and 4,376 substantial matches in those same proteomes. The instrument finds thousands of the promiscuous domain where it finds none of the family-defining one.

The absences hold at assembly level. Proteome absence is a statement about gene sets, so we re-asked the question of assemblies: 194 non-vertebrate genomes, 100.4 Gbp, sampled by five rules that sample most finely where the negative claims are. All 35 clade-level absences hold (Extended Data Fig. 1), and — the point of the design — all 35 are controlled. Zero genomes are uncontrolled: 115 are controlled across a kingdom boundary, 77 by their own target, one partially and one carries no control bait.

Making that true required abandoning the vertebrate sweep's control. There, the ryanodine receptors are a valid proof-of-search because every vertebrate has three; outside the animals the profiles find architecture-level ryanodine receptor in 2 of 6,928 proteomes, so a silent ryanodine receptor in a land plant is the correct answer and witnesses nothing. We therefore measured, per clade, which of six large, deeply conserved, multi-exon eukaryotic families is actually present in that clade's own proteomes and used it as the control. Apicomplexa takes the myosin head domain, present in 36 of 36 Aconoidasida and 23 of 23 Conoidasida proteomes, where the MIR domain occurs once per class; red algae take the SMC N-terminal domain at 12 of 12, where myosin reaches 8 %. A single control profile fixed in advance would have left both clades' absences resting on nothing.

Copy number outside the vertebrates is wide. Of the 194 genomes swept, 117 carry no IP_3 receptor, 43 carry one, 31 carry two to six, and three carry many more: the sponge *Dysidea avara* 8, the ciliate *Stentor coeruleus* 13 and the flatworm *Macrostomum lignano* 18 (Extended Data Fig. 1). The census's 62 *Macrostomum* protein records resolve to 18 real genes.

The plant and fungal records that do exist are genes, not contamination. All 99 plant and fungal IP_3 receptor records were chased individually against four independent lines of evidence: 47 are real genes, 52 are fragments, and none is a contaminant or lacks genome backing. The contamination test had full power — every record has a cross-kingdom BLAST hit, at 19.9–45.8 % identity (median 24.1 %), nowhere near the 95 % identity that marks an assembly contaminant.

Three paralogues, found in the DNA of 309 vertebrate genomes

The vertebrate scope is 309 assemblies — one per vertebrate order plus every species the protein record left in doubt — 552.5 Gbp, declared before the search and fixed. A 38-protein bait panel (30 IP_3 receptor, 8 ryanodine receptor) was aligned to each genome with a spliced aligner, and cells the alignment left empty were re-asked with a whole-genome translated search. All 309 completed with no failures; the ryanodine receptor positive control fired in every one of them, so no genome is excluded on control grounds. The sweep recorded 1,236 genome × cell results over 2,144 loci (Fig. 2 and Extended Data Fig. 2).

The sister family never once contested a locus. Across all 2,144 loci only one family's baits aligned at all. That separation is sharper at the genome than at the protein level, where a benchmark against curated decoys had promoted all six ryanodine receptors as novel IP_3 receptor paralogues until an explicit sister test was added; the family ambiguity is a property of comparing protein fragments, not of the two families.

Only four cells in 1,236 were called absent, and all four are cyclostome. *Petromyzon marinus* and *Myxine glutinosa* each carry *ITPR1* and no *ITPR2* or *ITPR3*. Both assemblies are contiguous enough to carry the gene and both have a firing control, so the call survives the usual gates — but the bait panel has no cyclostome-labelled *ITPR2* or *ITPR3* bait at all, and each of these genomes in fact carries three full-length family loci that all fall to the *ITPR1* bait. These are therefore cells whose paralogue cannot be assigned, not absences, and we treat them as such throughout.

The genomes hold genes the databases do not. Adding the sweep's gene models to the census contributed 1,058 models from 224 genomes. Of the IP_3 receptor models, **318 exist only as DNA** — no protein record of any kind — and **167 more sit inside an annotated gene that carries no family name**, so they are unreachable by any search that starts from a gene symbol.

How much the search missed, measured rather than estimated. Because no loss is reconstructed anywhere in this scope (below), every cell whose gene is independently known to be present and which the ledger did not grade *found* is a false negative of the method. That is 140 of 923 cells, **15.2 %**. The ryanodine receptor sister cell — present in every vertebrate, swept by the same aligner in the same assemblies, and using none of the same evidence — gives 42 of 309, **13.6 %**, indistinguishable from it (Fisher $p = 0.58$), so the rate is a property of the method and not of the paralogue calls.

Every miss is an assembly. A missed cell's median contig N50 is 23,460 bp against 3,396,515 bp for a found one; the odds of finding the gene rise 8.1-fold per tenfold of contig N50 (19.99-fold for the ryanodine receptor series); and chromosome-level assemblies miss 3 of 512 IP_3 receptor cells and 0 of 172 ryanodine receptor cells (Extended Data Fig. 3). We therefore report every downstream result twice, once over all 309 genomes and once over the 189 whose contigs are longer than the median measured gene span (142,212 bp). Above that bar the false-negative rate is 0.9 % on the IP_3 receptor series and 0.0 % on the ryanodine receptor series — a bar chosen from gene geometry with no error rate in it, and calibrated only afterwards.

Bait-panel design. Re-running the whole cell-assignment chain over 19 ablated panels reproduces the committed ledger 1,236 of 1,236 cells, so the ablation is exact rather than modelled. Phylogenetic breadth is nearly free: four human baits recover 782 of the 783 cells the full 38-bait panel recovers, and dropping any single clade band costs at most two cells. What matters is paralogue coverage — dropping one paralogue's baits costs 238–241 cells — and one bait alone recovers the gene at any identity above 0.5.

ITPR2 and *ITPR3* are sisters, and the surviving 2R links run through *ITPR1*

The tree. 134 representative proteins chosen by eight coded rules from the 18,065-record census — sampled per clade and per kingdom, never longest-per-species — were aligned with MAFFT L-INS-i (11,777 columns) and trimmed to 1,797 columns, 96.8 % of them parsimony-informative. Maximum likelihood under Q.insect+R7 (log-likelihood (-)215,452.0) recovers all three paralogue clades and the ryanodine receptor outgroup as monophyletic groups, with 69.5 % of the 131 internal nodes clearing both SH-aLRT ≥ 80 and UFBoot ≥ 95 (Fig. 3; the alignment itself, its identity structure and its per-sequence coverage are in Extended Data Fig. 4).

The sister question has an answer. The unconstrained tree groups *ITPR2* with *ITPR3*. Testing the three rooted arrangements against each other with the approximately unbiased test over 10,000 RELL replicates rejects *ITPR1+ITPR2* ($p = 1.8 \times 10^{-5}$) and *ITPR1+ITPR3* ($p = 1.65 \times 10^{-5}$) and does not reject *ITPR2+ITPR3* (p = 0.476) or the maximum-likelihood tree itself (p = 0.525); both surviving topologies carry the same pair (Extended Data Fig. 5). Because bootstrap support is optimistic under model violation and a four-kingdom alignment violates any single model by construction, we re-ran the search with an extra nearest-neighbour-interchange round on every bootstrap tree: no claim in the paper is weakened or lost.

The neighbourhood agrees, from evidence the alignment does not share. Flanking-gene sets around all 2,144 loci in 274 annotated genomes were compared against matched random windows drawn in the same two genomes, so annotation depth, naming convention and window rule are held constant. Within-paralogue Jaccard similarity runs 216- to 413-fold above its own matched null, while every cross-paralogue and every cross-family class sits at or below the null — the maximum mean Jaccard over 168,241 IP_3 × ryanodine receptor pairs is 0.0002 (Extended Data Fig. 6). A flank-consensus paralogue caller built on this, with its operating point measured rather than chosen, is 405 of 405 correct on loci whose paralogue the assembly's own annotation establishes, at a 0.008 false-call rate on 726 random windows, and it places 131 loci the sequence sweep could not.

Two-round duplication. Neighbourhood paralogy — not shared gene names, which after ~500 Myr are paralogues of each other rather than the same word — tests whether the three loci sit in duplicated blocks. Against 932 matched random neighbourhoods in the same genomes, which carry a link 2.6 % of the time, the *ITPR1* neighbourhood is paralogous to *ITPR2*'s in 141 of 175 genomes (80.6 %, 10 vertebrate classes; $p = 4.9 \times 10^{-118}$) and to *ITPR3*'s in 89 of 152 (58.6 %, 9 classes; $p = 9.4 \times 10^{-66}$), while *ITPR2* against *ITPR3* is 4 of 149 (2.7 %), exactly the background (p = 0.55) (Fig. 4). The three ryanodine receptors run through the identical instrument in the same genomes as a positive control.

Dating the individual links separates them. Of the two that survive in human, `GRM7`↔`GRM4` (*ITPR1–ITPR3*) has a Compara duplication node at Vertebrata and replicates in 84 genomes, whereas `BHLHE40`↔`BHLHE41` (*ITPR1–ITPR2*) dates to Opisthokonta and is therefore not a two-round ohnologue pair. So the pair the tree makes sisters is the pair whose neighbourhoods retain nothing shared — a record of which copies survived deletion, not of branching order, and we report it as a tension rather than an overturn.

Where the duplications sit on the vertebrate tree. Reconciling the gene tree against a hand-curated, literature-calibrated species tree — declared as an input, with a published age, an estimate spread and a stem age on every one of its 29 named nodes — places the split separating *ITPR1* from *ITPR2+ITPR3* on the **vertebrate stem**, older than crown Vertebrata (published estimates 480–615 Ma) with no upper bound this tree can set, and the split separating *ITPR2* from *ITPR3* on the **gnathostome stem**, bracketed 462–563 Ma (Extended Data Fig. 7). The placement does not depend on sister order: five topologies including both rejected arrangements give the same answer, and it survives collapsing every node below the tree's own support bar and re-rooting the vertebrate subtree at all 112 of its edges. What it does depend on is six cyclostome tips, whose branches are 0.96–1.04 times the median root-to-tip distance and so are not the long-branch artefact the objection would predict. An entirely independent instrument agrees: Dollo parsimony over 309 genomes, reading no gene tree at all, places the origin of *ITPR1* at Vertebrata and of *ITPR2* and *ITPR3* at Gnathostomata.

The teleost duplication doubled one paralogue and kept it. Across 73 teleost genomes above the contiguity bar, *ITPR1* is present at a mean of 1.97 copies and 97.3 % carry two, against 1.04 for *ITPR2* and 1.04 for *ITPR3* — while the same genomes carry 5.82 ryanodine receptors. The pre-3R ray-finned outgroups (bichir, gar, bowfin) carry 1/1/1 plus 3, and the lineages with a further whole-genome duplication carry 3/2/2 plus 8 (Extended Data Fig. 8). The two *ITPR1* copies are not a tandem array: 20 of 71 two-copy genomes place them on different contigs and the remainder a median 8.9 Mb apart. They partition the ancestral block disjointly in 45 of 49 genomes against a tetrapod reference and 46 of 49 against a pre-3R ray-finned one, and all 705 cross-anchor assignments agree ($p = 1.2 \times 10^{-212}$) — one ancestral duplication, not a series of lineage-specific ones.

One exon structure, three packings, and a sister family that shares none of it

The three paralogues are the same protein at 61–68 % identity, and the tree and the neighbourhood both say they are ohnologues. The gene itself is a third kind of character, and it is one no sequence alignment produces: where the introns fall. We measured it from the same spliced alignments the census was built on, for every locus in the 189 assemblies contiguous enough to carry the gene — 1,378 genes, 112,254 junctions.

The instrument is corroborated before it is used. 99.89 % of those 112,254 junctions read as a spliceable pair off the genome itself — 98.96 % canonical GT–AG and 1,035 minor sites — which the alignment score knows nothing about. And across 164 genomes carrying an independent annotation, 177,710 of 188,146 annotated coding edges (94.5 %) fall exactly on a boundary the alignment placed, more often for RefSeq annotations (94.9 %) than for submitter-deposited ones (91.1 %) (Extended Data Fig. 9). That is the whole-scope form of a check the two validated annotation failures below can each make only on themselves.

The exon count is conserved and the genomic span is not. All three paralogues are 57–58 coding exon genes carrying 8.0–8.3 kb of coding sequence, and they occupy 147 kb, 244 kb and 58 kb of chromosome respectively — a **4.2-fold spread of genomic span at a 3 % spread of coding length** (Extended Data Fig. 9). Mean exon length is 138–142 bp in all three. The ryanodine receptors, measured through the identical instrument in the same assemblies, are 104-exon genes over 14,910 bp: nearly twice the gene in both dimensions, at almost exactly the same mean exon length (144 bp). What differs between the IP_3 receptor paralogues is not how the protein is divided but how much intron is wrapped around it, and the ordering holds gene by gene inside single genomes rather than on average — *ITPR3* is shorter than *ITPR1* in 161 of the 182 genomes carrying both, and shorter than *ITPR2* in 146 of 181, at $q < 10^{-3}$ on paired sign tests corrected across the whole family of comparisons.

The three paralogues inherited one intron set; the sister family did not. An intron's position here is a pair — the alignment column its upstream exon ends in, and its phase — because two genes can be interrupted between the same two residues in different frames, which is not one ancestral intron. Scored inside each genome against an exact null that places each intron independently and uniformly on the columns both loci resolve, the three paralogues share a **median 46 to 49 positions where 0.55 is expected, an 85- to 88-fold enrichment that is significant in every one of the 181 to 183 genomes tested** — not on average, in each genome separately. Each paralogue has a core of 48–49 positions present in over 90 % of its own genomes.

And the control is the result. The ryanodine receptors carry every Pfam domain used to diagnose this family, and that shared architecture is what every search in this paper had to be built around. Measured through the same instrument in the same genomes, an IP_3 receptor and a ryanodine receptor share a **median of one intron position, in 0 of 188, 0 of 184 and 0 of 183 genomes at $p < 0.05$**. The two families share a fold, a pore and four Pfam signatures, and their exon structures have no ancestry in common. Whatever the domain architecture they share means, it was not inherited as a gene. The control that every search in this paper leans on is here made by a character no alignment of the proteins can see.

No paralogue is lost anywhere in 309 vertebrate genomes

The count is zero, and the deliverable is the map of fragility around it. Every genome × paralogue cell was re-stated under eight ordered positive tests of which exactly one is countable as a loss, and that one is reachable by construction — a self-test builds a contiguous, controlled, empty, spare-free cell and requires the loss state. Across 927 cells in 309 genomes: 783 hold one locus, 90 are truncated by the assembly, 7 partial, 43 are reassembled across contigs, 4 are the cyclostome cells whose paralogue cannot be assigned, and **none is absent** (Fig. 5). In the 189 assemblies contiguous enough to carry the gene, the minimum number of gene-equivalents any genome holds is 3.00.

Dollo parsimony over that matrix places **zero losses** on the family character and zero on each of *ITPR1*, *ITPR2* and *ITPR3* — from a routine that finds a constructed loss on a known edge, merges sister losses into their parent, and bounds rather than counts losses under a polytomy. What we report instead is the sensitivity map: 32 settings (an eight-rung evidence ladder × two decision rules on and off) × two codings × three branch-length schemes. The family-level coding manufactures a loss in **2 of 32** settings, worst case one genome in 309, and only by refusing everything except a complete locus *and* ignoring the contiguity bar; the paralogue-resolved coding manufactures one in **18 of 32**, up to 45 loss edges (Extended Data Fig. 10). Moving the reconstruction bar across the whole gap its own calibration measured changes no cell in 927; the branch-length axis changes the count in none of the 32 settings.

Continuous-time Markov models are **stated, not fitted**: with no variation in the character there is no transition to estimate, and every likelihood is monotone to its boundary on every model, axis and scheme, measured on a rate grid rather than asserted. All twelve fits at the operating point are refusals. There are likewise **no pseudogene fossils** to test for shared lesions: of 1,760 scored loci, 44 clear the frameshift-density bar and 44 of 44 are at full coverage with an intact model.

Selection is asymmetric, and three independent framings agree. From a codon alignment in which all 57 vertebrate tips carry a nucleotide sequence that provably encodes the exact protein aligned (2,459 codons after codon-aware trimming), the one-ratio ω is **0.0238 for *ITPR1*, 0.0415 for *ITPR3* and 0.0430 for *ITPR2***, and the ordering survives restricting to curated coding sequences (0.0212 / 0.0371 / 0.0318). Whole-tree two-ratio tests are significant for all three after correction: *ITPR1* foreground 0.0241 against background 0.0432 ($q = 2.9 \times 10^{-63}$), *ITPR2* 0.0435 against 0.0317, *ITPR3* 0.0455 against 0.0308. RELAX, which compares whole ω distributions rather than point estimates, returns k = 9.36 **intensified** for *ITPR1* and k = 0.91 and k = 0.84 **relaxed** for *ITPR2* and *ITPR3* (Extended Data Fig. 11). *ITPR1* — the paralogue that was doubled and kept after the teleost duplication, and the one whose neighbourhood carries both surviving two-round links — is also the one held about twice as tightly.

Three model-based results needed their fitted parameter rather than their p-value before they could be stated. M8 beats M7 in all three paralogues at $q \approx 1 \times 10^{-4}$, but its extra class sits at $\omega = 1.00000$, codeml's boundary, on 0.3–0.7 % of sites with 0, 0 and 1 sites reaching a Bayes empirical Bayes posterior of 0.95 — a small class of *unconstrained* sites, not positive selection; M2a versus M1a gives $2\Delta\ln L = 0.00$ in all three. Branch-site model A is significant on all three stems, but its foreground ω is pinned at codeml's 999 bound on the *ITPR2* and *ITPR3* stems with the likelihood flat above it, so only ***ITPR1*'s stem** is reported as a result: $\omega_2 = 5.53$ on 11.0 % of sites, stable across three of four independent starting values, with 8 sites at posterior ≥ 0.95. Three of twelve branch-site restarts converged below their own nested null, one per stem and at a different starting ω each time.

One decay signal, and it does not survive its control. Frameshifts and premature stops per kilo-aligned-residue were counted at every placed locus and compared within genomes, each *ITPR* cell against its own genome's other family loci. *ITPR3* carries an excess of lesions against identity-matched siblings (39:14, q = 0.0032); *ITPR2*'s apparent excess disappears under identity matching (q = 0.90) and was an alignment artefact of bait distance, which is the dominant confounder here (Spearman ρ = (-)0.397 against (-)0.077 for contig N50). Stratified by class the *ITPR3* signal is a bird result (25:2, $q = 4.5 \times 10^{-5}$ in Aves against 7:6, p = 1.0 in ray-finned fish), but 21 of its 27 pairs sit below the contiguity bar, so we report it as underpowered rather than as decay.

The comparison is saturated at synonymous sites. 88.8 % of all within-paralogue pairs are past dS = 1.5, with dN plateauing near 0.1 while pairwise dS runs past 50, so the pairwise matrix is a diagnostic and every ω quoted here is tree-based.

What the receptor cannot change, and what that says about its variants

Per-residue constraint was computed in each human paralogue's own numbering on four nested alignment layers: a **deep** layer of 249–265 orthologues of that one paralogue drawn from the genome sweep, a **vertebrate** layer, a **family-wide** layer of all 128 IP_3 receptor tips, and a **shallow** layer of that paralogue's representative sequences kept as the control for what the deep sets bought. Scores are sequence-weighted before any column statistic and gaps are treated as missing data, not as a twenty-first state.

The gate and the filter are the least changeable parts of the receptor, and the gate is identical between the paralogues. On both a divergence metric and a composition-free one, the gate and the selectivity filter rank at the top of every element in all three paralogues (gate mean JSD 0.799, 0.822, 0.822; filter 0.787, 0.797, 0.811, against a linker control of 0.721–0.726; Extended Data Fig. 12), and the gate's five lining residues are **100 % identical between all three human copies**. The most constrained whole domain is RIH-associated (0.798–0.804).

A 50-residue luminal loop is the least conserved sequence in the receptor, and it sits about fifty residues from the most conserved. Unresolved, the channel Pfam domain scored *below* the linker control ($p = 0.96$), which survived both a metric control and a gene-model control. The cause is a stretch of the channel domain that no annotation carries: we located it geometrically, as the contiguous run of channel residues whose Cα lies beyond the membrane on the luminal side of the measured axial span (*ITPR3* 2401–2450). It is the least constrained element in the receptor (mean JSD 0.452–0.485 against 0.721–0.726 for the linkers), the least conserved between paralogues (identity 0.13–0.31 against 0.64–0.70 whole-protein), and the one region where site-wise selection is weak (FEL median β 0.33–0.39, 27–33 % of sites purifying against 71–86 % protein-wide). With it separated out, the pore module clears the linker control ($p = 0.005$). The most and least constrained sequence in the receptor lie about fifty residues apart in the same Pfam domain (Fig. 6), and the next result turns on which of them counts as pore.

The ligand site is not in the domain named after the ligand, and what selection holds there is a pocket rather than a contact set. None of the ten IP_3 contacts measured on the 6DQN structure [22] (≤ 4.5 Å) falls inside PF08709, the signature called *Inositol 1,4,5-trisphosphate/ryanodine receptor*; all ten sit in MIR and in the N-terminal RIH domain, which the family shares with the ryanodine receptors — the same domains the crystal structures of the isolated binding core identified [23, 24] and the region whose architecture is conserved between the two families [25]. To ask what those residues are special *against*, we re-measured the site as a distance rather than a label: every residue within 15 Å of the ligand, all-atom, in six independent IP_3-bound human IP_3R_3 depositions. All six recover all ten of the published contacts — the positive control the measurement was required to pass — and agree on **two more**, Ala276 and Arg411, which sit inside 4.5 Å in a majority of structures and outside it in 6DQN alone. The ten contacts are more constrained than the receptor as a whole in all three paralogues ($p = 0.013$, 6×10^{-4}, 4.2×10^{-3}) and more constrained than the rest of the binding core in all three ($q = 0.048$, 0.017, 0.041). Against every *other* residue the structure places within 15 Å of the ligand they are more constrained in **none** ($q \geq 0.12$). Every shell out to 15 Å sits above the whole-protein mean and there is no step at 4.5 Å, and site-wise selection replicates it on an independent axis: **100 % of contact sites in all three paralogues are under detectable purifying selection**, and that share falls by 0.13, 0.25 and 0.18 to each paralogue's lowest shell (Extended Data Fig. 13). The fall is not monotone — the outermost shell sits slightly above the third in all three — so what the data show is a step down from the ligand's first two shells onto a floor rather than a gradient running all the way out. The constrained unit is a neighbourhood about fifteen ångström across, which a contact/not label cannot see.

The module that names the family is the less constrained of the two, and the answer reverses on a boundary nobody states. The IP_3-binding core is the one part of this receptor the ryanodine receptors do not share functionally, which makes it the obvious place to expect the tightest constraint. It is not. Measured per orthologue and paired inside each paralogue's own deep alignment — one core number and one pore number per sequence, 246–262 sequences per paralogue, so a generally divergent tip contributes a divergent core *and* a divergent pore and only the difference enters — the **pore module is ahead by about two identity points in *ITPR1* ((-)0.0239, 223 tips to 32, q = 4 × 10^{-34}) and *ITPR3* ((-)0.0199, 218 to 42, q = 2 × 10^{-28}), and level in *ITPR2*** ((-)0.0005, q = 0.13). That result depends entirely on one boundary. Leave the 50-residue luminal loop inside PF00520, which is how InterPro draws the domain and therefore how the comparison would be made by default, and **all three paralogues flip**: the ligand core wins by four to six identity points at q < 10^{-37} in every one. Neither answer is wrong about its own module; they are answers about different modules, and any version of this comparison that does not declare where the pore stops is not interpretable.

Losing the enzyme that makes the ligand does not relax the site that binds it. The lineages with a reduced upstream pathway were derived rather than read: phospholipase C profiles for both halves of the catalytic barrel were swept over every eukaryotic reference proteome in the scope — the 763 vertebrate and 2,764 other eukaryotic sets, 3,527 in all, the prokaryotes being irrelevant to the question — and **64 carry an IP_3 receptor and no phosphoinositide-specific phospholipase C** — concentrated in the oomycetes and the early-diverging fungi, every one of them holding a full-length receptor, in a gene set complete enough to have held the enzyme. Pooled over 36 against 450 records their ligand core looks relaxed (p = 9.7 × 10^{-6}), and that is a clade artefact: they sit at a median pore identity of 0.358 against 0.589 for the rest, so the comparison is largely distant against near. Matched on divergence — each record scored against phospholipase-carrying records within 0.03 pore identity of itself, all 36 matched — **the effect is gone**: median within-pair difference (-)0.0064 (95 % CI (-)0.016 to +0.015), 19 tips against 17, p = 0.87. This is a bounded null rather than an absent measurement. Measured through the same instrument at the same divergence, the ryanodine receptors — same pore, same diagnostic domains, no IP_3 site — shift by (-)0.062, and the matched test detects a shift that size essentially always and one half that size 59 % of the time (Extended Data Fig. 13).

Constraint separates the clinical variants, and the deepest layer is not the best classifier. 1,753 ClinVar missense records — the family's disease record runs from spinocerebellar ataxia and Gillespie syndrome at *ITPR1* [9, 11, 12] through anhidrosis at *ITPR2* [13] to neuropathy at *ITPR3* [14, 15] — were placed on the canonical sequences with every cited transcript's own coding sequence translated and checked residue by residue; none was dropped, and both of the literature's residue-level citations were recovered by a positive control the harvest could have failed. **1,546 (88 %) are variants of uncertain significance**, and *ITPR2*'s entire pathogenic missense record is a single variant. Scored as a classifier of pathogenic/likely-pathogenic against benign/likely-benign on the one fixed set of 44 versus 34 positions that every layer scores, the ranking is **family 0.872 > vertebrate 0.854 > deep 0.758 > shallow 0.684** (Extended Data Fig. 12) — against this task's own design, the 249–265-orthologue sets are the second-best of four, and the whole-family layer that ignores paralogue identity is the best. Site-wise selection agrees with constraint and adds nothing new: FEL finds 5,766 purifying sites across the three paralogues and **one** diversifying site, inside its own false-discovery budget.

Structures confirm the family call by shape, and show why they cannot do more. AlphaFold DB holds a usable (≥ 95 % covered) model for 20.9 % of the census's 8,319 UniProt-shaped IP_3 receptor records — but of the 5,861 at or above the family's 2,000 aa floor, **13 (0.2 %)** do, and of the 134 representatives every alignment and tree here stands on, **9** do. The median modelled record is 392 aa and the median unmodelled one 2,674 aa: the coverage is concentrated on fragments. A panel of 29 structures — 5 cryo-EM references resolved by query and assigned by the census rather than by entry title, a 5-state conformational panel, 3 size-matched negative controls and 16 predicted models — gives 20 of 20 agreement between the structural family call and the census, and no family call at all for any of the 3 negative controls (Extended Data Fig. 14). Calibration: two conformations of the same protein score a median TM of 0.78, two IP_3 receptors 0.43, an IP_3 receptor against a ryanodine receptor 0.39, and the negative controls 0.19 with 0 of 81 pairs reaching the 0.50

same-fold bar. Per-domain confidence explains what predicted structures are good for here: the IP_3-binding core is the best-modelled domain (median pLDDT 83.9) and the pore the worst (71.0), against 69.5 outside annotated domains.

The family is not badly recorded; the archive is

Every one of the 2,144 gene-scale loci was scored against the assembly's own annotation — same strand only, on coding blocks and never on gene spans — as complete, split, fragmentary, held only by a non-coding feature, or unannotated, with the ryanodine receptors scored the same way in the same assemblies through the same pipelines as the control.

The audit's own premise is contradicted by its control. IP_3 receptor loci are 73.9 % complete and 26.1 % failing; the ryanodine receptor control is 77.9 % and 22.1 %, and no overall difference survives multiple-testing correction (q = 0.13 over all scorable loci, q = 0.82 above the contiguity bar). One state does separate, and it is the family-specific one: an IP_3 receptor locus is 2.7 times more likely than a ryanodine receptor locus to be held *only* by a non-coding feature (42 against 16, q = 0.006) — a gene the annotation identifies and names correctly, files as a non-coding biotype, and therefore serves no protein for. Above the contiguity bar that excess is 4 against 5 (q = 1.0), so it is confined to assemblies too broken to carry the gene.

What decides whether the gene is recorded is the archive, not the gene (Fig. 7). RefSeq annotations are 98.8 % complete against submitter-deposited GenBank annotations at 37.5 %, with every state separating at $q < 10^{-300}$. Controlling for contiguity narrows the gap without closing it: 99.4 % against 63.8 %. Assembly contiguity is the second-largest effect — the IP_3 receptor failure rate falls from 26.1 % to 6.7 % across the bar. The completeness bar itself was inherited and validated rather than re-derived: over 1,077 correctly named, fully recovered loci a single annotated model covers a median 0.993 of the alignment, so the 0.50 bar sits at that distribution's 1.3 % point (Extended Data Fig. 15).

A "fragment" in the archive is an annotation that stopped where nothing splices. The 264 fragmentary and 27 split loci are a statement about coverage and say nothing about *where* the annotation stopped, and the difference is the whole claim: a model ending at a genuine junction has produced a plausible short gene, one ending inside an exon has produced a boundary no splicing machinery could make. Scoring each model's own internal termini against the alignment's exon boundaries and against the two genomic bases immediately outside them, **228 of the 291 loci carry at least one terminus inside an exon** and **3 are broken entirely at junctions the gene has** (Extended Data Fig. 9). Of 1,448 internal termini, 406 sit inside an exon and 353 inside an intron, against 689 (47.6 %) on a boundary the gene model carries. The verdict is one-sided by design — a mid-exon terminus falsifies the reading that the annotation found a real gene end, while a terminus on a boundary does not prove the pieces are separate genes — and on that one-sided reading almost every fragmentary record in this family is an annotation failure rather than a gene boundary.

On the protein side the family call is not in dispute; findability is. Of 11,402 full-length family records searched against the labelled bait panel, 4 disagree with the census on family by sequence and 5 carry a name from the wrong family, all non-vertebrate and all under 200 bits; 52 of 8,306 vertebrate symbols name the wrong paralogue. But **3,872 records carry a placeholder symbol and 2,395 carry none at all — 55.0 % of full-length family records have no usable gene symbol** — and 66 more are named for the superfamily rather than the family. All 15 reference proteomes that returned no family protein at all are gene-caller failures: every one of those species has a genome in the scope and every one of those genomes carries the gene.

The consequence, counted. Of 1,232 genes demonstrated to exist in these assemblies, **940 (76.3 %) cannot be reached by any protein-database search** — 386 in species with no reference proteome, 248 where only fragmentary records exist, and 286 where records exist but none resolves to that paralogue. This is not an artefact of the margin species the scope was extended for: 74.3 % for order representatives against 78.5 % for margin species.

Two failures validated to the exon, and both genes are transcribed. Where the annotation demonstrably could have delivered the gene — 382 loci passing five eligibility rules, of which the decisive one asks whether the same annotation builds genes that long anywhere else in the same genome — it gets it right at **375 of 382 (98.2 %)**, with a median annotation loss of zero and 360 loci at exactly zero. The seven failures fall in three genomes and cover all three failure modes. We validated one omission and one fragmentation at nucleotide resolution (Extended Data Fig. 16).

Nibea albiflora ITPR2 has 56 coding exons and **not one annotated gene model over any of them**, in a chromosome-level assembly with 84-fold headroom whose annotation gets *ITPR1* and *ITPR3* right. The locus splices into 2,673 codons with **zero internal stops**, against 14.2 expected under neutral drift at the observed divergence; all 55 introns carry a spliceable dinucleotide (54 GT-AG, 1 minor); 52 of 55 exon boundaries are shared by a majority of 35 independently annotated genomes; and it sits between `SSPN` and `BHLHE41`, this paralogue's two most conserved neighbours across 197 and 183 of 215 swept vertebrates. The same genome files every complete family gene as a pseudogene and names its *ITPR3* locus *ITPR2*. *Dissostichus eleginoides ITPR3* is one 68 kb gene called as three protein-coding models tiling residues 1–67, 52–467 and 468–1593, with the 3′ 41 % unmodelled; all 59 introns are GT-AG and 59 of 59 boundaries are shared by a majority of 40 genomes. Both species have **zero IP_3 receptor protein records in any database and three complete IP_3 receptor genes in their DNA.**

Reads settle it. 67 public RNA-seq runs (536 million reads) across three species were aligned against a per-species reference of spliced genomic coding sequence with a reversed, composition-matched decoy for every sequence. Each failed locus is detected in 13–31 of its runs and in 6–9 tissues, and **298 of the 314 junctions (94.9 %) that no annotated model spans are crossed by reads that read through them**, against 79 of 83 (95.2 %) of the annotated junctions in the same genes. The closed-set risk was measured rather than argued: every reference tiled exhaustively with synthetic reads gives 0 of 12,500 cross-mapped.

The audit ends in a list addressed to somebody else: **297 corrections** — assembly, coordinates, current state, current name, proposal and archived evidence — of which 52 are high priority and 18 are recorded as withheld, with the reason, rather than dropped.

Discussion

An ancient family that vertebrates cannot do without and other lineages have dropped repeatedly. The IP_3 receptor is present across the eukaryotic tree, in animals, amoebae, discobids, ciliates, oomycetes, dinoflagellates, haptophytes and green algae, and its duplication from the ryanodine receptors predates the animals: two unicellular holozoans carry both families at full length. Set against that background, the absences are the informative part, and they are not one event. Land plants have none, and their sister green algae do. The Dikarya have none, and neither do four other fungal phyla that branch apart from them, so at least five independent fungal losses are implied. Apicomplexa, Microsporidia, Rhodophyta, diatoms and one tapeworm class complete a list of lineages that lost a channel otherwise kept for a billion years. These are eukaryotes with reduced or re-routed calcium signalling — obligate parasites, walled autotrophs, yeasts — a pattern consistent with wider surveys of the eukaryotic calcium toolkit [19, 20, 26], which makes the correlation worth naming and not worth over-reading from a presence table. What can be said, and what previous surveys could not say, is that each absence is a claim about a declared space in which a comparably long, deeply conserved gene *was* found by the same instrument in the same assembly.

Complete retention within the vertebrates is the sharper result. In 309 genomes spanning every vertebrate order, no paralogue is lost: not once in 927 genome × paralogue cells, and in every assembly contiguous enough to carry the gene the genome holds at least three gene-equivalents. Because a count of zero is easy to obtain by not looking hard enough, the design was inverted — the search's own sensitivity was measured against an independent replicate (the ryanodine receptor cells, 13.6 % missed against 15.2 %), the loss state was shown to be reachable by construction, and instead of a robustness check we report the settings that manufacture a loss: the family-level count breaks in 2 of 32 and only by discarding two decisions made for reasons unrelated to the count. A gene family that a whole vertebrate lineage has never dispensed with, against a eukaryotic background of repeated loss, is a statement about vertebrate physiology rather than about the protein.

Retention is not symmetric, and the asymmetry has one direction. *ITPR1* is held about twice as tightly as its sisters by three independent framings (one-ratio ω, whole-tree two-ratio contrasts, and a distribution-level relaxation test), is the only paralogue whose stem carries an interpretable episodic signal, is the only one doubled and kept after the teleost genome duplication, and is the paralogue through which both surviving two-round paralogon links run. The one decay signal in the data points the other way, at *ITPR3*, and does not survive its own contiguity control. The obvious reading — that *ITPR1* carries the ancestral, least substitutable role and its sisters have been freer to diverge — is consistent with what is known of their expression and channel behaviour [6, 7, 8, 27] and with the phenotypes of the single-gene knockouts [28, 29], and this work supplies the evolutionary half of it rather than proving the functional half.

The sister question, and a tension worth keeping. *ITPR2* and *ITPR3* are sisters on sequence: the alternatives are rejected at $p < 2 \times 10^{-5}$ and the grouping survives a model-violation guard. Yet those are the two neighbourhoods that share no dated ohnologue link, while both surviving links attach *ITPR1* to one of them. The two instruments are measuring different things — a tree estimates the order of duplication, a retained ohnologue records which copies escaped deletion afterwards — so this is a tension rather than a contradiction, and the honest form of the result is to report both.

The two families share a fold and not a gene. The ryanodine receptors carry every Pfam signature used to diagnose this family, and that is the single fact that shaped every search in this paper. Intron positions are a character no protein alignment produces, and measured on them the two families have nothing in common: an IP_3 receptor and a ryanodine receptor in the same genome share a median of one intron position, in none of 183 to 188 genomes at $p < 0.05$, while the three IP_3 receptor paralogues share 46 to 49 in every genome tested. The shared domain architecture is real and it is old, but it was not inherited as a gene, and the practical consequence for anyone searching this superfamily is that domain content is the wrong evidence to separate them on — which is the conclusion this project reached by a different route at every earlier stage.

The part that names the family is not the part selection holds hardest. The IP_3-binding core is what distinguishes these receptors from the ryanodine receptors functionally and what the family's diagnostic Pfam signature is named for, so it is the natural place to expect the tightest constraint. Paired inside single orthologues against about 250 sequences per paralogue it is the *less* constrained of the two functional modules, and the finding that matters more than the direction is that the direction reverses on where the pore is taken to stop: fifty residues of luminal loop, which InterPro includes in PF00520, are enough to flip all three paralogues. A comparison between two protein modules is only as good as the boundaries drawn around them, and those boundaries are usually inherited from a database rather than declared. Within the site, what selection holds is a pocket about fifteen ångström across rather than the ten residues that touch the ligand — visible only because the site was measured as a distance in six independent structures rather than taken as a contact list from one. And losing the enzyme that makes IP_3 does not relax that pocket: matched on divergence the effect is a bounded null, against a positive control the same test detects almost always. Whatever holds the binding site in these lineages, the canonical route to its ligand is not required for it.

A gene the archive cannot find. Three quarters of the genes demonstrated here are unreachable by any protein-database search, more than half of the full-length records that do exist carry no usable gene symbol, and 318 IP_3 receptor genes have no protein record of any kind. The audit that was supposed to show this family is poorly recorded instead showed, through its own sister-family control, that it is recorded about as well as its neighbour, and that what predicts the record is the archive the assembly went to and how broken that assembly is. Two consequences follow. For this family specifically, any survey based on protein records — including every previous survey of its distribution — is working from about a quarter of what exists. For gene-family surveys generally — including the ones this family's own distribution has been read from [17, 18] — the result argues that the denominator has to be assemblies, that absence claims need a positive control in the same assembly, and that a sister family searched alongside is worth more than any threshold.

Limits. The vertebrate scope is one genome per order plus the species the protein record left in doubt, not every vertebrate genome; a paralogue lost in a single species inside a sampled order would not be seen. 120 of 309 assemblies cannot hold the gene on one contig, and every result is therefore reported twice, above and below that bar. The absence claims outside the vertebrates are claims about 6,928 reference proteomes and 194 genomes, sampled most finely where the claims are, and a clade represented by one genome is a clade whose absence rests on one genome. Synonymous saturation means every ω here is tree-based and no pairwise distance is interpretable. Predicted structures cover 0.2 % of the full-length records, so nothing structural in this paper rests on a model of a whole subunit. And the deepest constraint layers built here are not the best variant classifier: the family-wide layer is, which is a result about what conservation-based variant scoring should use, and a caution about assuming that more orthologues of the same paralogue is the same thing as more signal. The module and pocket comparisons run on those same deep alignments, which are vertebrate by construction, so the ligand core's position relative to the pore is a vertebrate statement; and everything within 15 Å of the ligand is also inside the fold that holds it, so no measurement here separates constraint on ligand binding from constraint on domain packing. That would take a mutational or binding dataset; sequence will not do it.

Methods — search and scope

All analyses are scripted, and every table, figure and report in the deposit is regenerated from committed inputs by a named script. Tool versions are recorded in `results/toolchain_manifest.txt`: MAFFT v7.526, HMMER 3.4, BLAST+ 2.16.0+, miniprot 0.18-r281, trimAl v1.5.rev1, IQ-TREE 2.3.6, PAML 4.10.10, pal2nal v14, HyPhy 2.5.101, TM-align 20240303, HISAT2 2.2.3, sra-tools 3.4.1, NCBI datasets 18.35.0. Where a tool's output depends on thread count it was run at a pinned thread count and the command line, version, runtime and SHA-256 of every input and output recorded; MAFFT L-INS-i and IQ-TREE are not reproducible on this machine at automatic thread selection, and both were pinned for that reason.

The family definition lives in one module: the Pfam signatures PF08709 (IP_3-binding core), PF02815 (MIR), PF01365 (RIH) and PF08454 (RIH-associated), the 2,000–4,000 aa size band, the human reference and sister-family panels, and the known-name substrings. Nothing else in the code names a gene.

Separating the two families is a positive test at every stage. Because RYR1 carries all four IP_3 receptor-diagnostic Pfam signatures, no stage assumes a family: the census call requires the *complete* IP_3 receptor architecture with none of the ryanodine receptor-specific signatures, with length as support only; profile assignment requires the winning profile to beat the loser by more than 10 % of its own bit score; genomic loci are assigned to a family by alignment score before any paralogue question is asked; and structures require the winner to clear TM-align's own 0.50 same-fold bar before a family call is made at all. The rule is audited against evidence it never sees — gene symbols for the census call (6,191 labelled records, 0 disagreements), the architecture call for profile assignment (11,875/11,876 agreement outside the seeds).

Protein-level enumeration. The four family signatures were walked to exhaustion through the InterPro API with every raw page archived and the census re-parsed from the archive, then joined to one streamed UniProt query carrying full Pfam architecture, length and lineage. A PF08709-only census would have missed 2,914 records, 758 of them called IP_3 receptor across 385 taxa, so the enumeration uses the union of the four.

Profile sweeps. Two profile HMMs were built from census-derived seed sets under five enforced selection rules — a seed whose census call, architecture or length has drifted since the manifest was written aborts the build — and swept over 763 vertebrate reference proteomes (14,414,821 proteins, 6.97 G residues) and, in a second sweep, 6,928 archaeal, bacterial and non-vertebrate eukaryotic reference proteomes (63,144,898 proteins, 24.93 G residues) whose four eukaryote groups partition Eukaryota against the vertebrate set. Every search ran at a relaxed reporting threshold with the primary $E \leq 10^{-5}$ call taken as a filter on the same output, so the two sensitivities come from one search; the equivalence of that design was tested rather than assumed, and the test found two real effects (conditional E-value normalisation, and two-significant-figure printing at the threshold) neither of which moves a call. A match spanning fewer than 200 profile match states is counted but not enrolled, a floor measured from this family's own domain coordinates; without it 12,874 vertebrate proteins carrying only a SPRY domain were called ryanodine receptor.

Completeness. jackhmmer was iterated to convergence from single sequences in seven runs across vertebrate, invertebrate, protist, plant and fungal databases, with three coded kill rules evaluated per round on that round's own inclusion list. In all four non-vertebrate groups the iterated model settles on exactly the single pass's IP_3 receptor count, so iteration finds no receptor the single pass missed. Completeness was also measured in the other direction: each protein-census record the sweep failed to return was looked up in the 8.8 GB sequence database itself, and none was present and missed.

Genome scope. The vertebrate denominator is 309 assemblies, declared before the sweep: the best assembly of each of 161 vertebrate orders, ranked by annotation status, RefSeq category, assembly level and scaffold N50, united with 169 margin species derived from the committed census tables by four stated rules (no protein hits in the reference proteome; fewer than three paralogues; only fragmentary records; literature anchor). The non-vertebrate denominator is 194 assemblies chosen by five rules from 24,596 NCBI eukaryote reference assemblies less the vertebrates, under one principle — sample most finely where the negative claim is — with per-phylum and per-class representatives, one per class in every clade the proteome sweep found empty, 14 literature anchors and the highest-copy species per class.

Genomic sweep. A 38-bait panel (30 IP_3 receptor, 8 ryanodine receptor as positive control) derived from the census by seven enforced rules was aligned to each vertebrate genome with miniprot; empty cells were re-asked by whole-genome tblastn and attributed family-first. The maximum intron setting is a threshold on the call, not a performance knob — too small a value splits a gene and a split gene reads as a fragment — so it was measured: no IP_3 receptor gene in an 11-species panel has an intron over the 200 kb default (widest 152,216 bp) while the ryanodine receptor control exceeds it twice. The rescue attribution margin inherited from a related project (0.333) was re-measured at loci whose paralogue the assembly's own annotation establishes, where 7 of 9 complete loci fall below it, and reset to 0.22; the reset was then validated on the evidence it acts on, 53 of 53 rescue fragments agreeing with the annotation over margins 0.227–0.445. The non-vertebrate sweep uses a 108-bait panel with per-group intron and contiguity settings measured the same way, and identity to the nearest bait was retired as a call gate there after it was shown not to separate confirmed from contradicted loci (Youden J 0.71); the profile call carries it instead.

Positive controls for absence. In the vertebrates the ryanodine receptors are the control and fired in all 309 genomes. Outside the animals they are not a control, because architecture-level ryanodine receptor occurs in 2 of 6,928 non-metazoan proteomes; six candidate control profiles — all large, deeply conserved, multi-exon eukaryotic families — were therefore measured over each clade's own swept proteomes and each clade takes the one that is present in it. A genome that returns neither its own control nor a cross-kingdom one supports no absence claim.

Contiguity. The contiguity bar is the median measured IP_3 receptor gene span, 142,212 bp: an assembly whose contigs are shorter cannot carry the gene on one contig. It was chosen from gene geometry with no error rate in it and calibrated only afterwards, where it gives a 0.9 % residual false-negative rate on the IP_3 receptor series and 0.0 % on the ryanodine receptor sister series while retaining 189 of 309 genomes. Every result is reported above and below it.

Methods — alignment, phylogeny and downstream analysis

Representatives and alignment. 134 representatives were chosen from the 18,065-record census by eight coded rules, per clade and per kingdom, never longest-per-species: a longest-first pick reliably selects chimeric gene models. No rule ranks or filters on sequence identity, because identity was retired as a call gate outside the vertebrates; a paralogue label is vertebrate-only, because four non-vertebrate records carry a type number by annotation transfer; and each group's length target is its own measured median rather than a global one, since the family's size varies far more outside the vertebrates than within. MAFFT L-INS-i (`--thread 1`) gave 11,777 columns and trimAl `-automated1` kept 1,797 (15.3 %, 7.7 % gaps), 96.8 % of them parsimony-informative. A ragged alignment is a hard failure: a silent MAFFT failure degrades to a star alignment with no other symptom.

Phylogeny. ModelFinder's exhaustive scan was started, measured at 11 of up to 1,232 models in 11 min, projected at ~21 h and abandoned for a two-stage greedy scan with both stages committed; `Q.insect+R7` was chosen (the free-rate models are not optional here — `LG+R5` beats `LG+I+G4` by 682 BIC units). IQ-TREE 2.3.6 with 1,000 ultrafast bootstrap and 1,000 SH-aLRT replicates, threads and seed pinned, gave log-likelihood (-)215,452.0. Sister hypotheses were tested with the approximately unbiased test over 10,000 RELL replicates on three constrained maximum-likelihood trees, each constraining only the taxa its hypothesis is about — a constraint naming every tip pins tips outside the groups it declares and makes all three hypotheses fail together. The search was repeated with an extra nearest-neighbour-interchange round per bootstrap tree as a model-violation guard, and every clade claim re-asked of that tree from the same committed tip sets.

Synteny. Flanking genes were read from each assembly's own annotation under two window rules (($\pm$)10 coding genes; the 10 nearest *informative* symbols each side) and three symbol vocabularies, the third stripping trailing digits so that a two-round ohnologue pair is visible at all. Every real locus pair is scored against matched random-window control pairs drawn in the same two genomes, seeded per accession, and compared by a sign test with ties dropped and counted. The paralogue caller's operating point was chosen by maximising call rate minus random-window false-call rate, not call rate alone.

Duplication. Human paralogy came from a pinned Ensembl BioMart archive (Ensembl Genes 116) with the archive's registry committed beside the map, so a re-run scores the same windows against the same gene tree. Every IP_3 *and* ryanodine receptor gene is removed from every window before testing, and the null is a permutation over *real* genomic windows at the matched gene count. Links carry their Compara duplication node, which is what separates a two-round ohnologue pair from an older duplication whose copies happen to lie in these blocks. The 309-genome replication uses the synteny task's own control sampler unchanged.

Reconciliation. The species tree is an input, not a result: 31 species, 29 named internal nodes, each with a published crown age, the spread of published estimates, a stem age and its source, validated by a checker that fails on an uncalibrated node, an age inversion or a species mismatch. Duplications were called by the non-binary LCA rule and losses per Zmasek & Eddy; the familiar binary rule reads a support-collapsed four-way node as a speciation. Five topologies × three tip variants were reconciled, and the vertebrate subtree re-rooted at all 112 of its edges.

Gene architecture. Exon blocks come from the same spliced alignments the census was built on, read in target order so a minus-strand gene's first block is the one carrying residue 1. What counts as an intron was calibrated against the genome rather than chosen: block gaps were binned and scored by their splice dinucleotides, and the floor was put at the smallest gap the genome calls spliceable. Intron positions are a pair — the alignment column the upstream exon ends in, and the classical phase — carried into one frame through a bait-to-human-reference-to-column map whose anchor test requires all 14 residues measured on the reference structure to land in one column in all three paralogues. The shared-intron null is exact: each intron of one gene placed independently and uniformly on the columns both loci resolve, keeping its own phase, which makes the match count a Poisson-binomial whose upper tail needs no simulation; a seeded permutation is run beside it and both are committed. Every comparison is paired inside one genome.

Selection. Every vertebrate tip carries a coding sequence that provably encodes the exact protein aligned: three independent routes each generate *candidates*, and the first that validates by translation wins, because a UniProt entry lists every transcript of its gene and in both human *ITPR1* and *ITPR2* the first is not the aligned isoform. Every translation disagreement is masked to NNN. PAL2NAL output was cross-checked nucleotide-by-nucleotide against an independent in-house codon mapping, and trimAl columns chosen on the protein were applied codon-aware, whole triplets only. Selection sets are the tree's own extended paralogue clades, cross-checked against the committed clade table, because a foreground that is not a clade does not fail — it silently marks a larger one. Branch-site model A was restarted from four initial ω by construction; the branch-site likelihood-ratio test is reported against a 50:50 mixture as well as ${\chi^2}_1$; and Benjamini-Hochberg correction runs across the whole 12-test family.

Constraint and variants. Conservation was computed per residue in each human paralogue's own numbering on four layers, sequence-weighted before any column statistic, with gaps excluded from the distribution and the occupancy each score is conditional on reported beside it. Pfam element coordinates were measured per accession and not transferred; the structural elements (selectivity filter, gate, IP_3 contacts) were transferred from PDB 6DQN with an anchor test that aborts rather than writing a coordinate. The luminal loop has no annotation and was located geometrically from the structure's own membrane span. ClinVar records were placed only after each cited transcript's own translated coding sequence was aligned to the canonical and the reference amino acid required to match. Classifier performance is reported on three contrasts, the decisive one restricted to the positions every layer scores.

The ligand site. The two functional modules are defined here rather than taken from Pfam, twice each: a primary definition derived from measurement (the ligand core as the minimal span holding every measured IP_3 contact; the pore as PF00520 less the geometrically located luminal loop) and a sensitivity definition taken from how the field draws the same region (the published binding core; PF00520 as InterPro draws it). Every test is run under all four combinations. Each definition is checked against something it does not contain — the core must hold all ten contacts and no filter or gate residue, the pore both filter and both gate residues and no contact — and a definition that fails raises rather than being written. The pocket was measured all-atom, not on Cα, within 15 Å of the ligand in six independent IP_3-bound depositions, with recovery of the ten published contacts required as a positive control. The module comparison is paired per orthologue, one core and one pore number per sequence, with a tip required to resolve half of *both* modules to enter. Phospholipase C presence requires both halves of the catalytic barrel in one protein; the lineage comparison is matched on pore identity within 0.03, and its power is reported against the shift the ryanodine receptor control produces on the same instrument.

Structures. References were resolved by an RCSB query over the union of the family's Pfam signatures — a PF08709-only query misses this project's own IP_3 receptor reference — and assigned to a family by the census, never by entry title. Every structure is reduced to its largest chain before scoring, because both families are homotetramers and an assembly is ~11,000 residues. AlphaFold DB coverage is reported three ways, and only the third — resolved residues over the census length — detects that the database answers a canonical accession with an isoform (181 residues of a 2,701-residue *ITPR2*).

Expression. Per-species references are spliced *genomic* coding sequence, deliberately not frameshift-corrected, with three housekeeping anchors built by the same pipeline and a reversed, composition-matched decoy for every sequence. Runs were selected by keyword and verified against each run's own sample attributes. Alignment ran with spliced alignment disabled, because the reference is already coding sequence and a junction read is contiguous in it.

Negative controls. Every stage from the bait screen onward runs a suite of constructed negative controls before it writes anything, and refuses to write if one fails: 11 for synteny, 12 for the codon alignment, 14 for constraint, 15 for the loss instrument, 21 for the loss counts and 21 for gene architecture, 31 for duplication, 32 for the methods results, 44 for the annotation audit and 44 for the ligand site, and 47 for the annotation-bug validation. Each suite was itself mutation-tested by breaking a rule deliberately and requiring the suite to catch it. Several caught real defects, including a within-protein control that had swept in the 225-residue domain the ligand question is about, a hash-seeded non-determinism in a ranked table, and a self-test that was overwriting the committed table it was checking.

Reproducibility. Reports are rendered from committed tables by script and contain no hand-written numbers; figures are drawn only from committed tables and are copied, never re-plotted, into the manuscript; and every load-bearing number in this paper is declared in a claims ledger with its source table and the operation that recovers it, re-verified on every build (`manuscript/claims_check.tsv`).

Figure legends

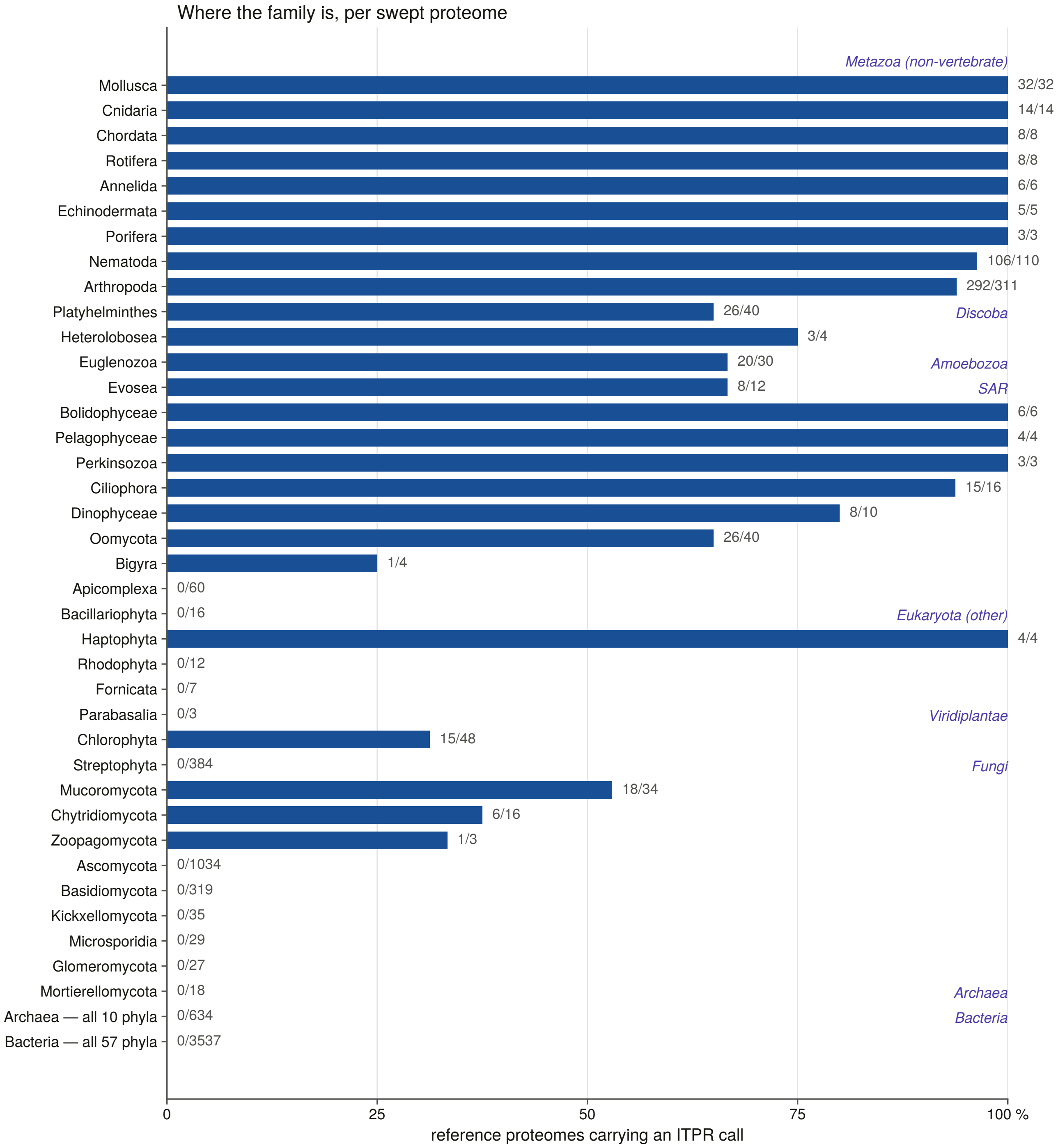


Fig. 1 | The IP_3 receptor family across the eukaryotes. The fraction of each lineage's swept reference proteomes carrying at least one IP_3 receptor call, with the count over the denominator beside each bar and lineages grouped by the sweep group they sit in. Denominators are proteomes, not records, so a well-sequenced phylum cannot outvote a sparse one; the italic headings are the kingdoms and supergroups the sweep partitioned Eukaryota into, not ranks. Bars at zero — Apicomplexa 0/60, Bacillariophyta 0/16, Rhodophyta 0/12, Fornicata 0/7, Parabasalia 0/3, Streptophyta 0/384, Ascomycota 0/1,034, Basidiomycota 0/319, Kickxellomycota 0/35, Microsporidia 0/29, Glomeromycota 0/27, Mortierellomycota 0/18, and all 10 archaeal and all 57 bacterial phyla — are absences within a search that recovers the family in the same kingdom (Chlorophyta 15/48, Mucoromycota 18/34) and recovers the promiscuous MIR domain in the empty lineages themselves.

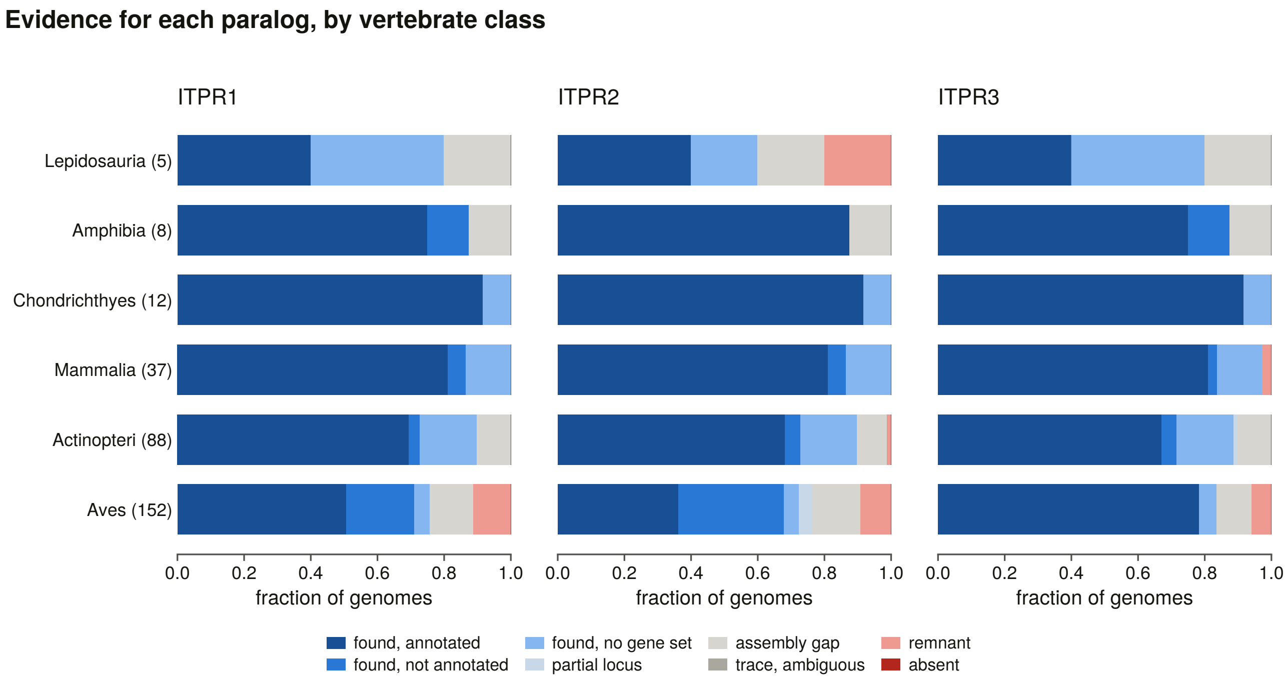


Fig. 2 | The three-paralogue census across 309 vertebrate genomes. Evidence class for each of *ITPR1*, *ITPR2* and *ITPR3* in every genome, stacked as a fraction of the genomes of each vertebrate class that the scope samples more than once — six classes and 302 of the 309 genomes (class sizes in brackets). Darkest blue is a locus found and correctly annotated; the next two blues are a locus found in an assembly whose annotation misses it or that has no gene set at all; the palest blue is a locus recovered only in part; grey is an assembly gap and the darker grey an ambiguous translated trace; salmon is a remnant recovered only by translated search. The `absent` colour appears nowhere because the four cells that carry it — both cyclostomes' *ITPR2* and *ITPR3* — are the only two genomes of their classes and those classes are not drawn; S15 re-states those four cells as paralogue-unassignable, for the reason given in the text. Aves carry the most non-blue cells of any class and are the least contiguous class in the scope; the largest non-blue *fraction* is Lepidosauria's, on five genomes.

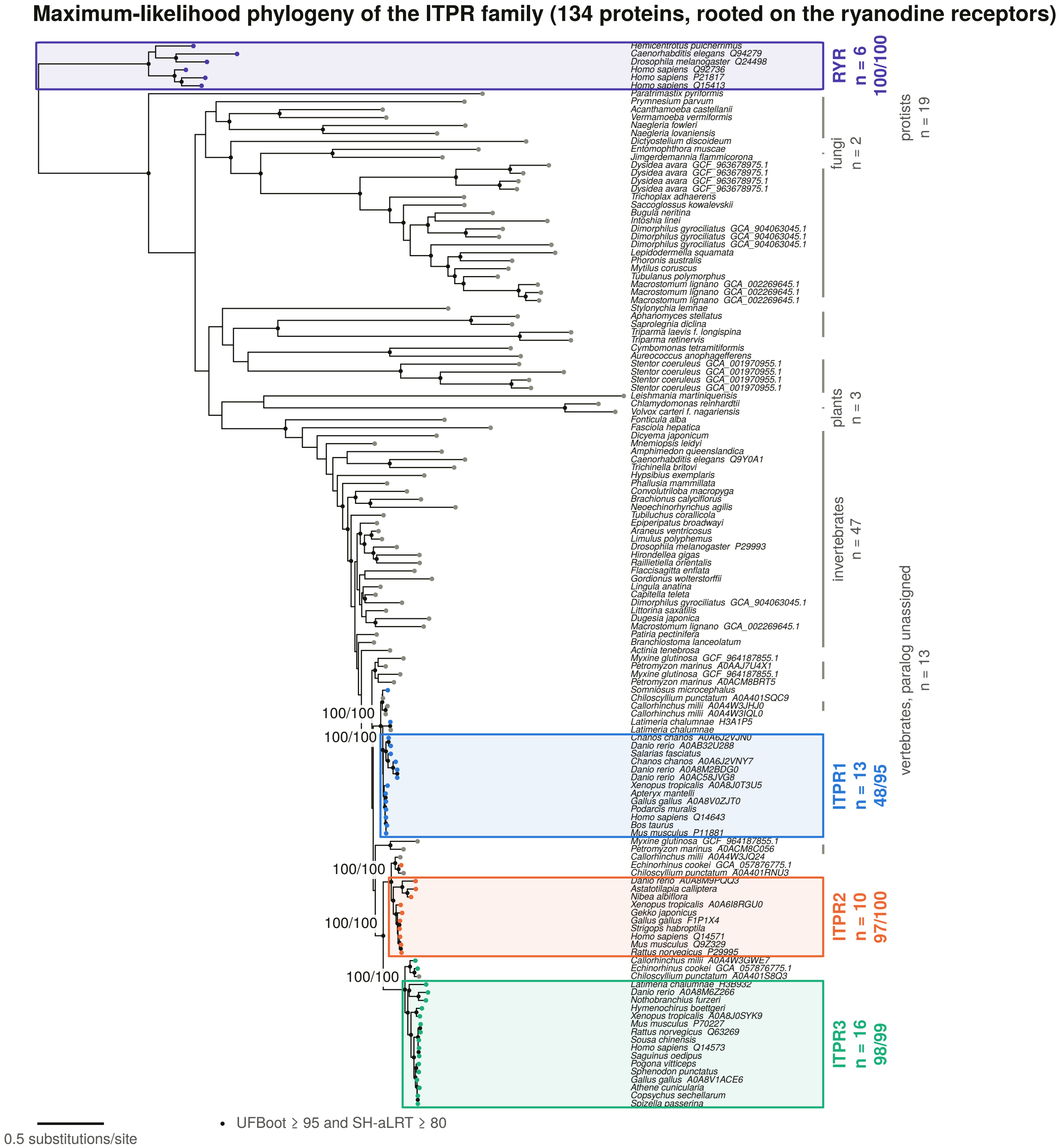


Fig. 3 | One family, three vertebrate paralogues, rooted on the ryanodine receptors. Maximum-likelihood phylogram of 134 representative proteins over 1,797 trimmed alignment columns under Q.insect+R7, rooted on the six ryanodine receptor sequences. Coloured boxes mark the three vertebrate paralogue clades and the outgroup, each labelled with its tip count and its SH-aLRT/UFBoot support; the vertical bars on the right name the taxonomic grades of the tips outside them. Filled circles mark nodes clearing SH-aLRT ≥ 80 and UFBoot ≥ 95. The five `100/100` labels on the backbone are the two ancestors of each boxed clade: the three extended paralogue clades — each core clade plus the basal vertebrate tips the tree nests with it — an inner *ITPR1* node of 15 tips, and, deepest, the 32-tip node uniting *ITPR2* and *ITPR3*. Scale bar 0.5 substitutions per site.

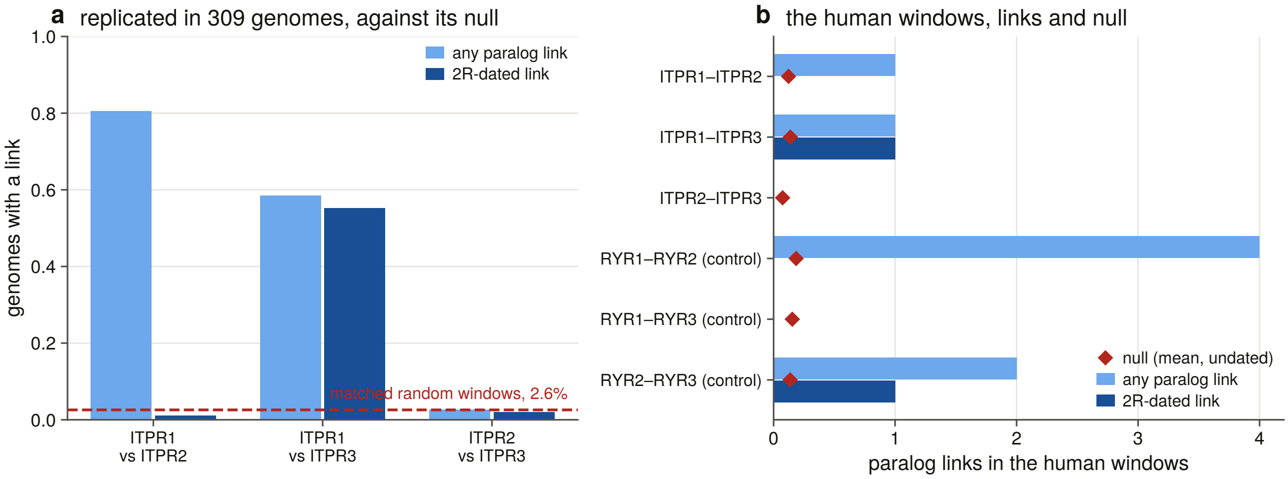


Fig. 4 | Where the three vertebrate paralogues came from. (**a**) The fraction of genomes in which each pair of IP_3 receptor neighbourhoods carries a paralogy link, replicated across 309 genomes, with the dark bars restricted to links whose Compara duplication node dates them to the vertebrate two-round event. The dashed line is the rate in 932 matched random neighbourhoods drawn in the same genomes (2.6 %). *ITPR1–ITPR2* and *ITPR1–ITPR3* are far above it and *ITPR2–ITPR3* is exactly on it. (**b**) The same test in the single human genome, where the counts are small and the null (red diamonds) is a mean rather than a rate; the three ryanodine receptor pairs run through the identical instrument as a positive control. The contrast between the two panels is the point: the human genome measures the instrument's ceiling, and the 309-genome replication measures the signal.

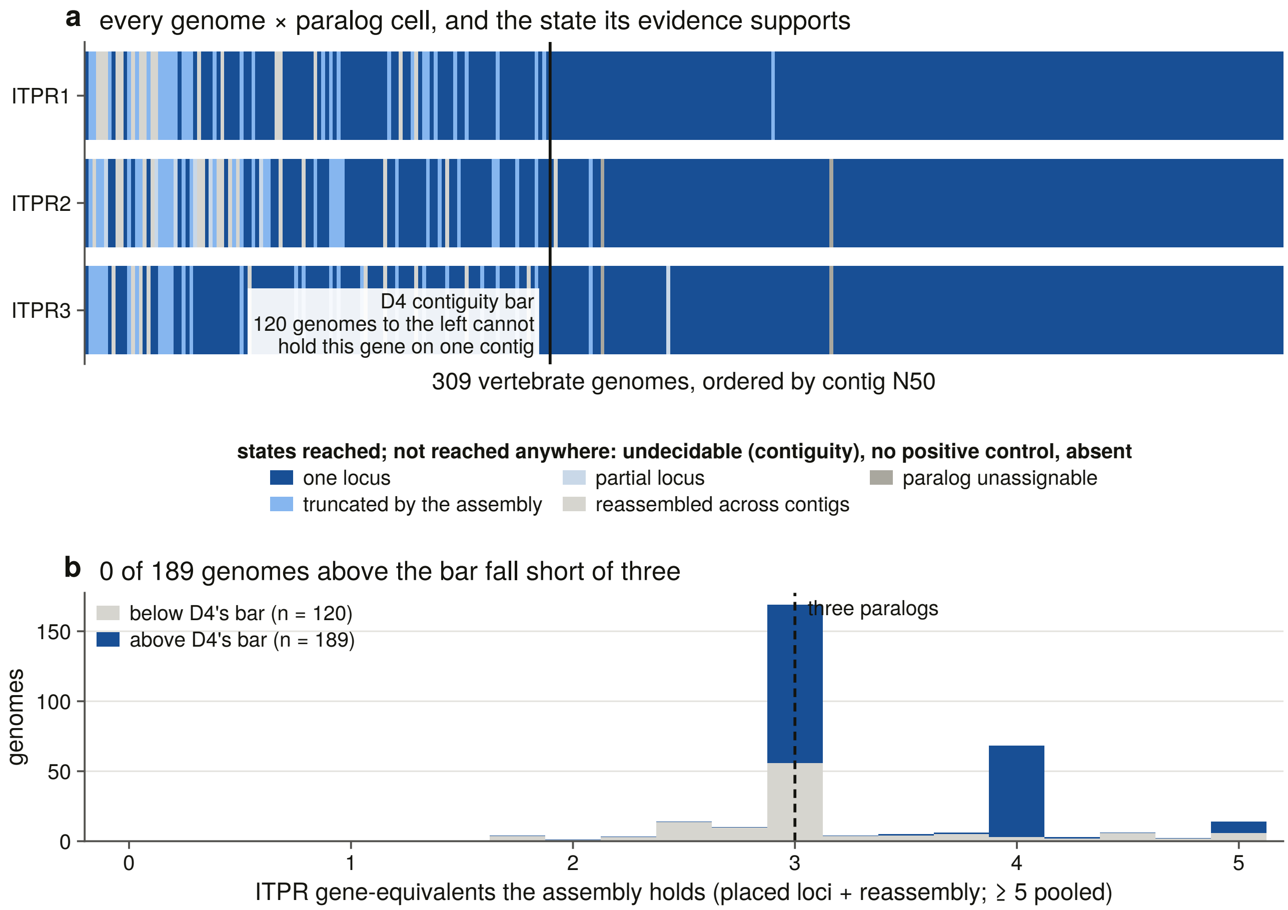


Fig. 5 | Three paralogues, retained in every vertebrate genome searched. (**a**) Every genome × paralogue cell, 309 genomes ordered left to right by contig N50, coloured by the state its evidence supports. The vertical line is the contiguity bar: the 120 genomes to its left cannot carry the gene on one contig. Three states are reached nowhere in the matrix and are named in the legend rather than drawn — undecidable for contiguity, no positive control, and absent. (**b**) The number of gene-equivalents each assembly holds, counting placed loci plus references reassembled across contigs, split by side of the bar. No genome above the bar falls short of three.

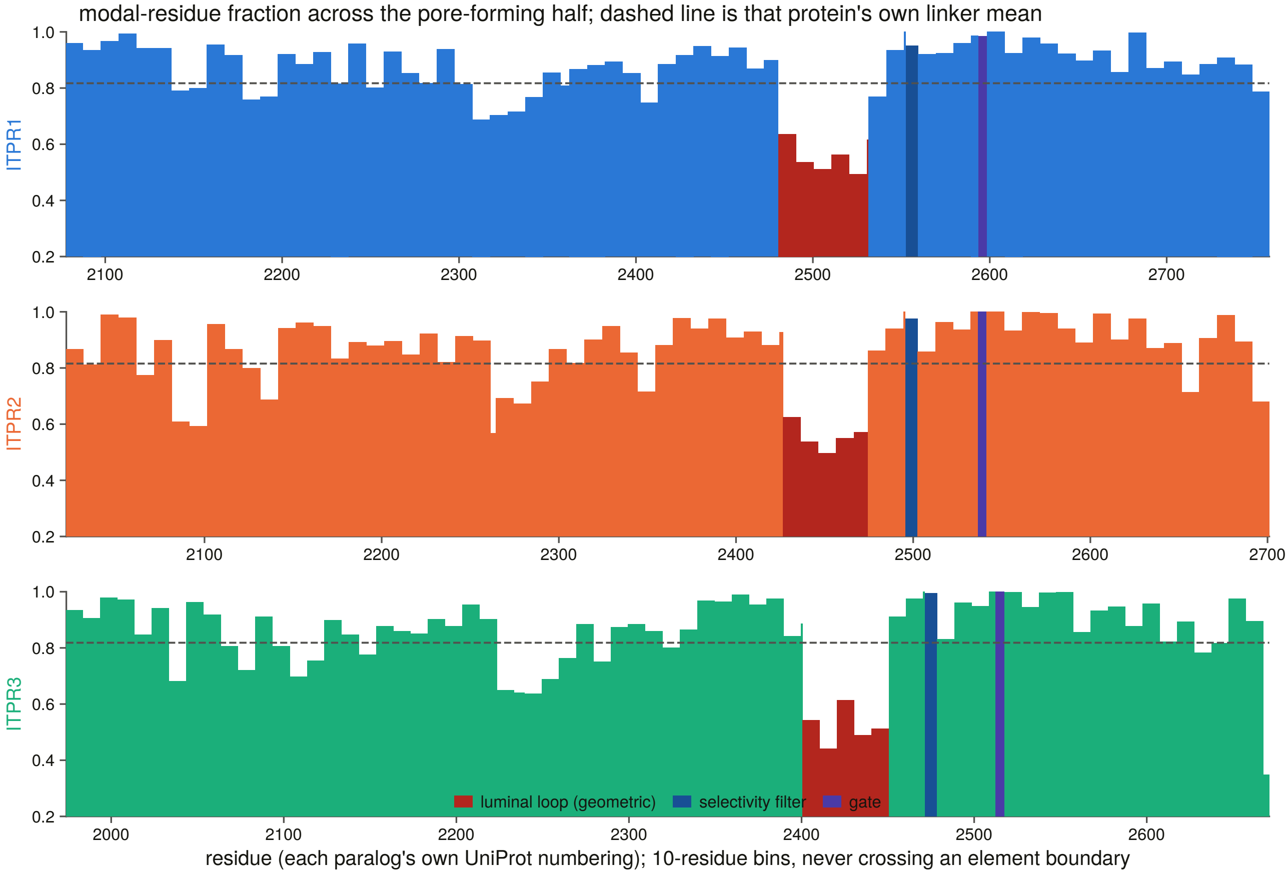


Fig. 6 | The pore-forming half: what cannot change, and what can. Constraint along the C-terminal half of each human paralogue in its own UniProt numbering, as the fraction of aligned sequences carrying the modal residue, in 10-residue bins that never cross an element boundary. The dashed line is that protein's own linker mean. Dark red marks the geometrically located luminal loop, which falls far below the linker control in all three; the two narrow bars to its right are the selectivity filter and the gate, which are at or near the top of the whole protein in all three. The most and the least constrained sequence in the receptor lie about fifty residues apart inside the same Pfam domain.

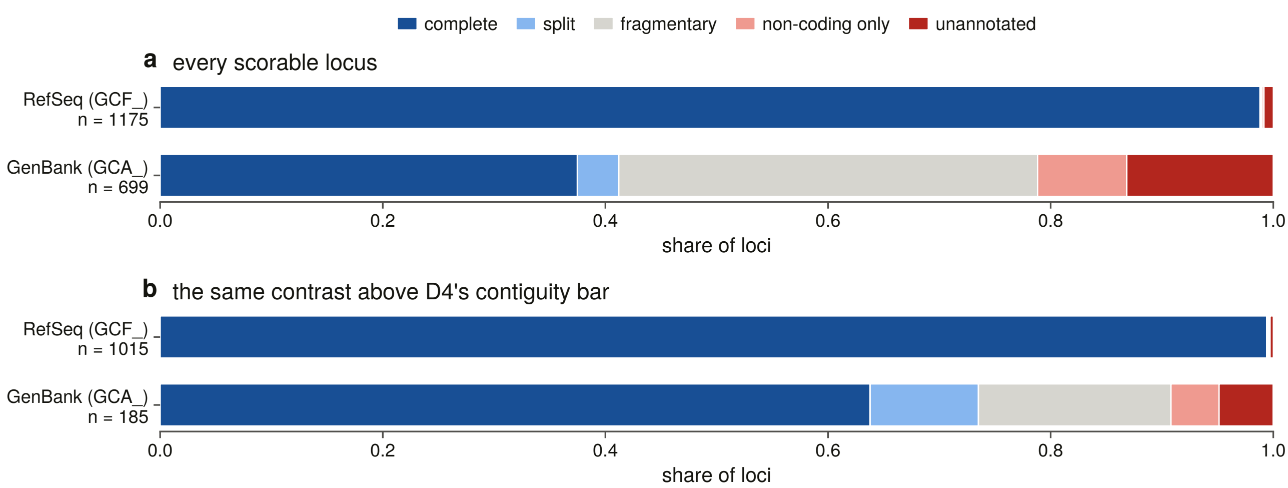
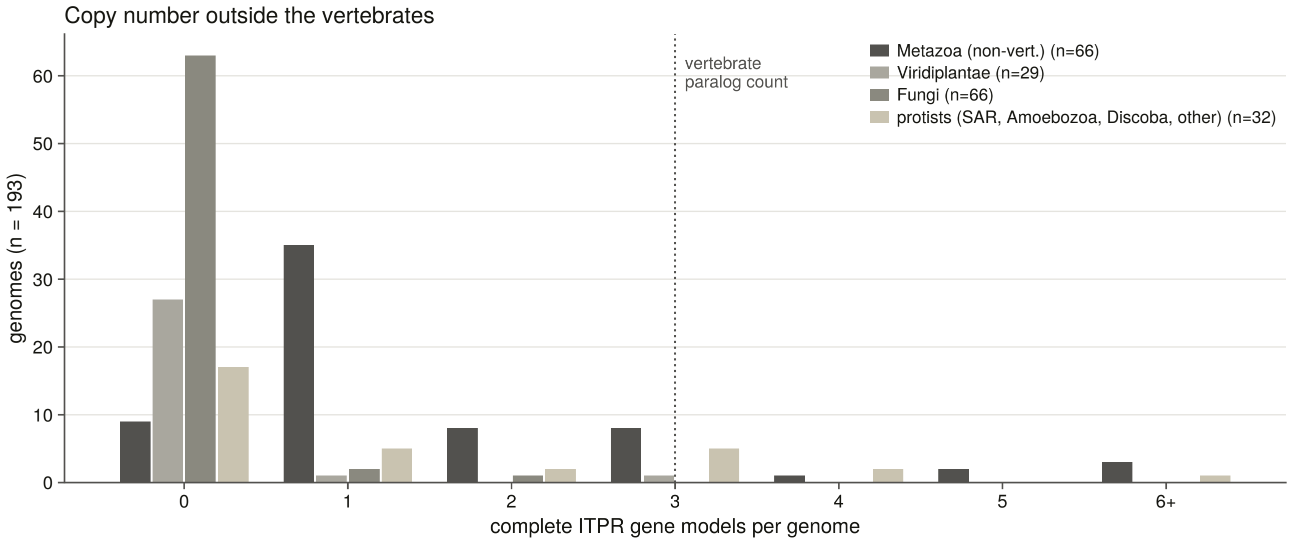
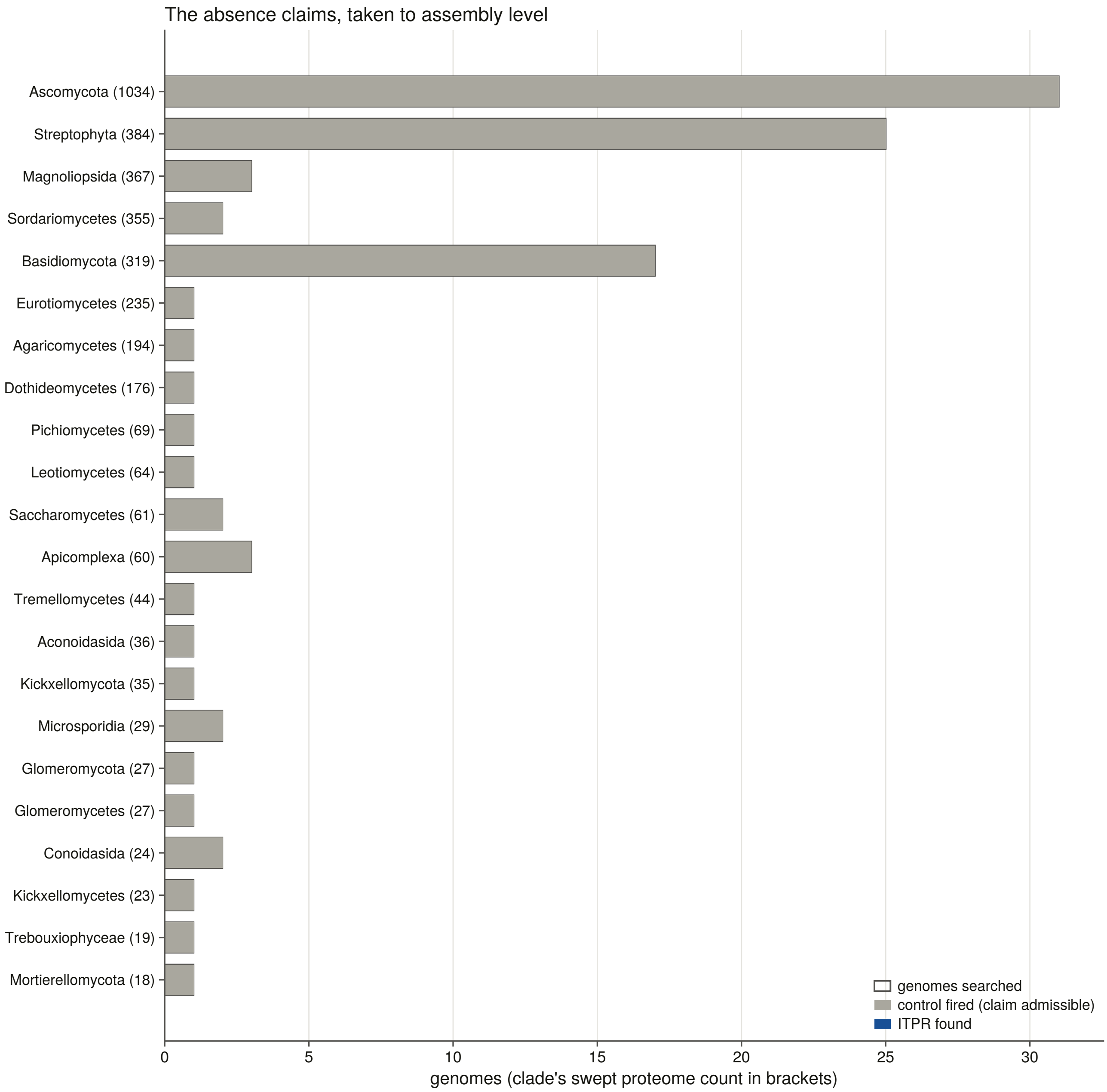
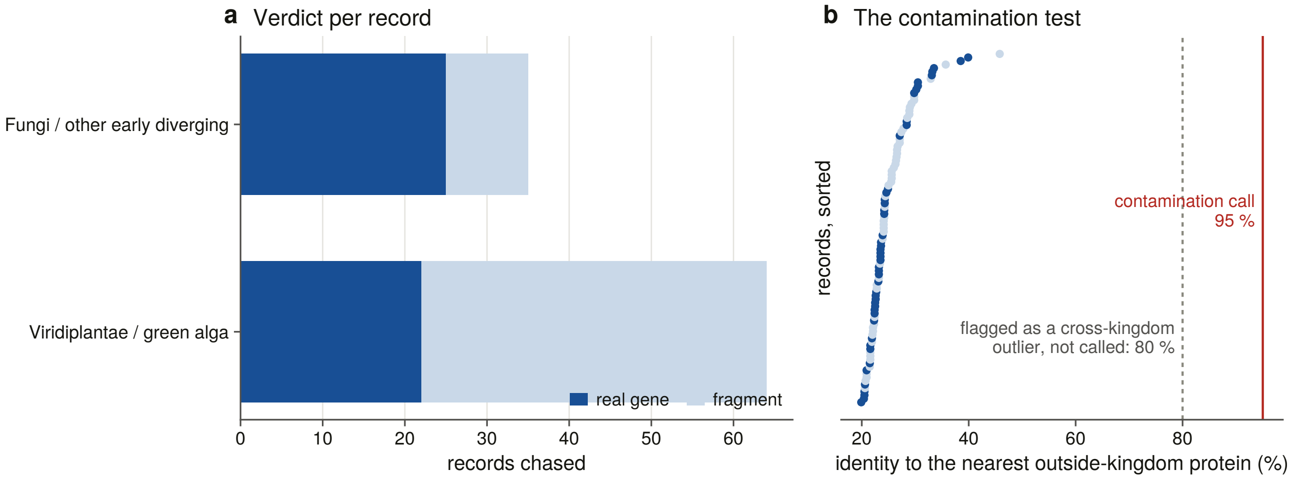


Fig. 7 | How the family is recorded, and by whom. Annotation state of every scorable IP_3 receptor locus, split by whether the assembly's annotation came from RefSeq or from the submitter through GenBank. (**a**) All 1,874 scorable loci: RefSeq is 98.8 % complete and GenBank 37.5 %. (**b**) The same contrast restricted to loci on a contig long enough to hold the gene, which removes assembly quality as an explanation and leaves 99.4 % against 63.8 %. The archive an assembly is deposited in predicts whether its IP_3 receptor is recorded far better than anything about the gene.

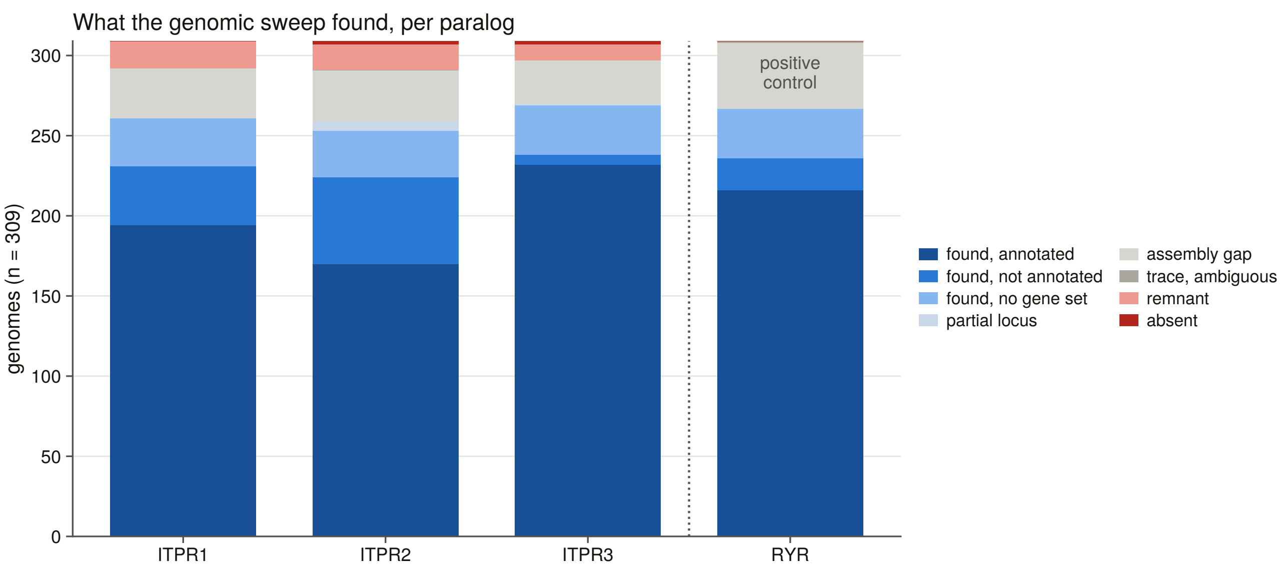
Extended Data figure legends

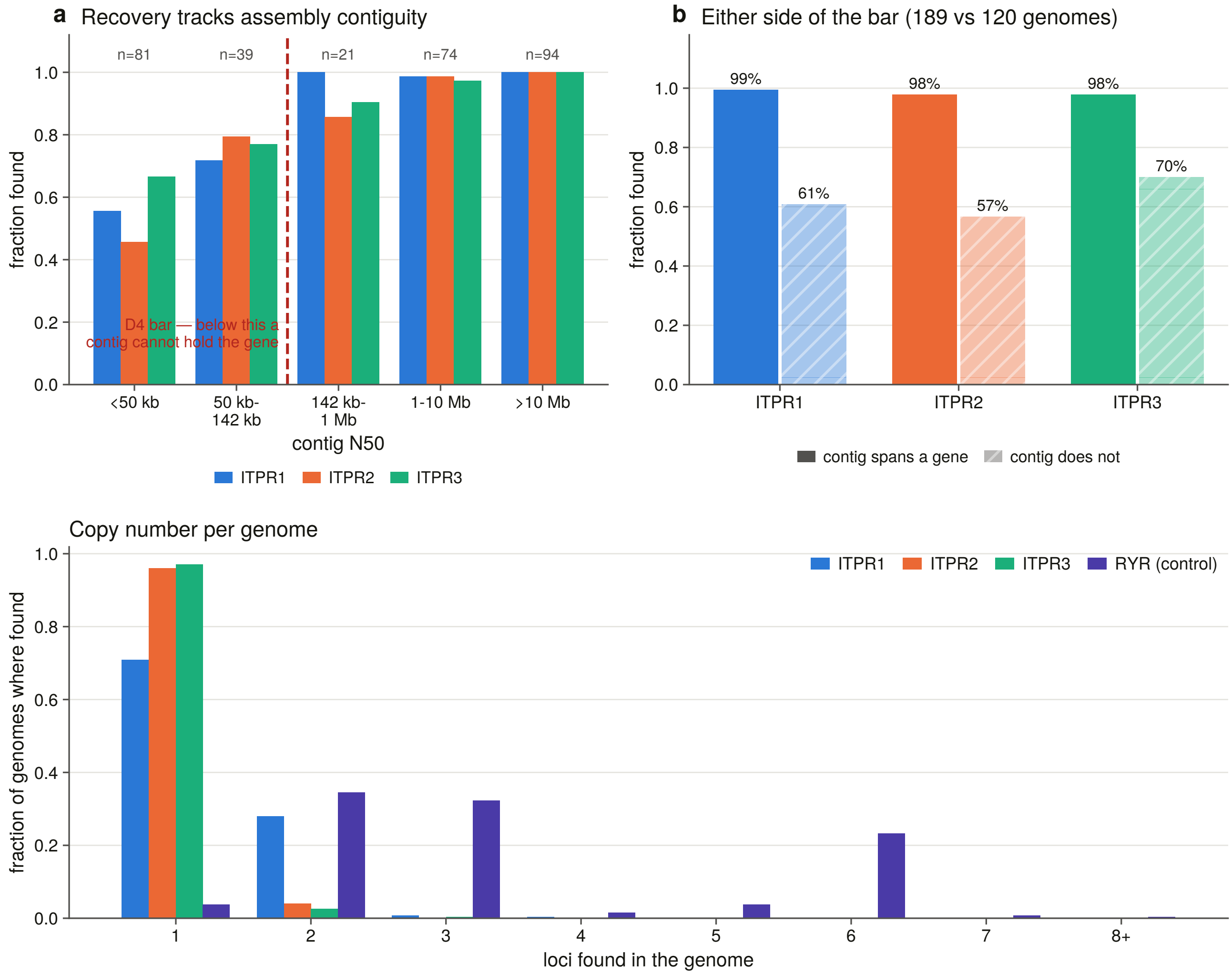




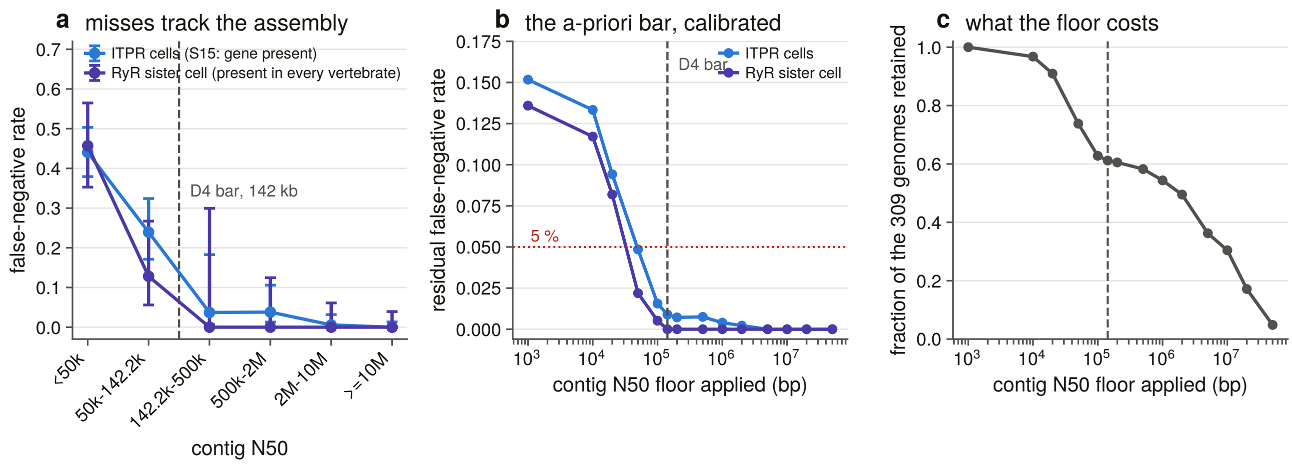


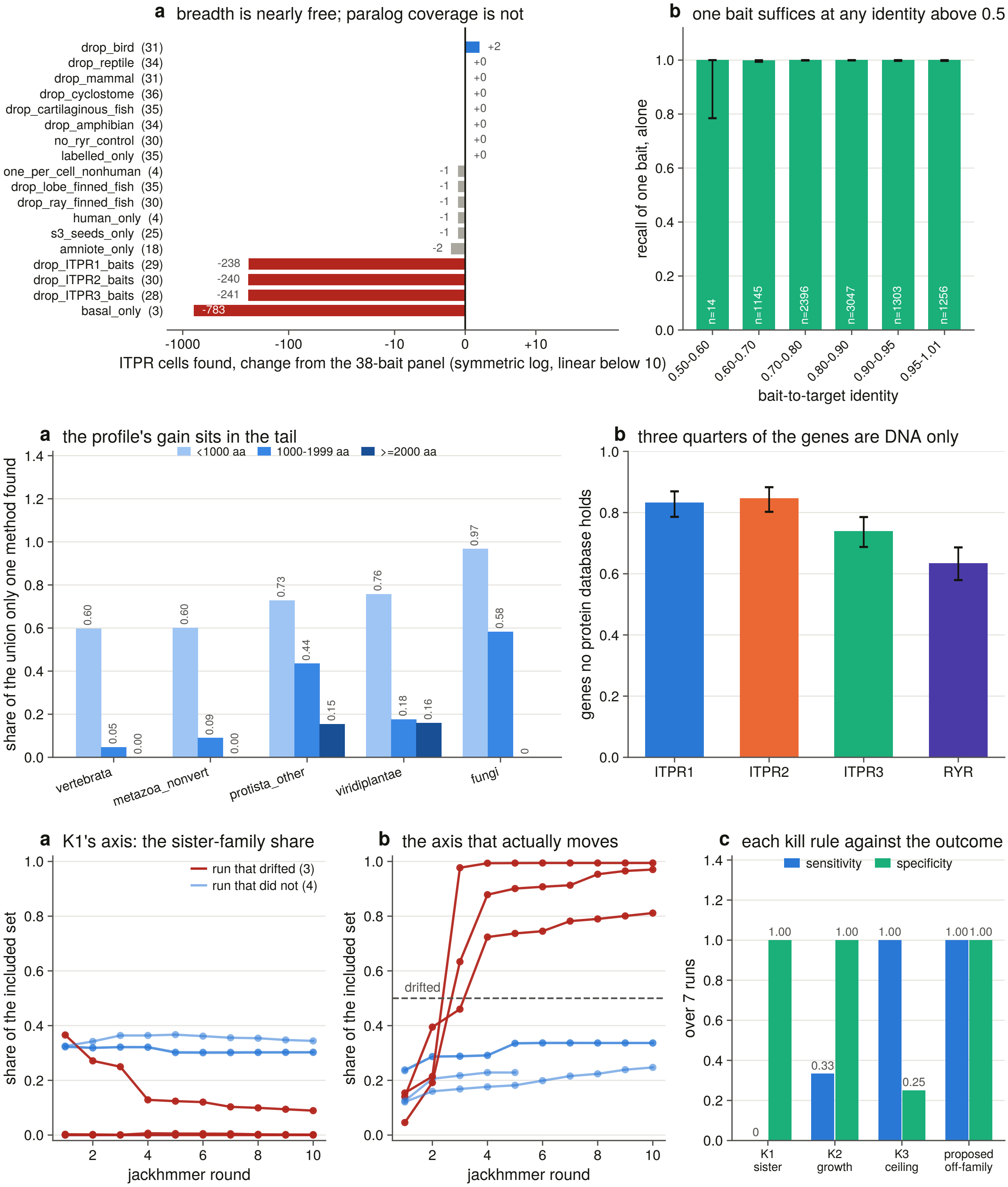
Extended Data Fig. 1 | Copy number and controlled absence outside the vertebrates. (**a**) Complete IP_3 receptor gene models per genome across the 193 controlled non-vertebrate genomes, grouped by kingdom, with the vertebrate paralogue count marked. Most fungal and plant genomes carry none; most non-vertebrate metazoan genomes carry one; nine genomes carry more than three and four carry six or more, the largest being *Macrostomum lignano* at 18. (**b**) Every absence clade taken to assembly level, the 22 with the largest proteome denominators shown: the open outline is genomes searched, the grey bar is genomes in which a measured positive control recovered a comparably long, deeply conserved gene, and the blue bar — invisible because it is zero everywhere — is genomes with an IP_3 receptor. The grey bar filling the outline in every row is the result. (**c**) All 99 plant and fungal records chased individually: verdict per record, and each record's identity to its nearest protein outside its own kingdom, sorted. Every record sits between 19.9 % and 45.8 %, far below the 95 % that marks an assembly contaminant and below the 80 % that would flag one as a cross-kingdom outlier; both lines are drawn.



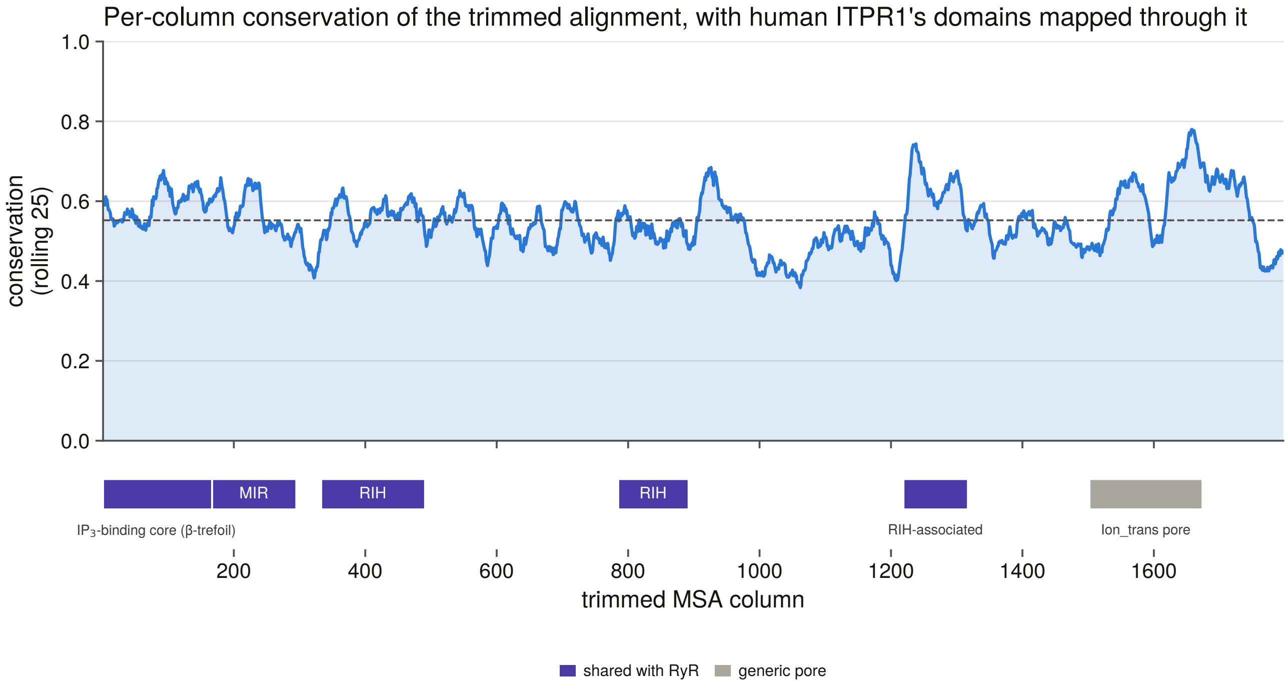


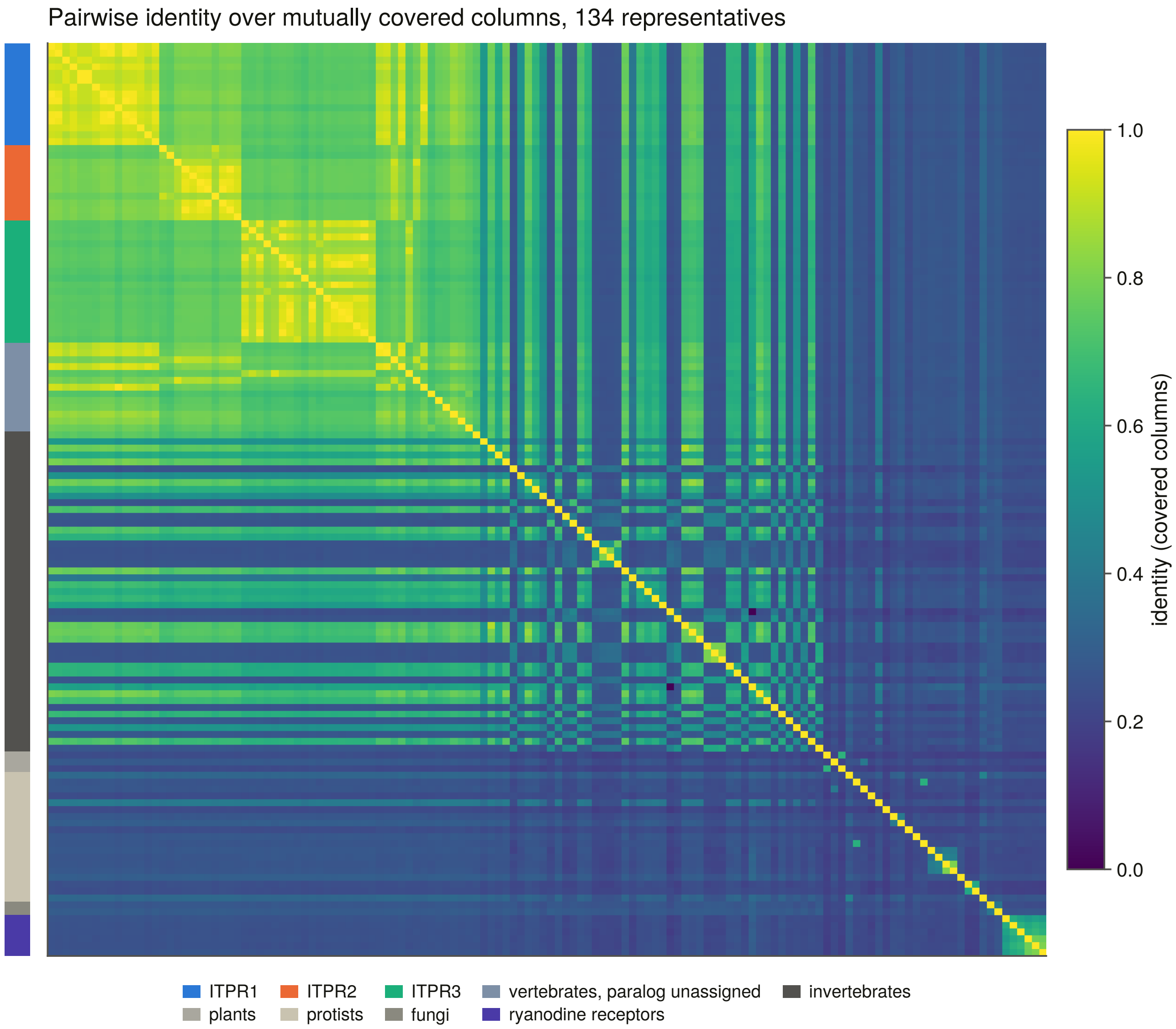
Extended Data Fig. 2 | The vertebrate genomic sweep and its contiguity confounder. (**a**) What the sweep found for each paralogue and for the ryanodine receptor positive control across all 309 genomes. (**b**) Recovery against assembly contiguity, binned on the contiguity bar rather than across it, and the same comparison as a two-bar contrast either side of it: 98–99 % of cells are found above the bar against 57–70 % below it, and below the bar the ordering is *ITPR3* > *ITPR1* > *ITPR2*, which is the order of their gene spans. (**c**) Loci found per genome for each paralogue, with the ryanodine receptors beside them: the *ITPR1* second copy in the ray-finned fish is the only substantial multi-copy signal in the family, against two to six copies for the control, whose three genes the sweep most often recovers as two or three loci.

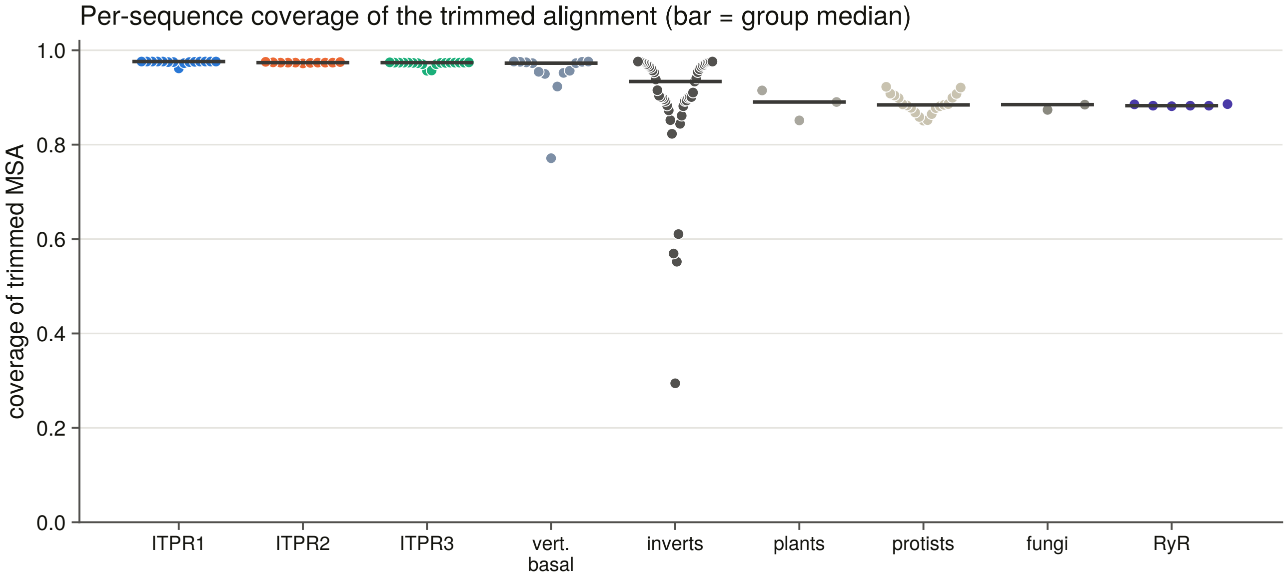




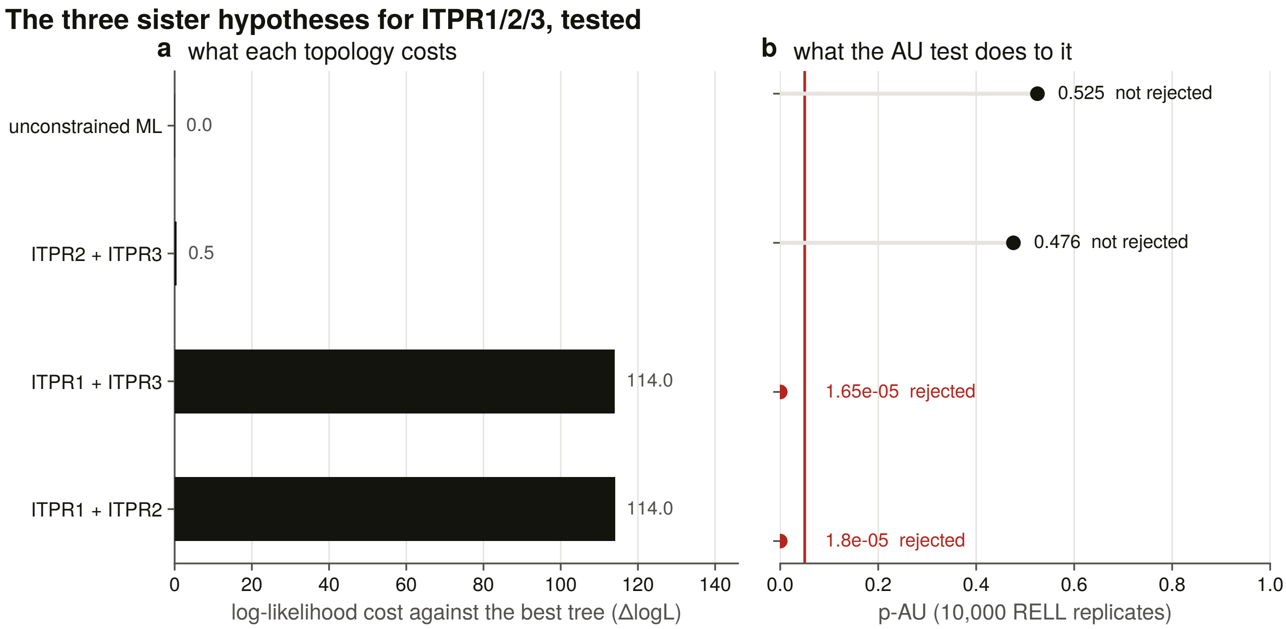
Extended Data Fig. 3 | What the search methods were worth. (**a**) The false-negative rate against contig N50 for the IP_3 receptor cells and for the independent ryanodine receptor sister series, binned on the contiguity bar; the residual rate at every possible floor, with the 5 % line drawn; and what each floor costs in genomes retained. (**b**) The bait-panel ablation as a change from the full 38-bait panel on a symmetric-logarithmic axis, and the recall of a single bait against its identity to the target. (**c**) Where a profile HMM earns its place — almost entirely below 1,000 residues in the two well-sampled groups, and at every length in the protists; the fungal and plant panels above 1,000 residues rest on a dozen to two dozen records each — and the fraction of demonstrated genes that no protein database holds. (**d**) The kill criterion written for iteration drift, measured: the sister-family share it acts on, the off-family share that actually moves, and each rule scored as a classifier of the outcome over seven runs.

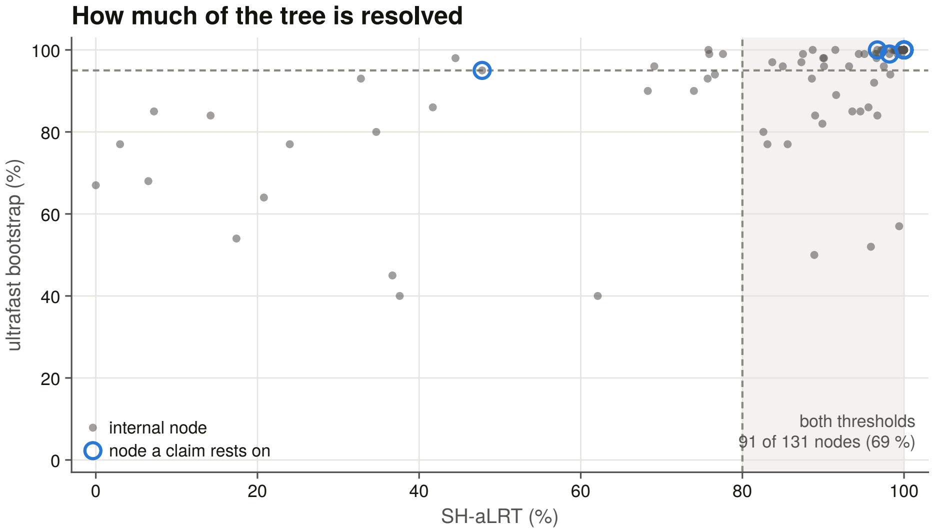


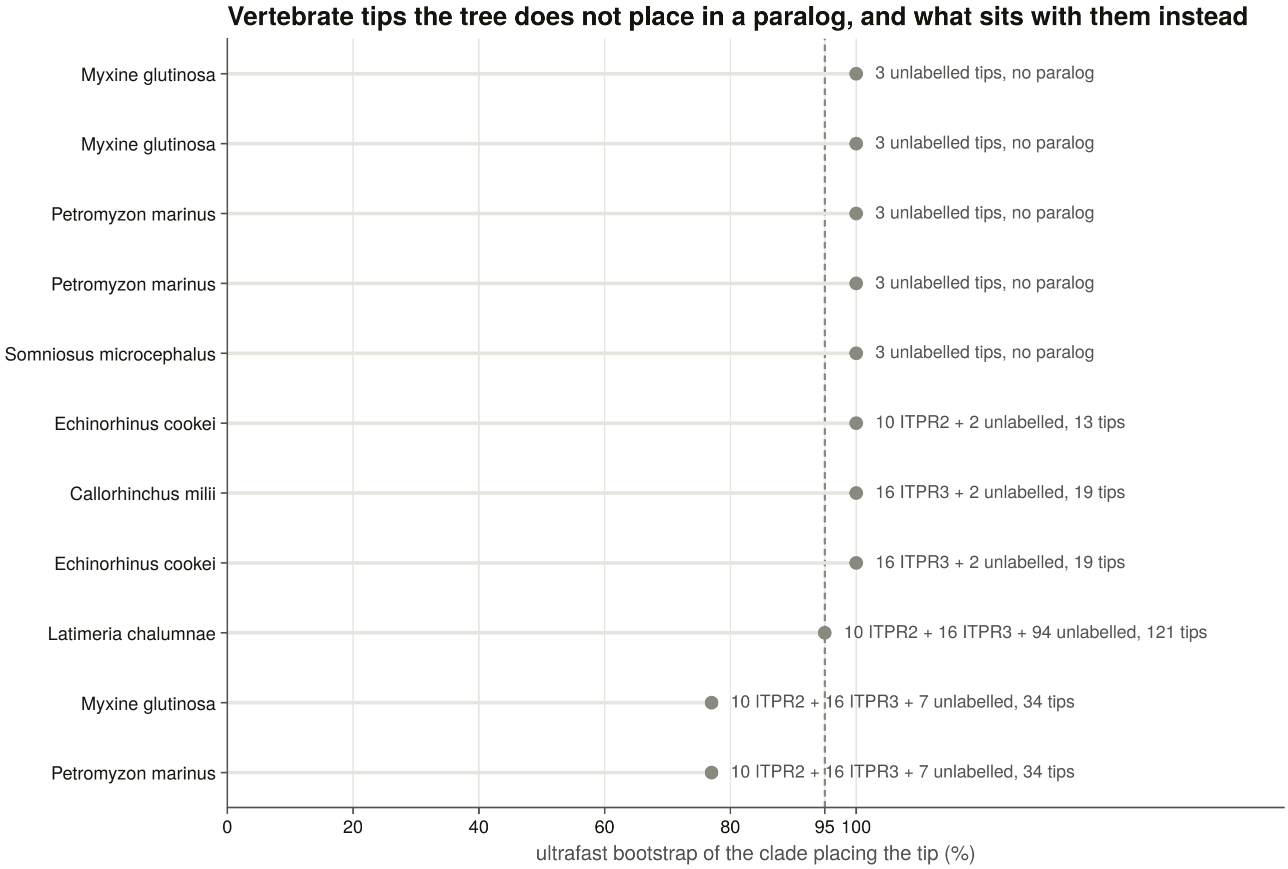




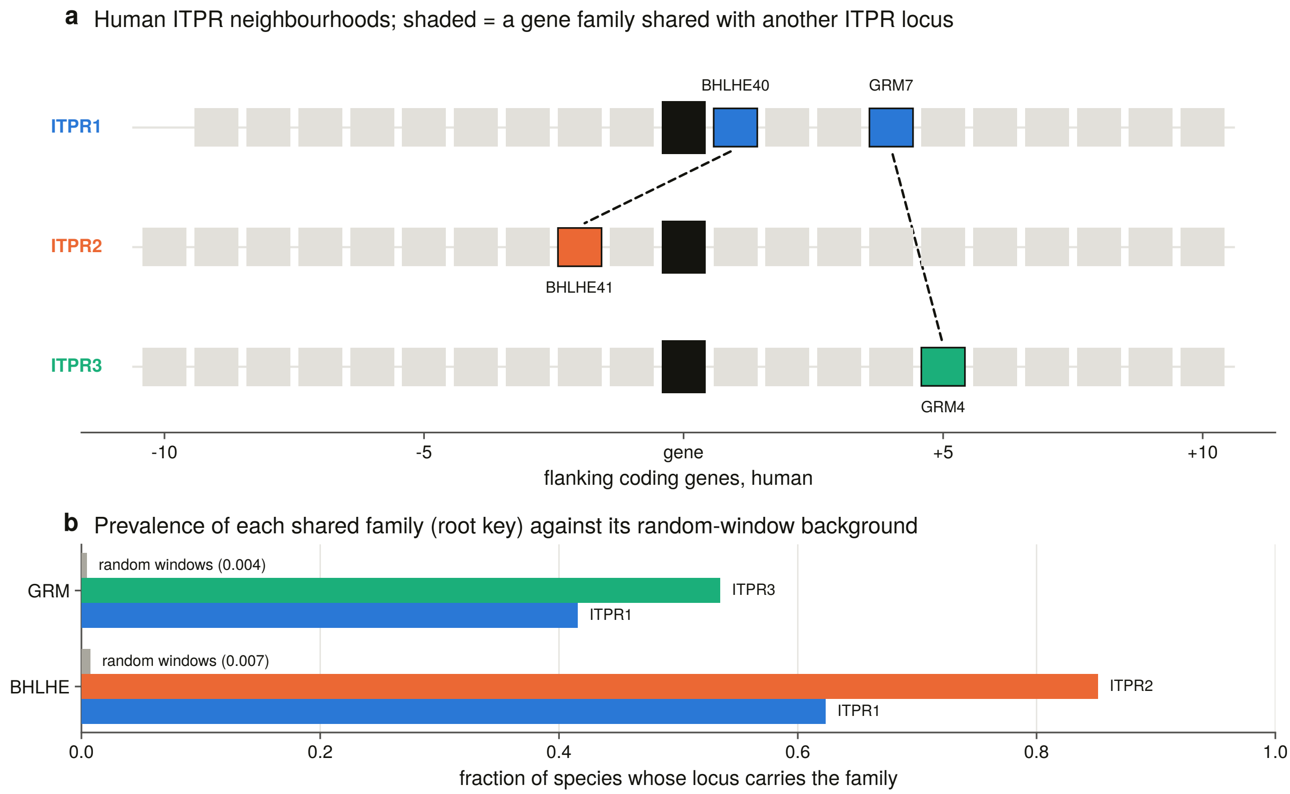
Extended Data Fig. 4 | The representative alignment every downstream result stands on. (**a**) Per-column conservation of the 1,797-column trimmed alignment, drawn as a rolling mean of 25 columns with the alignment-wide mean dashed, and human *ITPR1*'s Pfam architecture mapped through the alignment rather than scaled onto it, the domains coloured by whether they are shared with the ryanodine receptors or are the generic pore. (**b**) All-against-all identity over mutually covered columns for the 134 representatives, ordered and side-barred by group; the three bright blocks at top left are the vertebrate paralogues, and the whole cross-family field is dark. (**c**) Per-sequence coverage of the trimmed alignment by group. Median coverage is 0.96 and one tip of 134 falls below half.

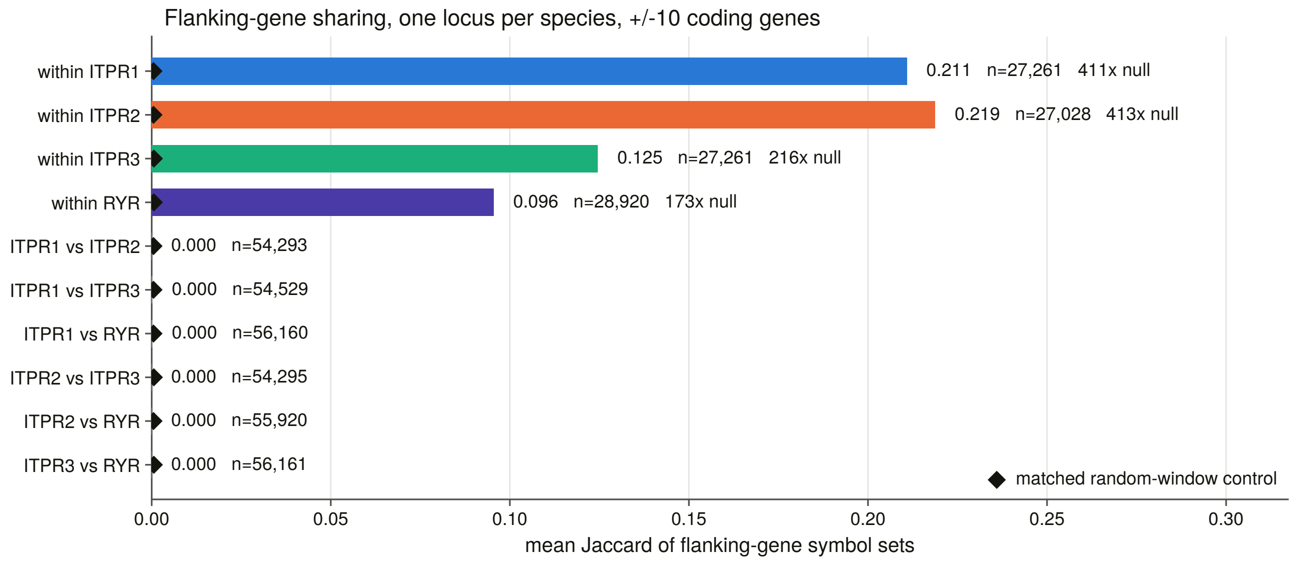


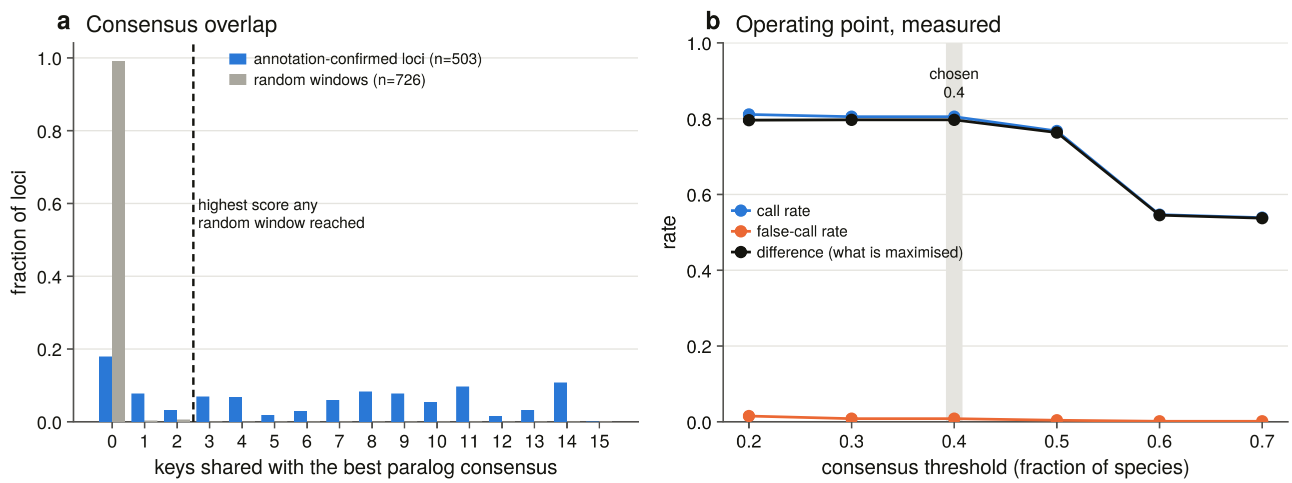


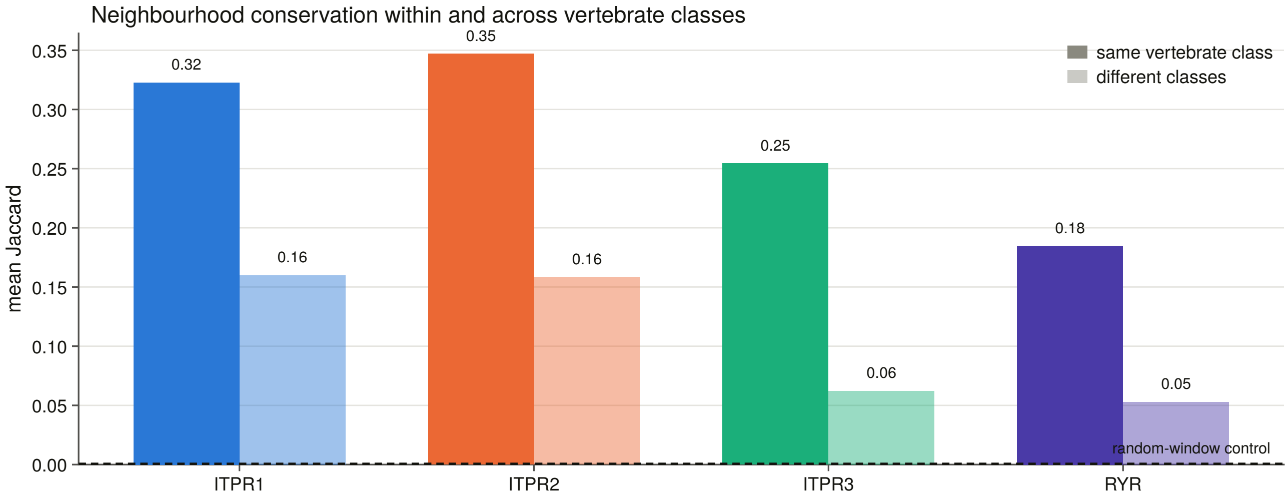


Extended Data Fig. 5 | The sister test, node support and paralogue placement. (**a**) Each of the three rooted sister hypotheses as a constrained maximum-likelihood tree: the log-likelihood cost against the best tree, and the approximately unbiased p-value over 10,000 RELL replicates. *ITPR1+ITPR2* and *ITPR1+ITPR3* each cost 114 log-likelihood units and are rejected; *ITPR2+ITPR3* costs 0.5 and is not. (**b**) SH-aLRT against ultrafast bootstrap for all 131 internal nodes, with the joint threshold region shaded and the nodes any claim in the paper rests on ringed; 91 of 131 nodes (69 %) clear both. (**c**) The eleven vertebrate tips the tree declines to place inside a paralogue clade, the support of the clade that does place each, and that clade's composition. Reciprocal best hits upheld the database name for all five names the tree disputes, so these are tree uncertainty rather than annotation error.

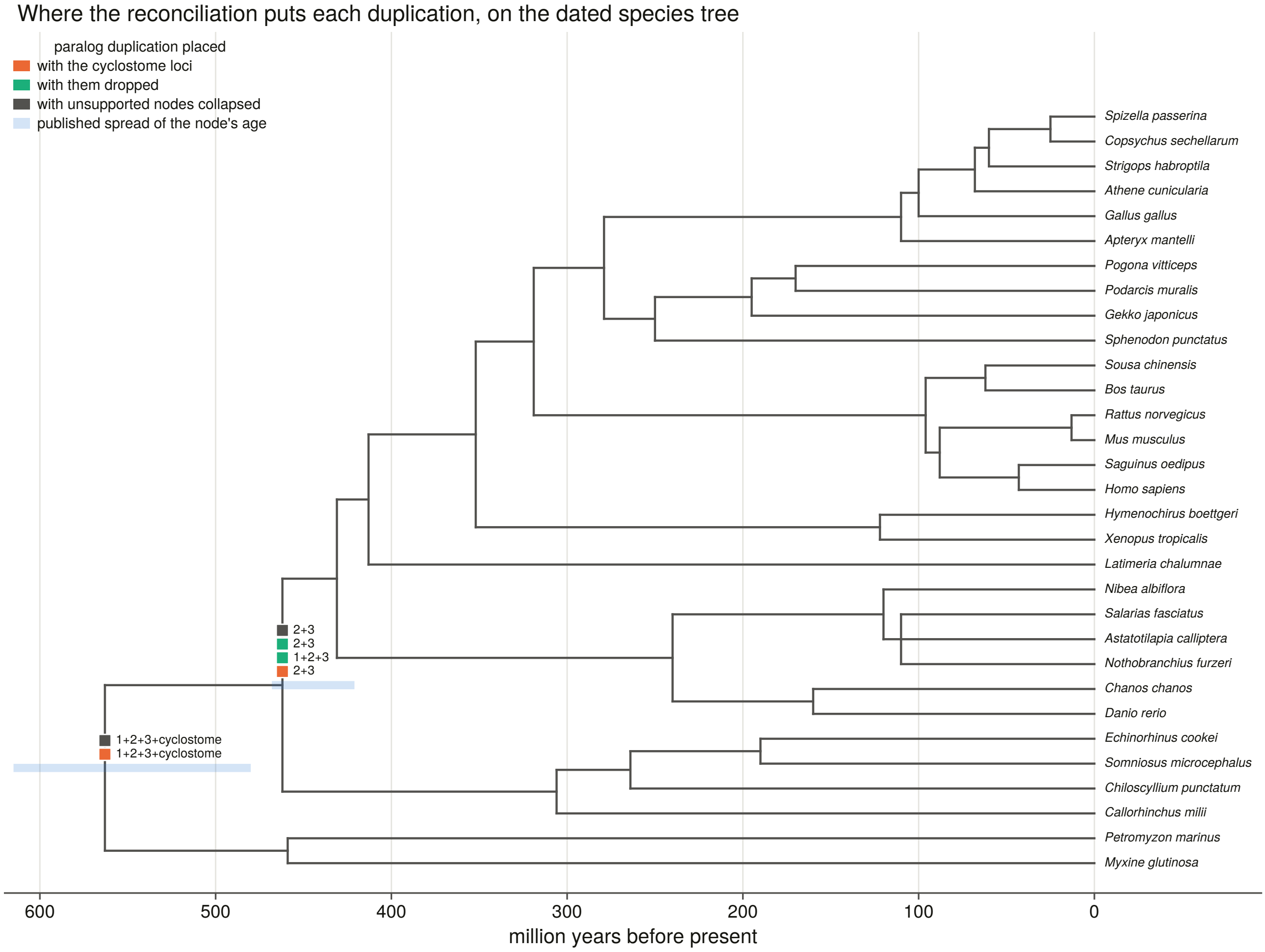

a Human ITPR neighbourhoods; shaded = a gene family shared with another ITPR locus
ITPR1
ITPR2
ITPR3
BHLHE40
GRM7
BHLHE41
GRM4
-10
-5
gene
+5
+10
flanking coding genes, human
b Prevalence of each shared family (root key) against its random-window background
GRM
BHLHE
random windows (0.004)
ITPR3
ITPR1
random windows (0.007)
ITPR2
ITPR1
0.0
0.2
0.4
0.6
0.8
1.0
fraction of species whose locus carries the family

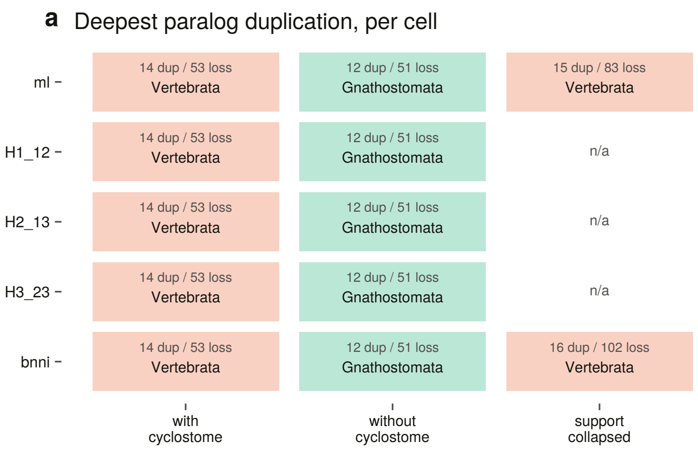

Flanking-gene sharing, one locus per species, +/-10 coding genes
within ITPR1
0.211 n=27,261 411x null
within ITPR2
0.219 n=27,028 413x null
within ITPR3
0.125 n=27,261 216x null
within RYR
0.096 n=28,920 173x null
ITPR1 vs ITPR2
0.000 n=54,293
ITPR1 vs ITPR3
0.000 n=54,529
ITPR1 vs RYR
0.000 n=56,160
ITPR2 vs ITPR3
0.000 n=54,295
ITPR2 vs RYR
0.000 n=55,920
ITPR3 vs RYR
0.000 n=56,161
matched random-window control
0.00
0.05
0.10
0.15
0.20
0.25
0.30
mean Jaccard of flanking-gene symbol sets

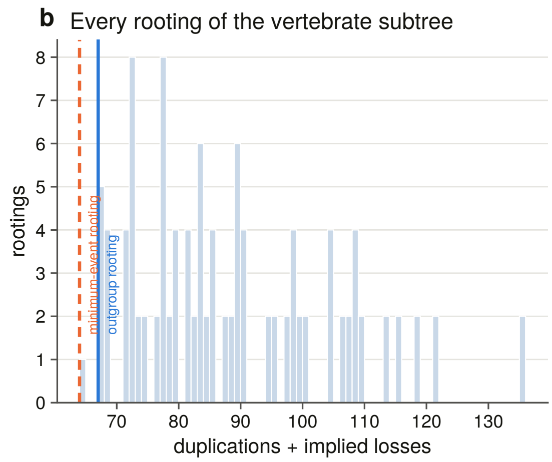

a Consensus overlap
annotation-confirmed loci (n=503)
random windows (n=726)
highest score any
random window reached
fraction of loci
1.0
0.8
0.6
0.4
0.2
0.0
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
keys shared with the best paralog consensus
b Operating point, measured
chosen
0.4
call rate
false-call rate
difference (what is maximised)
rate
1.0
0.8
0.6
0.4
0.2
0.0
0.2
0.3
0.4
0.5
0.6
0.7
consensus threshold (fraction of species)

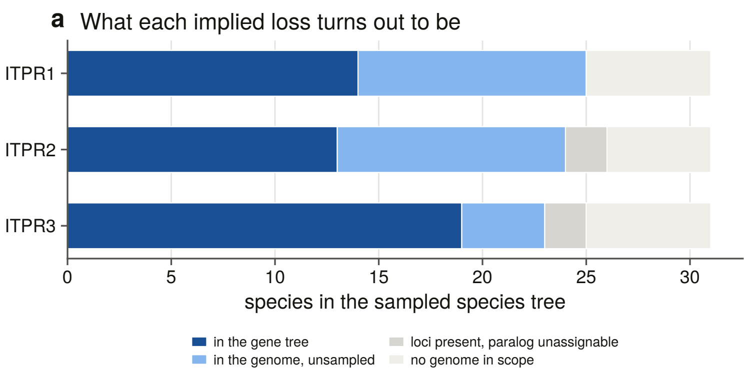


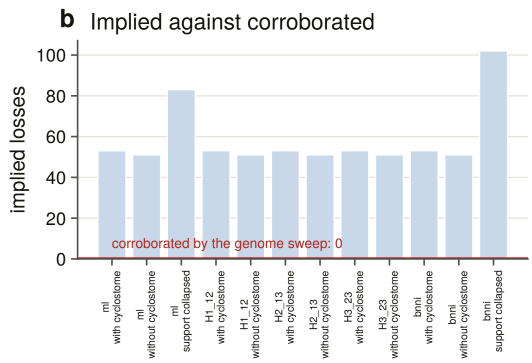
Extended Data Fig. 6 | Each paralogue has its own genomic neighbourhood. (**a**) The human *ITPR1*, *ITPR2* and *ITPR3* neighbourhoods drawn as gene tracks at (±)10 coding genes, with the two gene families shared between two IP_3 receptor loci linked, and the prevalence of each shared family across species against its random-window background beneath. (**b**) Mean Jaccard similarity of flanking-gene symbol sets by pair class, one locus per species, each against its own matched random-window control (black diamonds). Within-paralogue classes run 173- to 413-fold above the null; every cross-paralogue and cross-family class is at the null. (**c**) The consensus paralogue caller: how many consensus keys a locus shares, for annotation- confirmed loci against random windows, and the threshold sweep that chose the operating point by maximising call rate minus false-call rate. (**d**) Neighbourhood conservation within against across vertebrate classes. *ITPR3*'s neighbourhood is the one that does not travel.

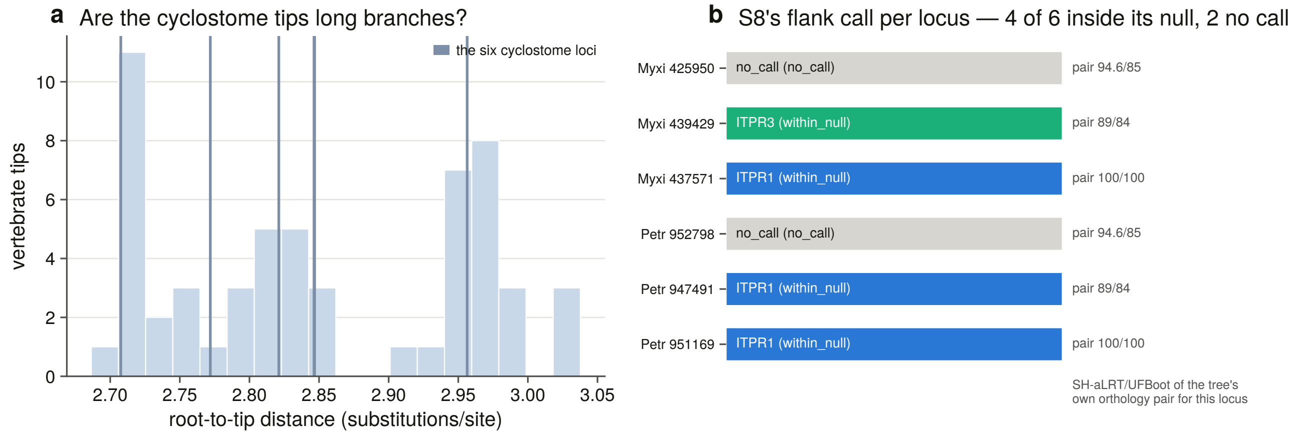

Where the reconciliation puts each duplication, on the dated species tree
paralog duplication placed
with the cyclostome loci
with them dropped
with unsupported nodes collapsed
published spread of the node's age
Spizella passerina
Copsychus sechellarum
Strigops habroptila
Athene cunicularia
Gallus gallus
Apteryx mantelli
Pogona vitticeps
Podarcis muralis
Gekko japonicus
Sphenodon punctatus
Sousa chinensis
Bos taurus
Rattus norvegicus
Mus musculus
Saguinus oedipus
Homo sapiens
Hymenochirus boettgeri
Xenopus tropicalis
Latimeria chalumnae
Nibea albiflora
Salarias fasciatus
Astatotilapia calliptera
Nothobranchius furzeri
Chanos chanos
Danio rerio
Echinorhinus cookei
Somniosus microcephalus
Chiloscyllium punctatum
Callorhinchus milii
Petromyzon marinus
Myxine glutinosa
2+3
2+3
1+2+3
2+3
1+2+3+cyclostome
1+2+3+cyclostome
600
500
400
300
200
100
0
million years before present


a Deepest paralog duplication, per cell
ml
H1_12
H2_13
H3_23
bnni
14 dup / 53 loss
Vertebrata
12 dup / 51 loss
Gnathostomata
15 dup / 83 loss
Vertebrata
n/a
16 dup / 102 loss
Vertebrata
with cyclostome
without cyclostome
support collapsed

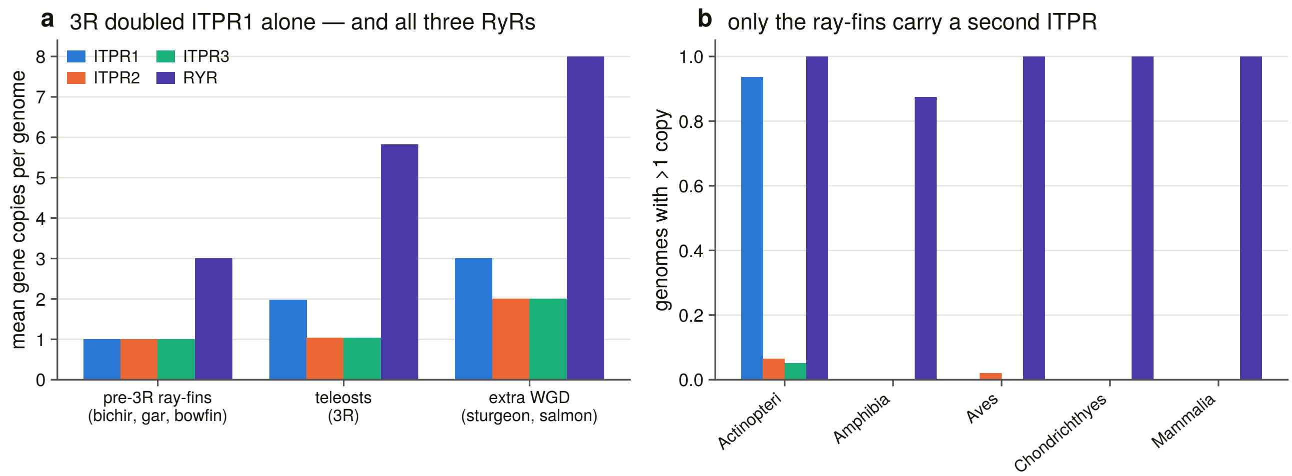

b Every rooting of the vertebrate subtree
rootings
minimum-event rooting
outgroup rooting
0
1
2
3
4
5
6
7
8
70
80
90
100
110
120
130
duplications + implied losses

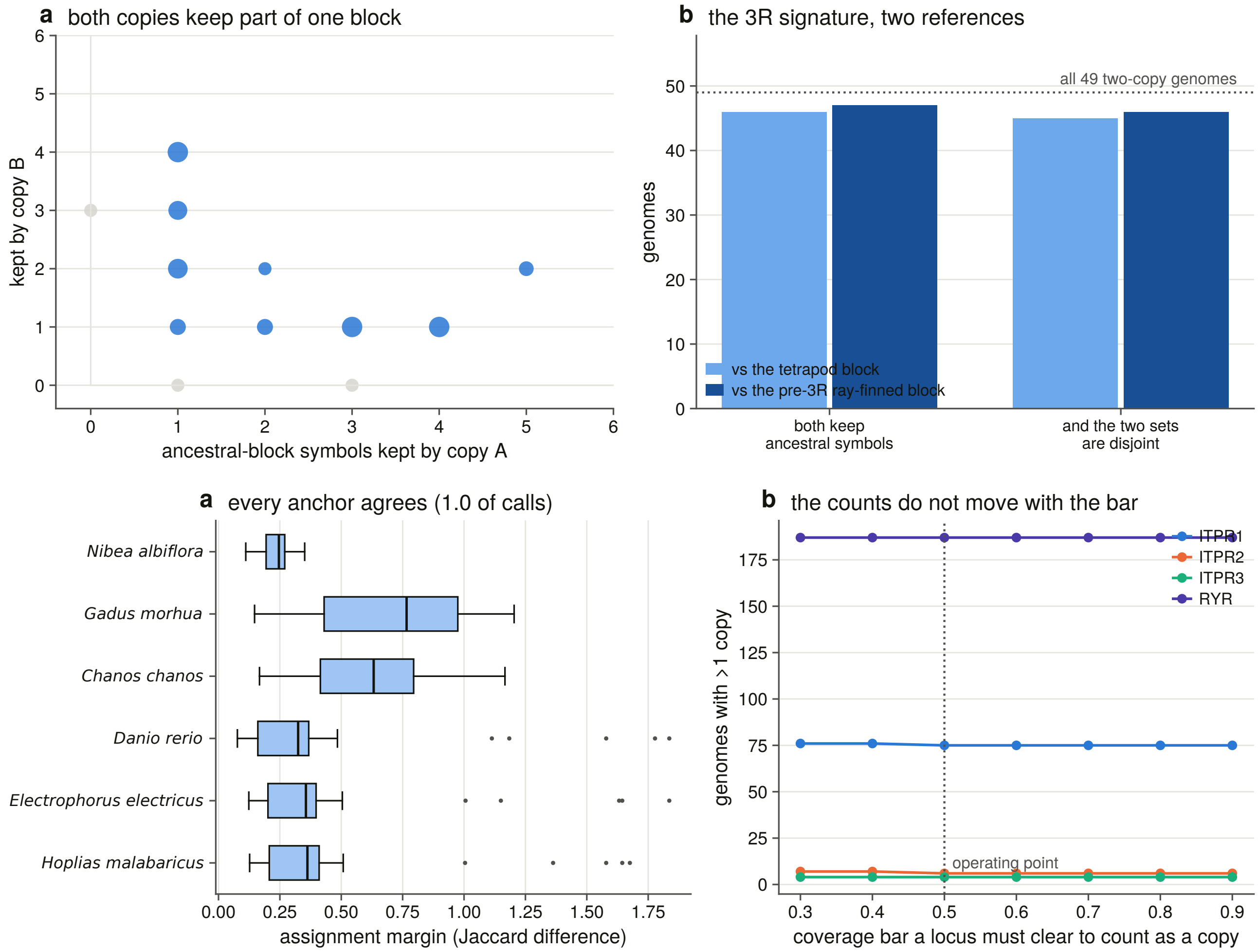

a What each implied loss turns out to be
ITPR1
ITPR2
ITPR3
0
5
10
15
20
25
30
species in the sampled species tree
in the gene tree
in the genome, unsampled
loci present, paralog unassignable
no genome in scope


b Implied against corroborated
implied losses
0
20
40
60
80
100
corroborated by the genome sweep: 0
ml with cyclostome
ml without cyclostome
ml support collapsed
H1_12 with cyclostome
H1_12 without cyclostome
H2_13 with cyclostome
H2_13 without cyclostome
H3_23 with cyclostome
H3_23 without cyclostome
bnni with cyclostome
bnni without cyclostome
bnni support collapsed

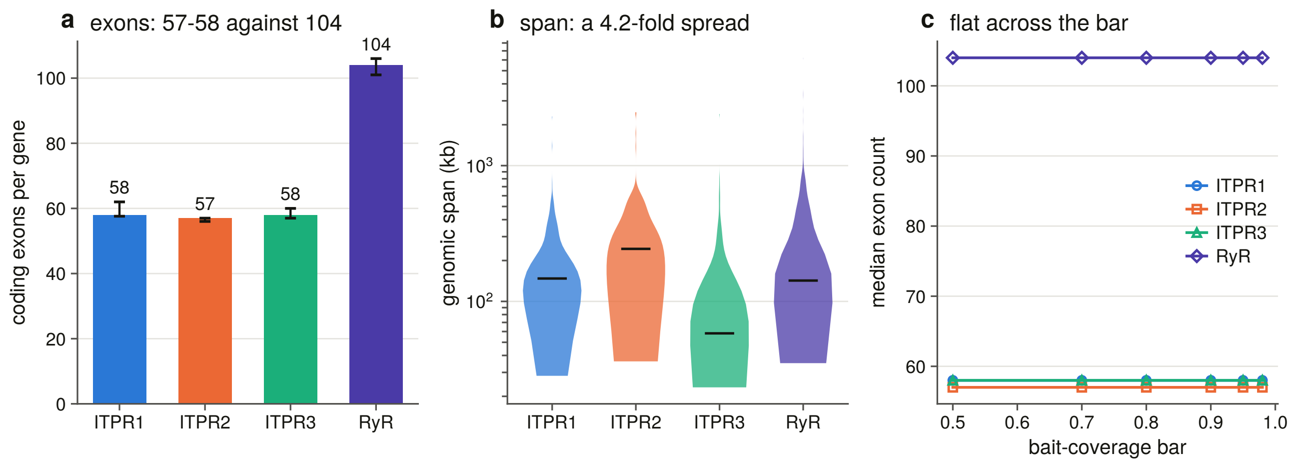


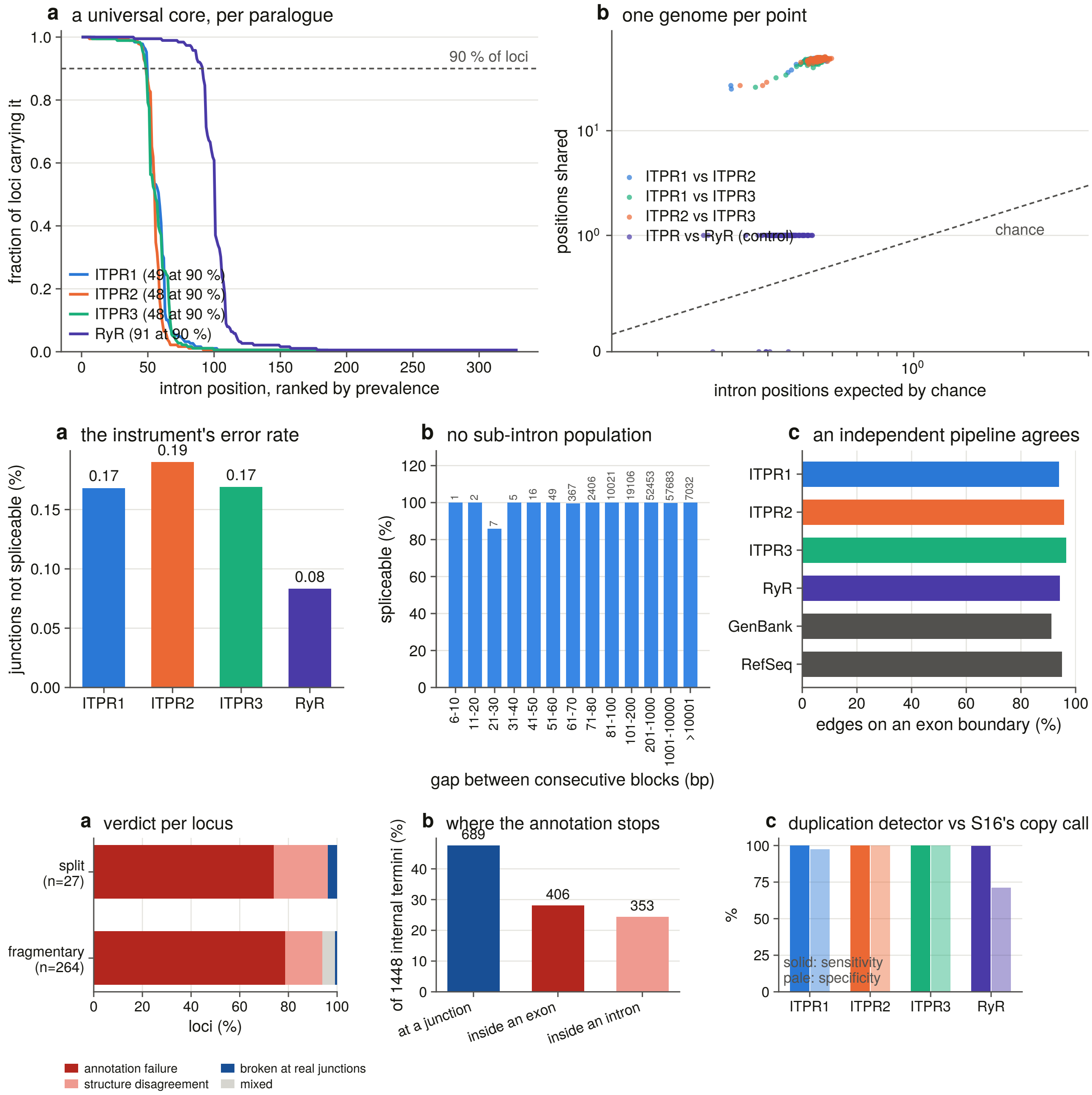
Extended Data Fig. 7 | Dating the duplications that made *ITPR1*, *ITPR2* and *ITPR3*. (**a**) The hand-curated, literature-calibrated species tree on a linear time axis, with each reconciliation variant's duplication placement marked at the node it maps to and the published age spread drawn as a band on the nodes a placement uses. Every internal node carries a calibration; only those two are banded, because a band on all 29 would obscure the tree. (**b**) The full matrix — five topologies by three tip variants — showing the deepest paralogue duplication in each cell, and the distribution of total events over all 112 rootings of the vertebrate subtree with the outgroup rooting and the minimum-event rooting marked. (**c**) What each implied loss turns out to be when asked of the genomes — 51 or 53 in the ten variants that keep the tree's own resolution, 83 and 102 in the two that collapse its unsupported nodes: none is corroborated. (**d**) Root-to-tip distances for all 57 vertebrate tips with the six cyclostome loci marked — two of them 0.0004 substitutions per site apart and drawn as one line — and each of those loci beside the independent flanking-gene call for it: four fall inside their own null and two have too few informative neighbours to call at all.



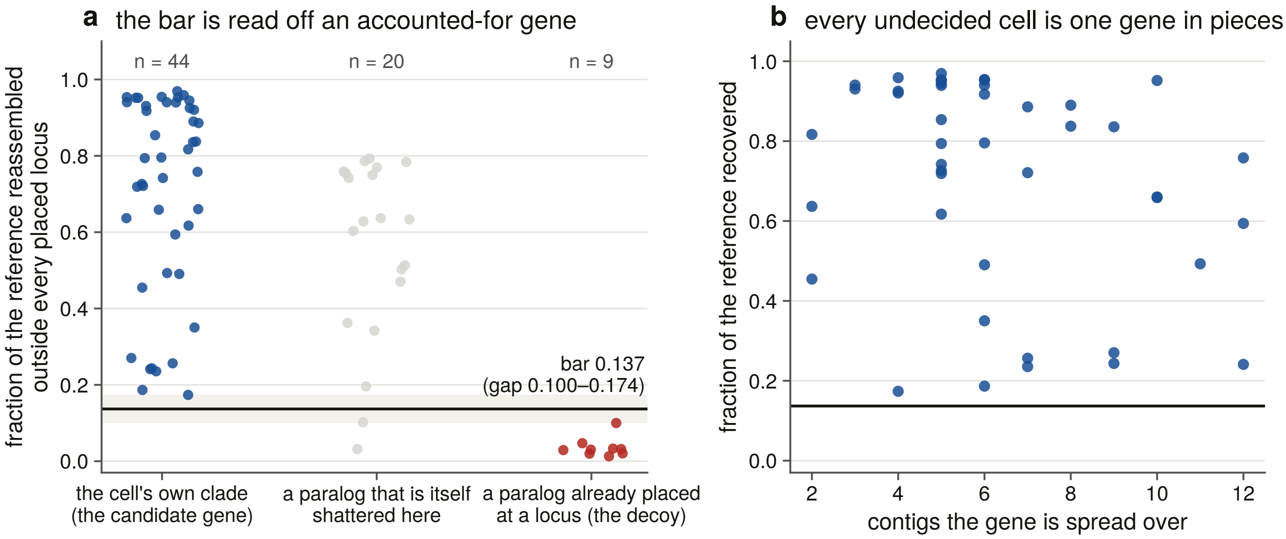


Extended Data Fig. 8 | The teleost genome duplication and the ancestral block. (**a**) Mean gene copies per genome for the three paralogues and the ryanodine receptor control in the pre-3R ray-finned outgroups, in the teleosts, and in the lineages with a further whole-genome duplication; and the fraction of genomes carrying more than one copy, by vertebrate class. (**b**) For each two-copy teleost genome, the number of ancestral-block gene symbols each *ITPR1* copy retains: points off both axes are genomes in which both copies keep part of one block, and the bar chart counts genomes meeting that criterion and the stricter one that the two copies' symbol sets are disjoint, against a tetrapod and a pre-3R ray-finned reference. (**c**) The cross-anchor assignment margin per reference genome, and the copy counts recomputed at seven coverage bars.

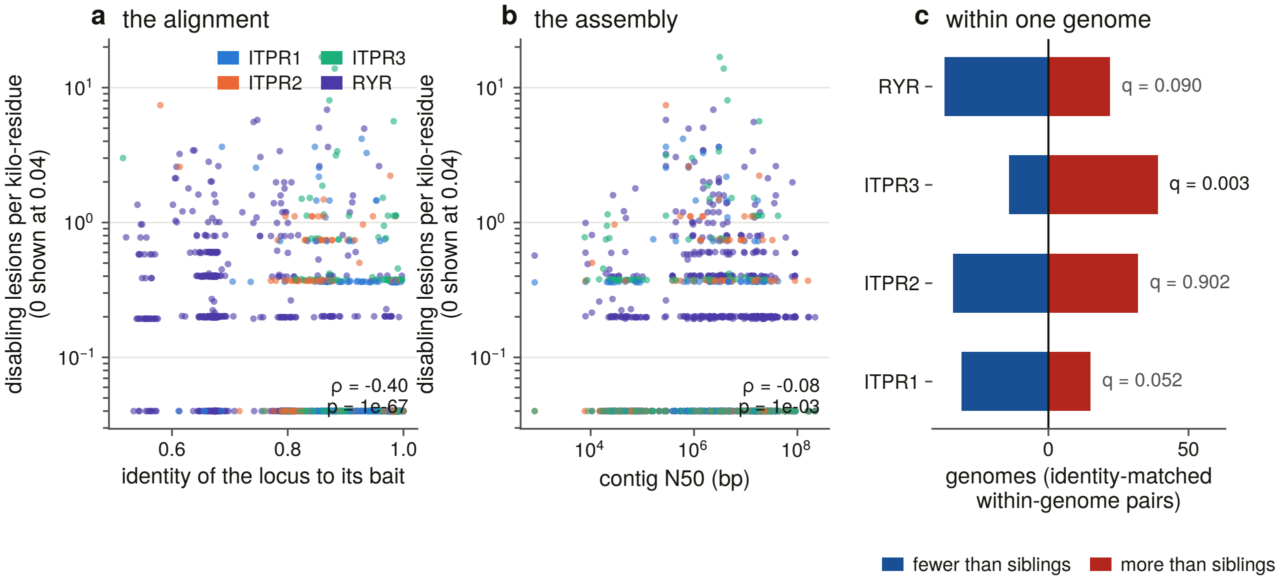




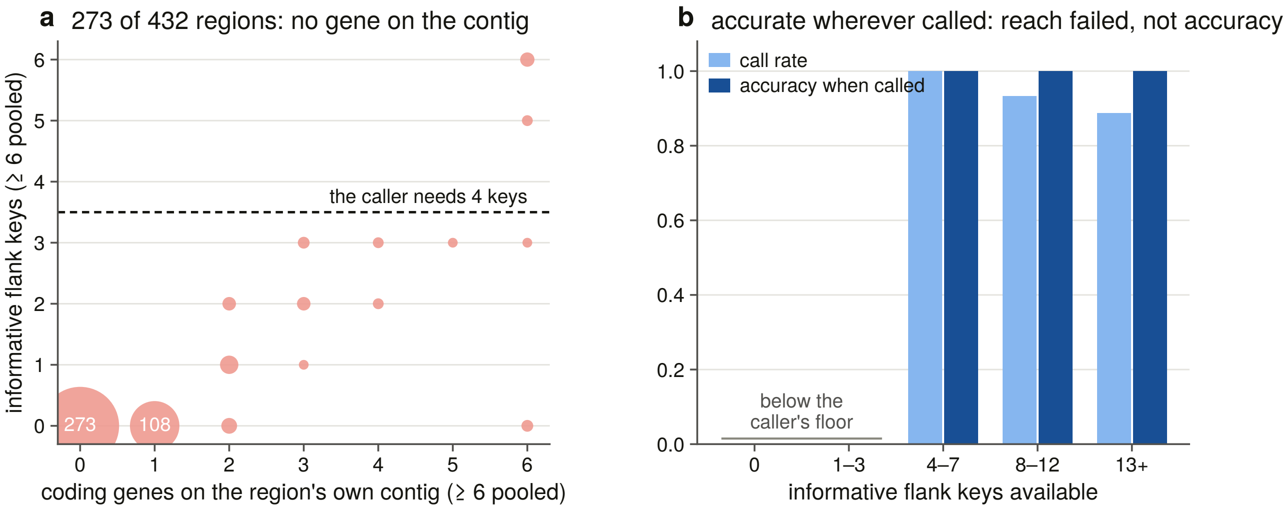
Extended Data Fig. 9 | The gene: exon structure, shared introns, and where a fragmentary annotation stops. (**a**) Coding exons per gene (median with 10th–90th percentiles), genomic span per locus on a logarithmic axis, and the median exon count recomputed at six coverage bars, for the three paralogues and the ryanodine receptor control across the 189 contiguous assemblies. The exon count does not move across the bar; the span differs 4.2-fold between paralogues at a 3 % difference in coding length. (**b**) Intron positions ranked by prevalence within each paralogue, with the 90 % line drawn, and shared positions against positions expected by chance, one point per genome per pair, with the identity line. Every IP_3 receptor pair sits far above the line in every genome; every IP_3 receptor against ryanodine receptor pair sits on it. (**c**) The instrument's own error rate: the splice dinucleotides read off the genome at all 112,254 junctions, the gap distribution that sets what counts as an intron, and agreement with 188,146 coding edges annotated by an independent pipeline. (**d**) The verdict on every split and fragmentary locus, where the annotation's internal model termini sit relative to the gene model, and the duplication detector scored against the copy call it never sees.



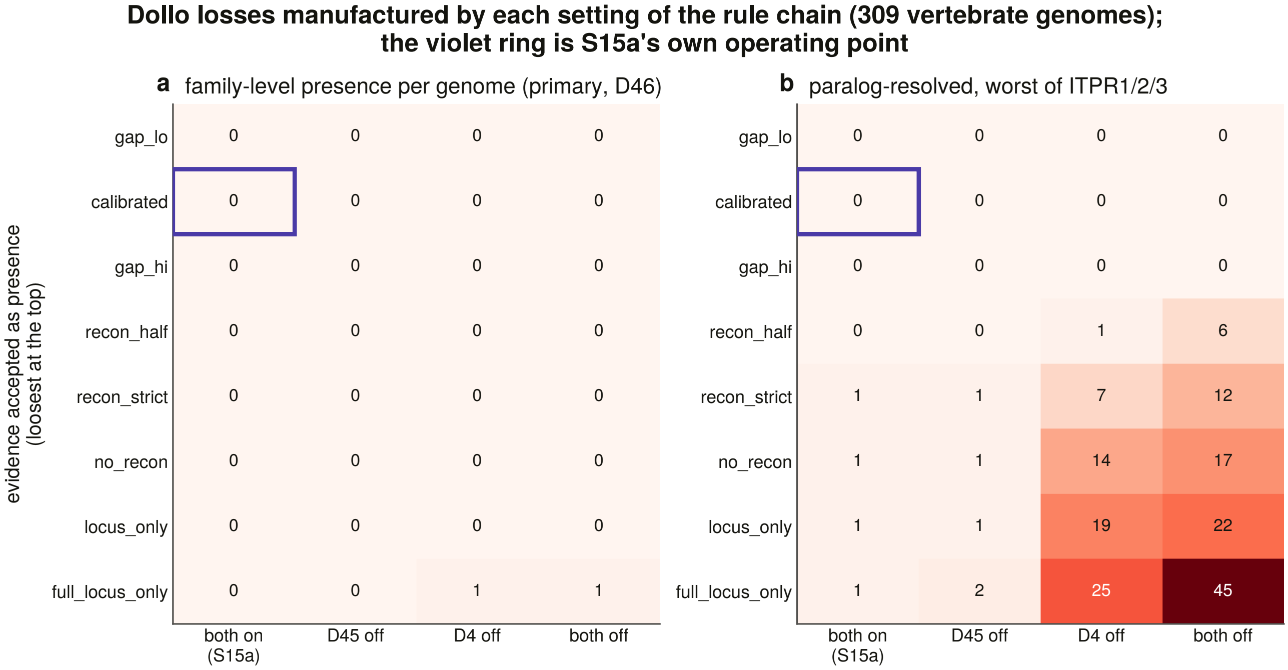
a the bar is read off an accounted-for gene
n = 44
n = 20
n = 9
fraction of the reference reassembled outside every placed locus
1.0
0.8
0.6
0.4
0.2
0.0
bar 0.137
(gap 0.100–0.174)
the cell's own clade (the candidate gene)
a paralog that is itself shattered here
a paralog already placed at a locus (the decoy)
b every undecided cell is one gene in pieces
fraction of the reference recovered
2
4
6
8
10
12
contigs the gene is spread over



a the alignment
ITPR1
ITPR3
ITPR2
RYR
disabling lesions per kilo-residue (0 shown at 0.04)
10^1
10^0
10^{-1}
ρ = -0.40
p = 1e-67
0.6
0.8
1.0
identity of the locus to its bait
b the assembly
ρ = -0.08
p = 1e-03
10^4
10^6
10^8
contig N50 (bp)
c within one genome
RYR
ITPR3
ITPR2
ITPR1
q = 0.090
q = 0.003
q = 0.902
q = 0.052
0
50
genomes (identity-matched within-genome pairs)
fewer than siblings
more than siblings



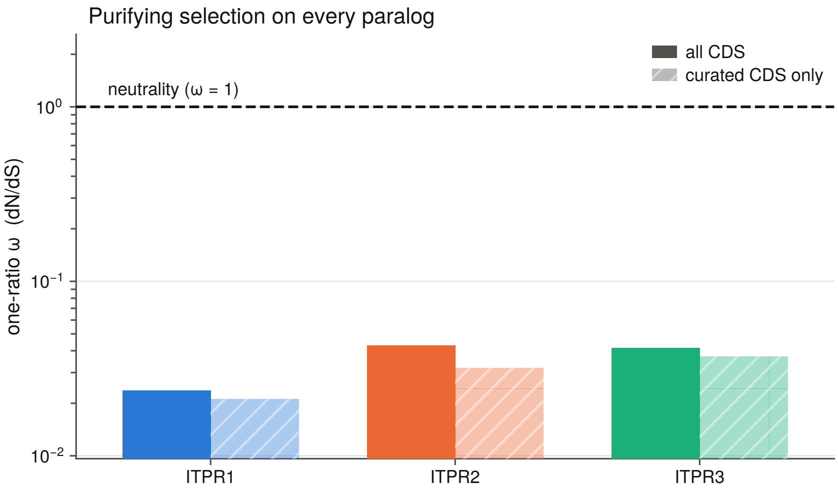
a 273 of 432 regions: no gene on the contig
informative flank keys (≥ 6 pooled)
the caller needs 4 keys
273
108
coding genes on the region's own contig (≥ 6 pooled)
b accurate wherever called: reach failed, not accuracy
call rate
accuracy when called
below the caller's floor
0
1–3
4–7
8–12
13+
informative flank keys available

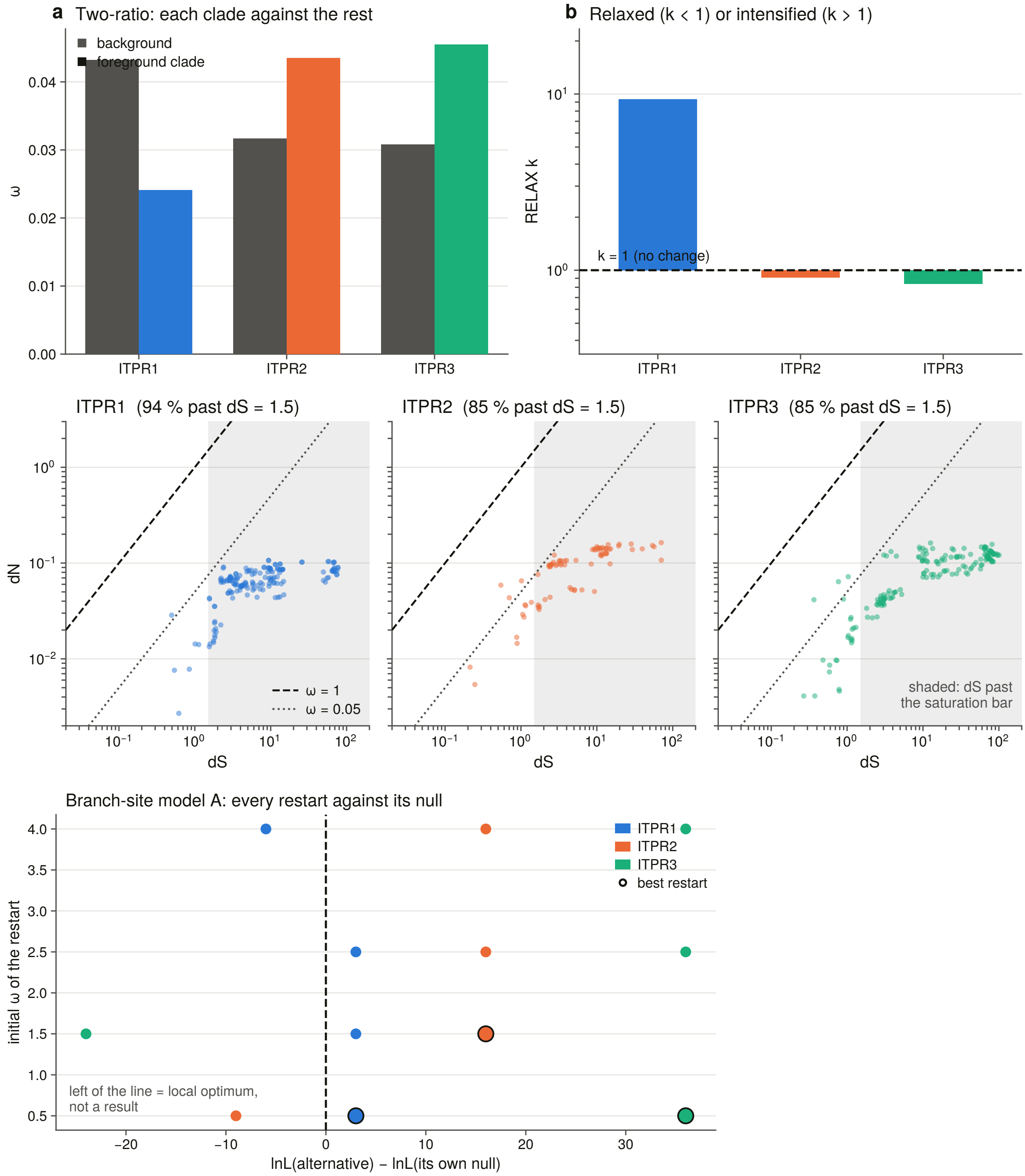


a	both on (S15a)	D45 off	D4 off	both off
gap_lo	0	0	0	0
calibrated	0	0	0	0
gap_hi	0	0	0	0
recon_half	0	0	0	0
recon_strict	0	0	0	0
no_recon	0	0	0	0
locus_only	0	0	0	0
full_locus_only	0	0	1	1

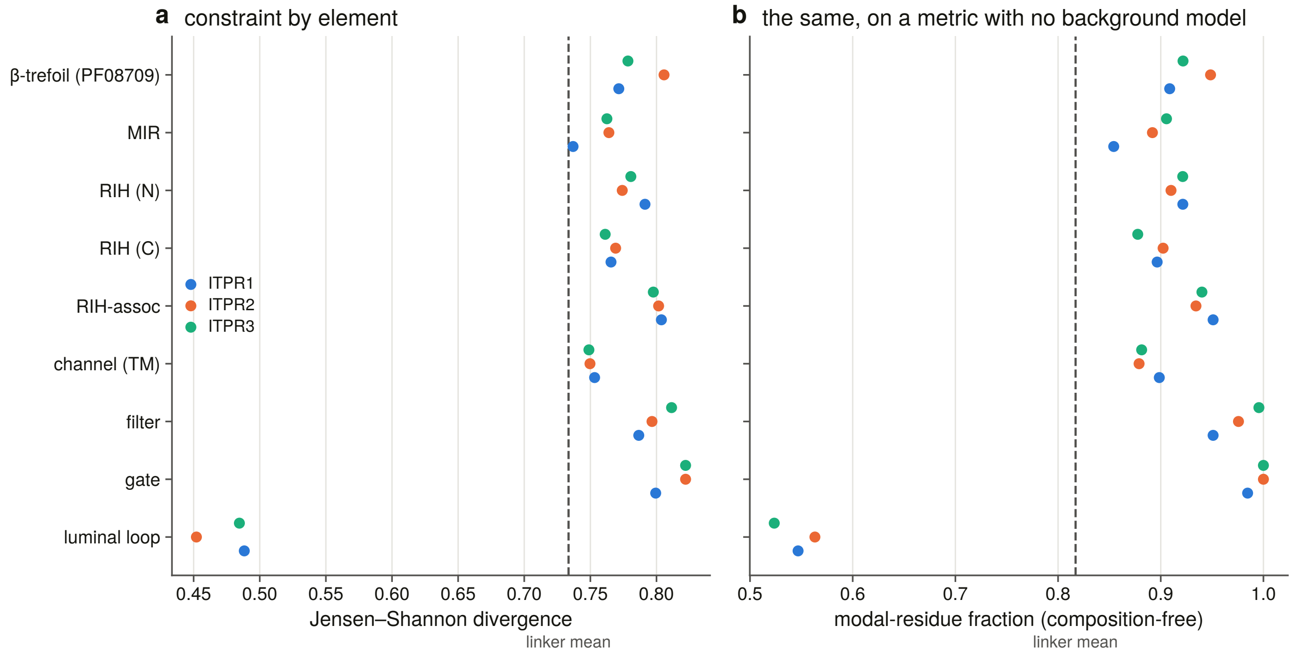
b	both on (S15a)	D45 off	D4 off	both off
gap_lo	0	0	0	0
calibrated	0	0	0	0
gap_hi	0	0	0	0
recon_half	0	0	1	6
recon_strict	1	1	7	12
no_recon	1	1	14	17
locus_only	1	1	19	22
full_locus_only	1	2	25	45

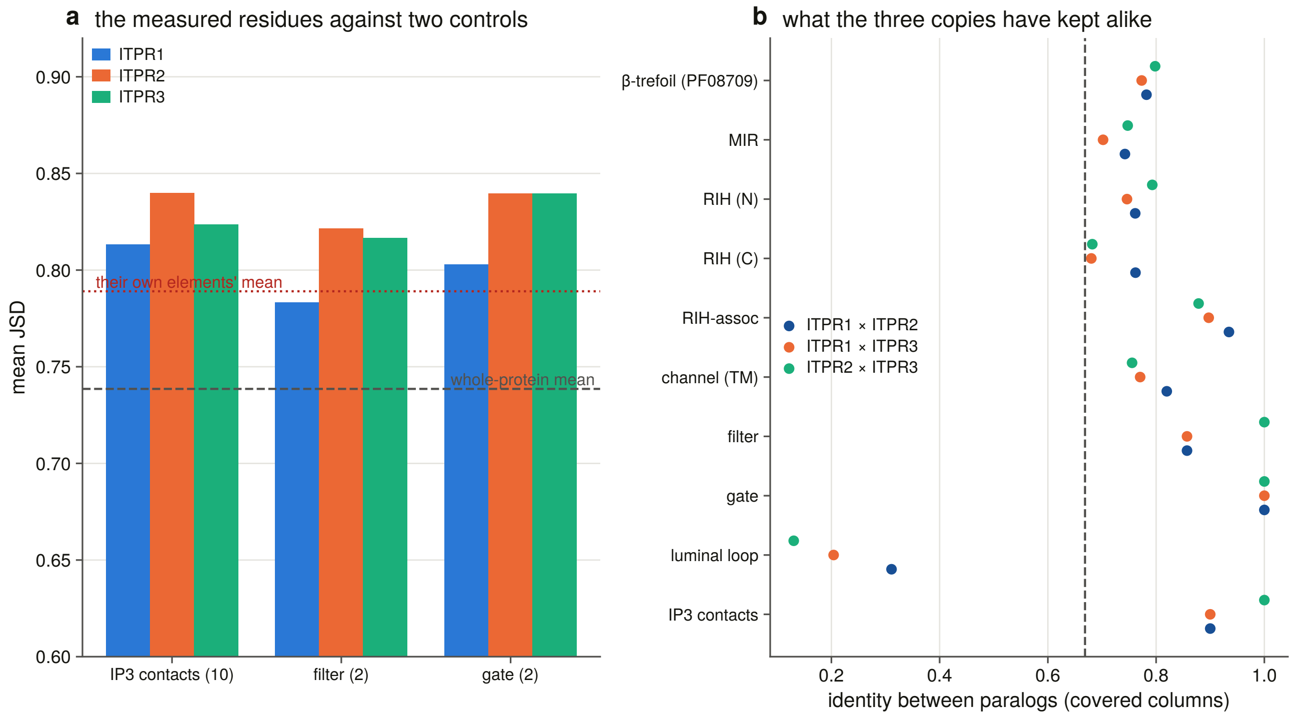
Extended Data Fig. 10 | The loss instrument, and what it takes to manufacture a loss. (**a**) The calibration: how much of a reference protein is reassembled outside every placed locus, for candidate genes, for paralogues that are themselves shattered in the same genome, and for the decoy — a paralogue whose gene the aligner has already placed elsewhere in that genome. The two informative populations separate completely; the bar sits at the midpoint of the gap and both gap edges are committed. Beside it, how far each undecided cell's reference is spread across contigs, which is what that coverage is a coverage of. (**b**) Lesion density against bait identity and against contig N50, showing that the confounder is the alignment and not the assembly, and the within-genome paired sign tests that follow from it. (**c**) Why the synteny route could not be used: 273 of 432 rescue regions sit on a contig carrying no annotated gene, while the caller is accurate at every key count it can act on. (**d**) The sensitivity matrix: Dollo losses under every combination of coding, evidence bar and the two decision rules. Zero is drawn as an explicit zero, never as an empty cell.

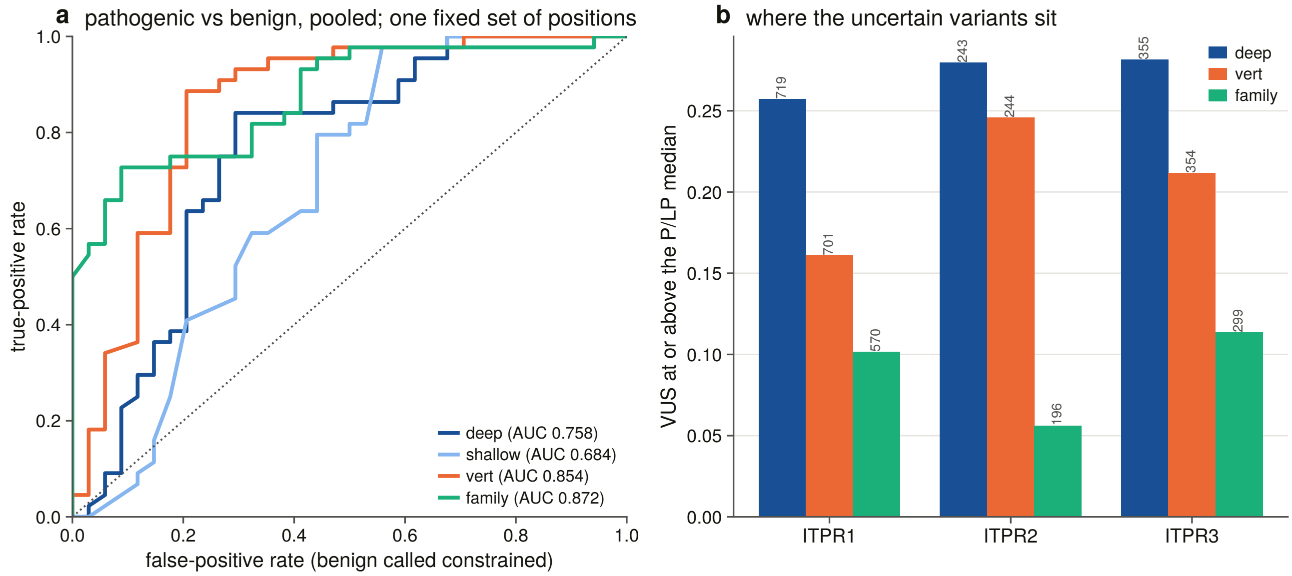




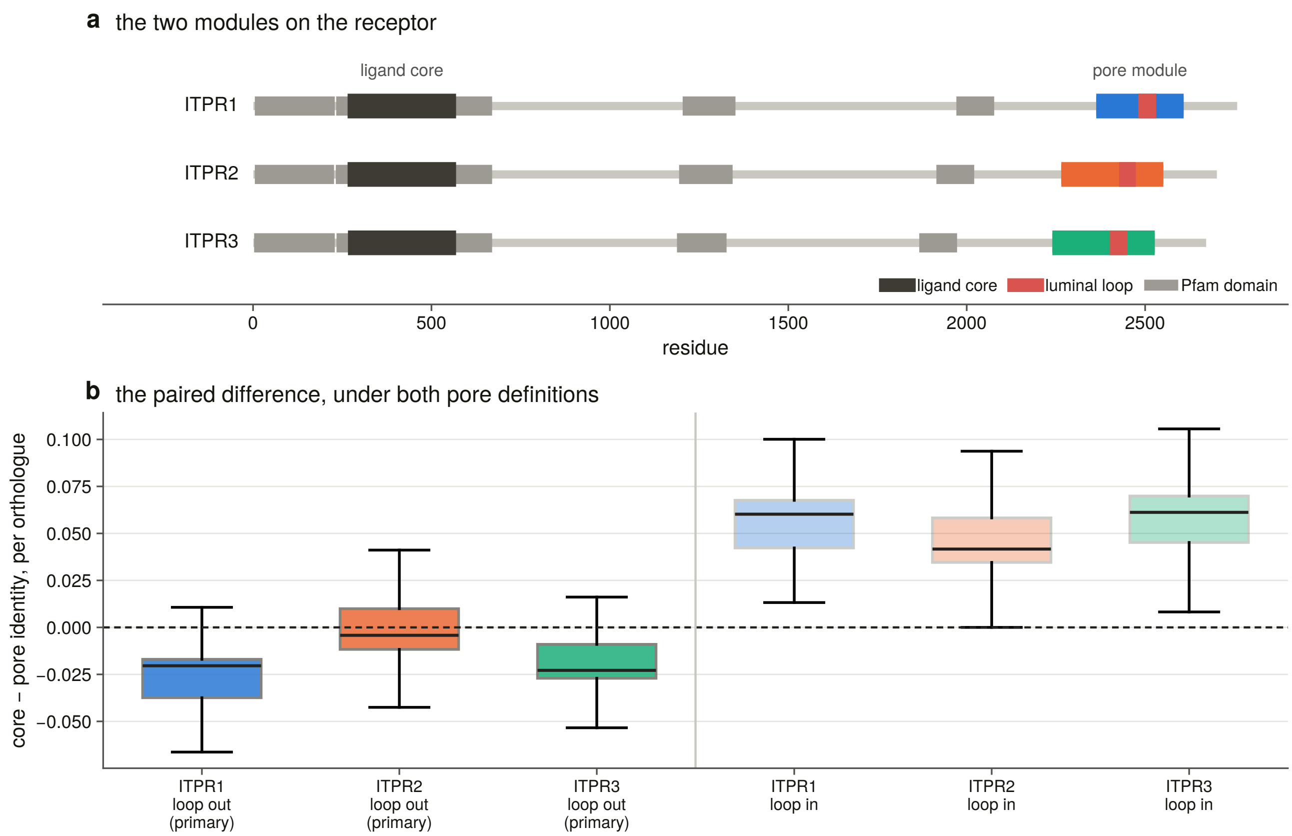
Extended Data Fig. 11 | Selection across the three paralogues. (**a**) One-ratio ω per paralogue on a logarithmic axis, with the estimate from curated coding sequences only beside each. (**b**) Whole-tree two-ratio contrasts, each clade against the rest of the tree, and the RELAX selection intensity parameter k for the same three contrasts. (**c**) dN against dS for every within-paralogue pair on log–log axes with the neutral diagonal and the saturation bar drawn; 85–94 % of pairs are past the bar, which is why every ω quoted is tree-based. (**d**) Every branch-site restart against its own nested null: points left of the line reached a lower optimum than the null they are tested against and are local optima, not results — one per stem, at a different starting ω each time.

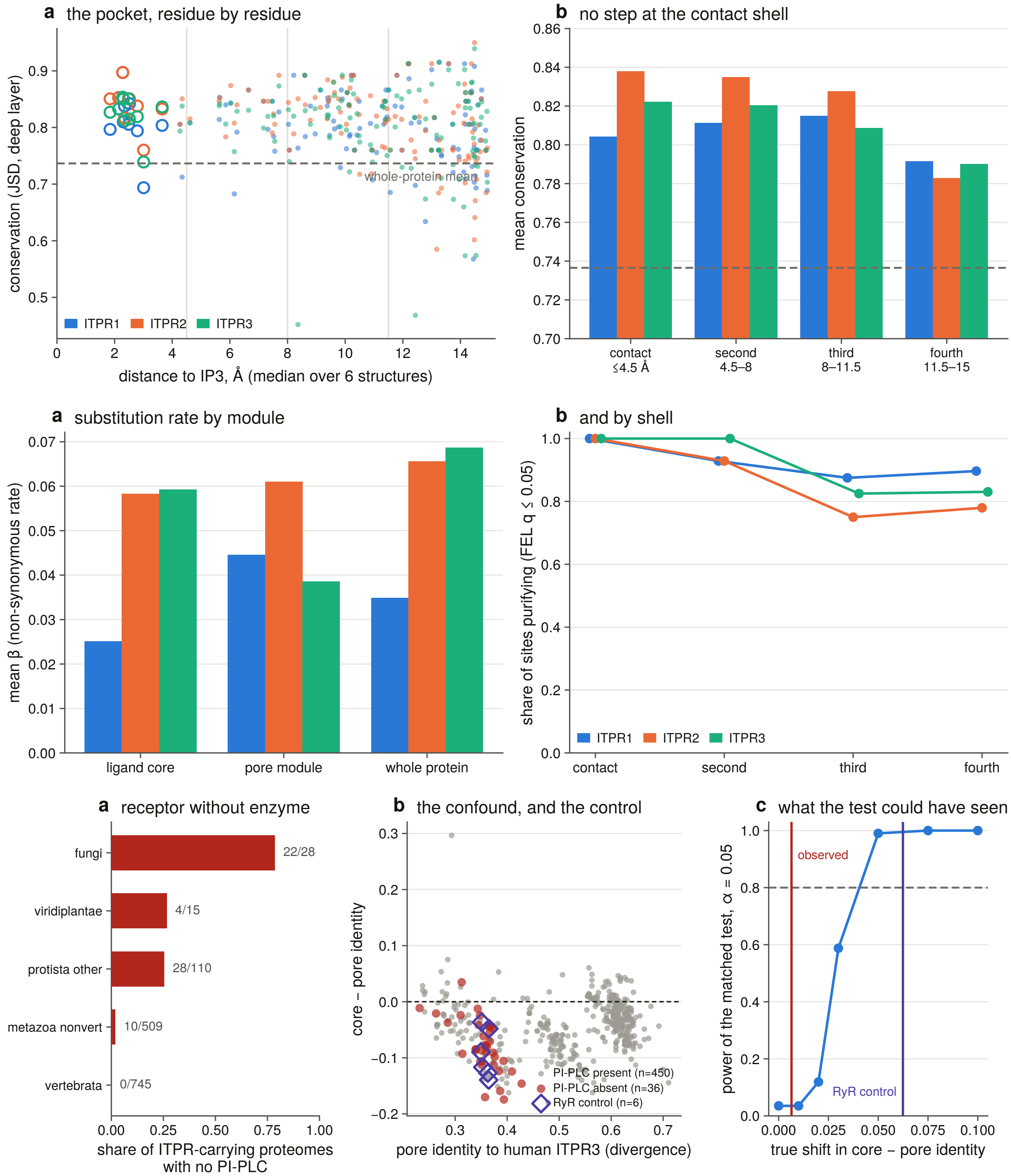

a constraint by element
b the same, on a metric with no background model
β-trefoil (PF08709)
MIR
RIH (N)
RIH (C)
RIH-assoc
channel (TM)
filter
gate
luminal loop
ITPR1
ITPR2
ITPR3
0.45 0.50 0.55 0.60 0.65 0.70 0.75 0.80
Jensen–Shannon divergence
linker mean
0.5 0.6 0.7 0.8 0.9 1.0
modal-residue fraction (composition-free)
linker mean


a the measured residues against two controls
ITPR1
ITPR2
ITPR3
mean JSD
0.60 0.65 0.70 0.75 0.80 0.85 0.90
their own elements' mean
whole-protein mean
IP3 contacts (10)
filter (2)
gate (2)
b what the three copies have kept alike
β-trefoil (PF08709)
MIR
RIH (N)
RIH (C)
RIH-assoc
channel (TM)
filter
gate
luminal loop
IP3 contacts
ITPR1 × ITPR2
ITPR1 × ITPR3
ITPR2 × ITPR3
0.2 0.4 0.6 0.8 1.0
identity between paralogs (covered columns)

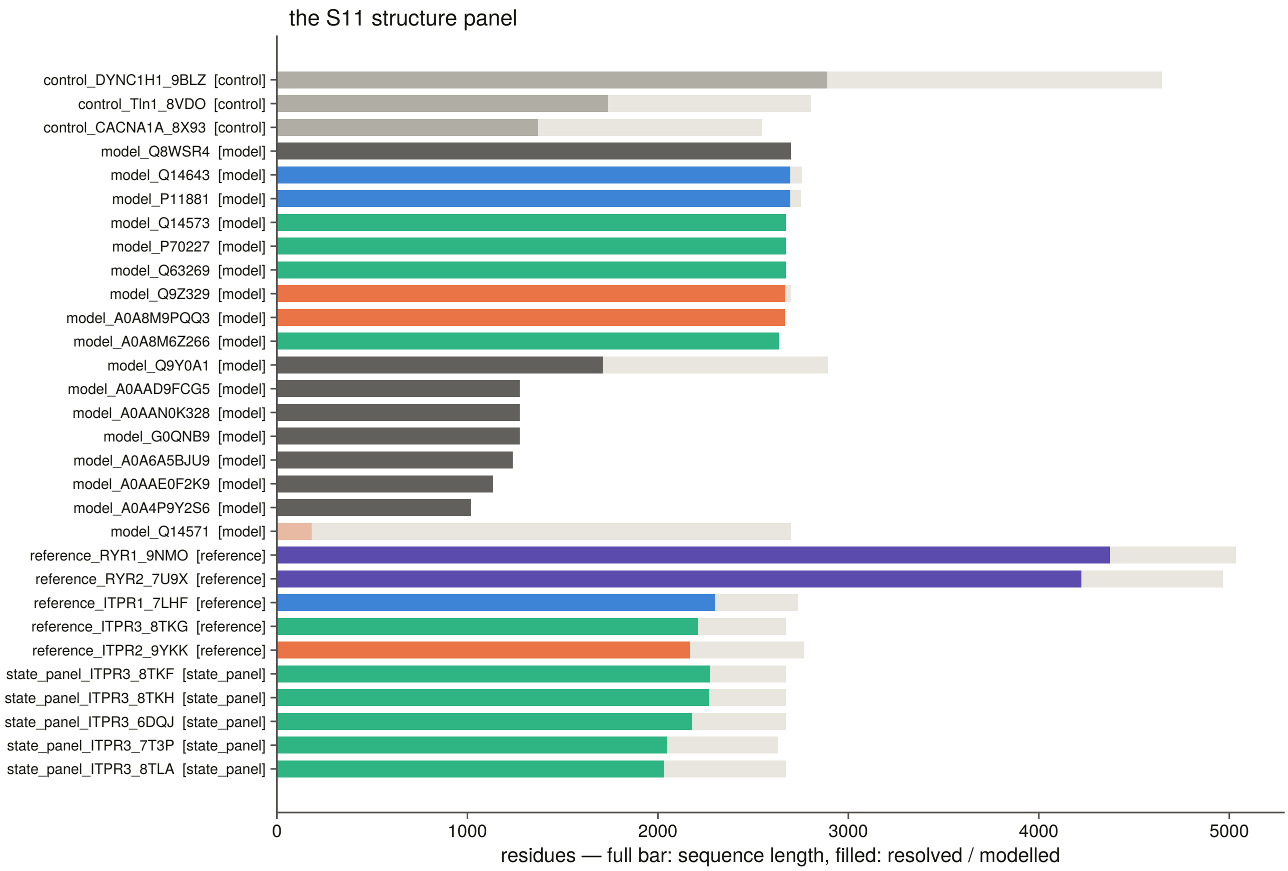


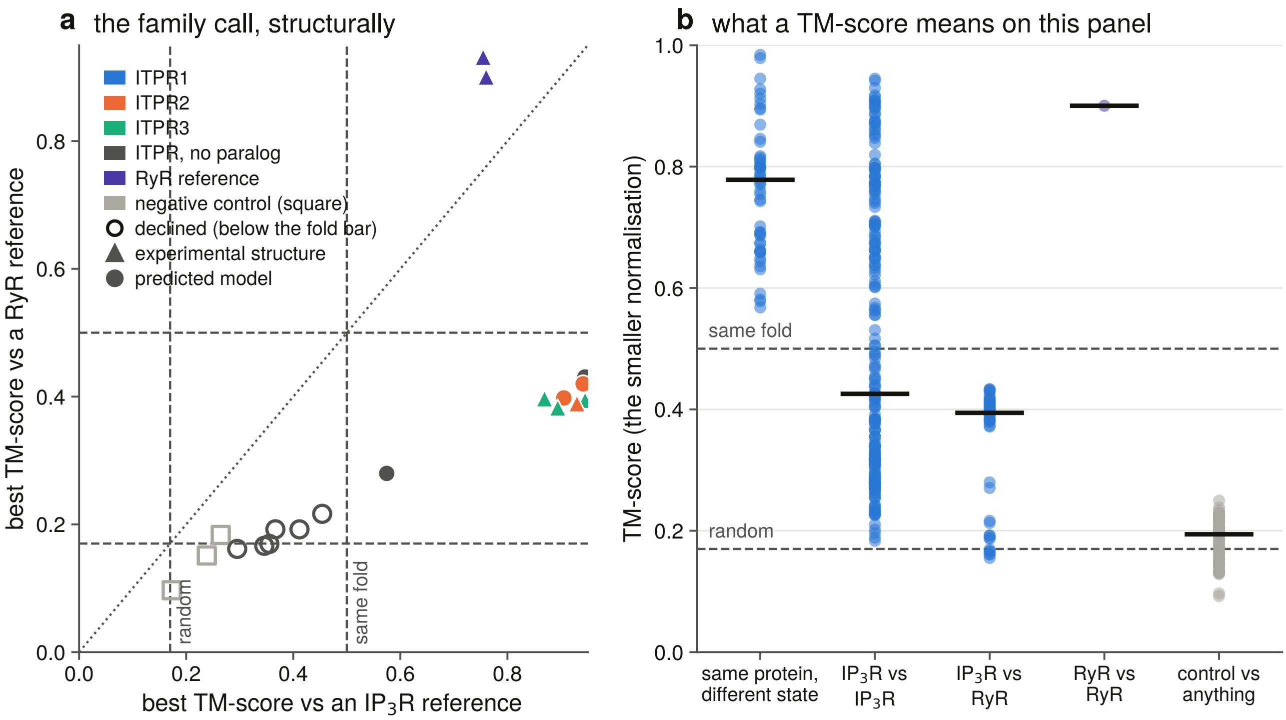
Extended Data Fig. 12 | Constraint by element, at the ligand site, and as a variant classifier. (**a**) Per-element constraint in all three paralogues on a divergence metric and on a composition-free one, with the linker control's mean over the three proteins drawn as one dashed line per panel; the two metrics agree, including on the luminal loop. (**b**) The ten measured IP_3 contacts, the two filter-lining residues and the two gate-lining residues against two controls — the whole protein and the rest of their own elements — and, beside it, between-paralogue identity per element, where the gate is at 1.00 and the luminal loop at 0.13–0.31. (**c**) Receiver-operating curves for all four constraint layers as classifiers of pathogenic against benign missense variants on one fixed set of positions, and where the uncertain variants fall relative to the pathogenic median on each layer.

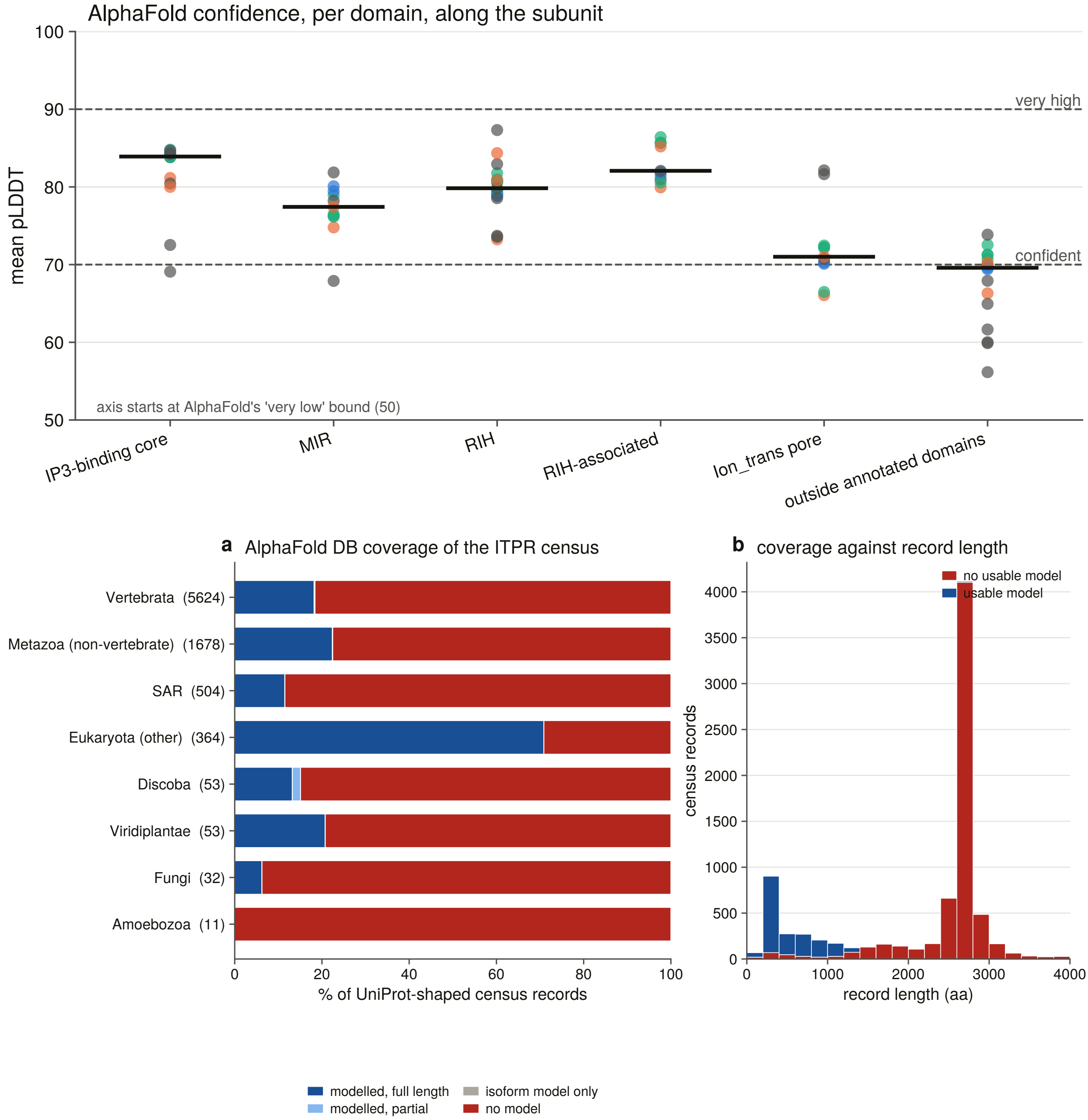




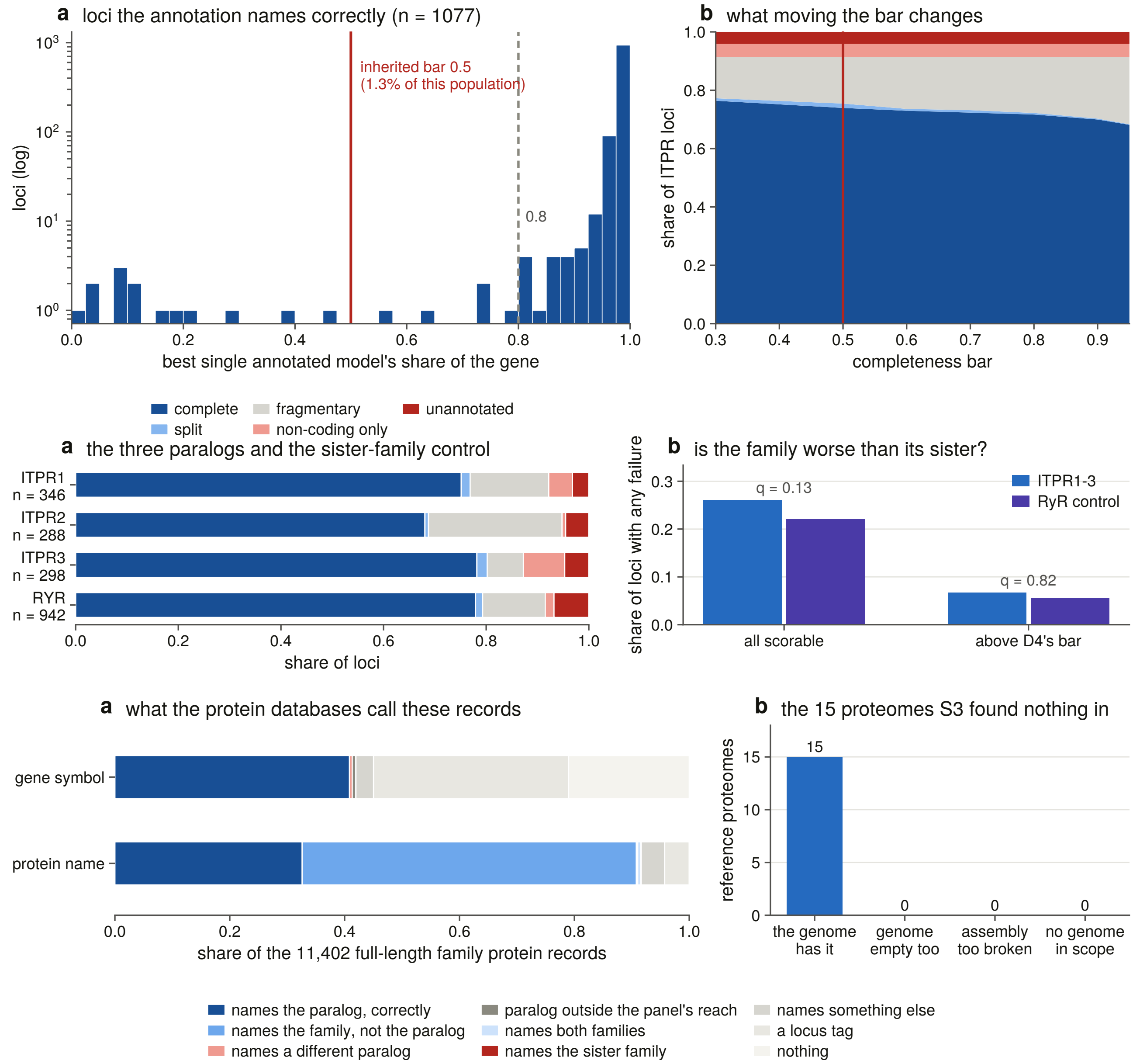
Extended Data Fig. 13 | The ligand core against the pore, and the lineages that lost the enzyme upstream. (**a**) The two modules drawn on the receptor, and the paired per-orthologue difference between them under both pore definitions. With the luminal loop excluded the pore is more conserved in *ITPR1* and *ITPR3*; with it included — as InterPro draws PF00520 — all three reverse. Zero is marked, and both definitions are on one axis because the answer turns on the choice. (**b**) Constraint against distance from the ligand, one point per pocket residue across six IP_3-bound depositions, with the 4.5 Å contact bar drawn: there is no step at it. (**c**) The same two modules and the same four shells measured on site-wise substitution rate rather than on column conservation. (**d**) The lineage test: the 64 proteomes carrying a receptor and no phosphoinositide-specific phospholipase C, the divergence confound that makes the pooled comparison uninterpretable, the same comparison matched on divergence, and the power the matched test has against the shift the ryanodine receptor control actually produces.



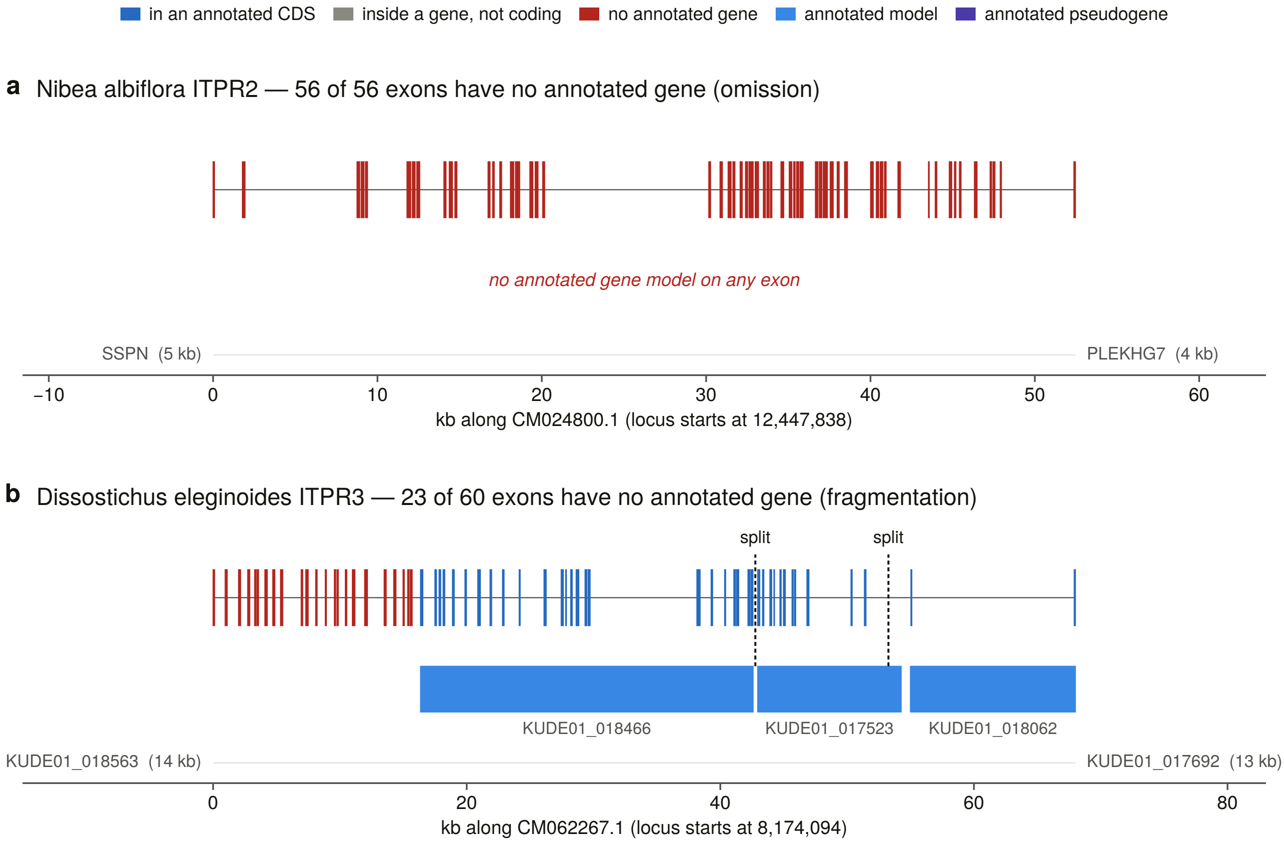




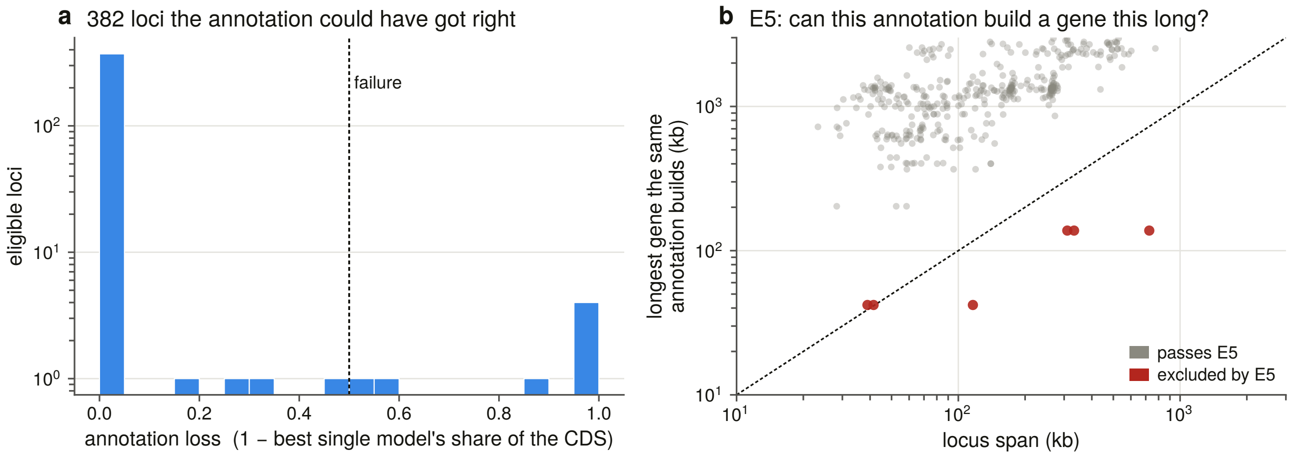
Extended Data Fig. 14 | Predicted and experimental structures across the family. (**a**) The panel: sequence length and resolved or modelled residues for every reference, state-panel member, negative control and predicted model. Twenty-nine of the thirty are long enough to score; the thirtieth is the record AlphaFold DB serves for human *ITPR2*, a 181-residue isoform, drawn here because that is the coverage result. (**b**) The structural family call: best TM-score against an IP_3 receptor reference against best against a ryanodine receptor reference, with TM-align's own random and same-fold bars drawn rather than described; open symbols are declined for falling below the fold bar, and marker shape is the role: triangle experimental, circle predicted, square control. Beside it, what a TM-score means on this panel — conformation is worth 0.22, and no control pair reaches the fold bar. (**c**) Mean AlphaFold confidence per domain: the IP_3-binding core is the best-modelled domain and the pore the worst. (**d**) AlphaFold DB coverage of the census by group, and against record length, where the modelled mass sits below ~1,300 residues and the peak at a full-length subunit is almost entirely unmodelled.



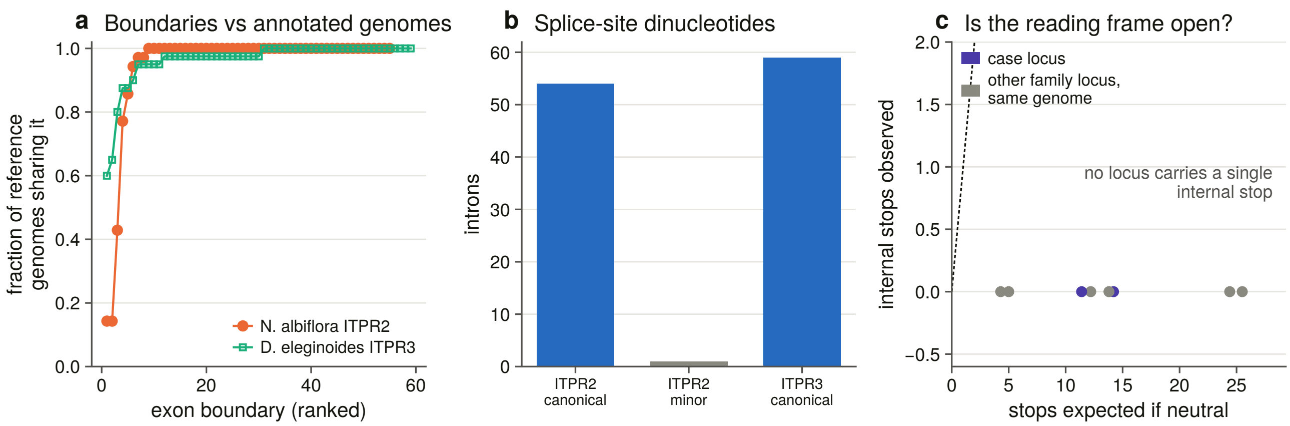
Extended Data Fig. 15 | The annotation audit against its own controls. (**a**) The completeness bar validated rather than re-derived: over 1,077 loci the annotation names correctly, the best single model's share of the gene, with the inherited 0.50 bar marked at that distribution's 1.3 % point; and every state recounted across bars from 0.30 to 0.95. (**b**) Annotation state for each paralogue and for the ryanodine receptor control, and the family against the control with and without the contiguity restriction: the difference does not survive correction either way. (**c**) What the protein databases call the 11,402 full-length family records, by gene symbol and by protein name, and the resolution of the 15 reference proteomes that returned nothing — all 15 are gene-caller failures.



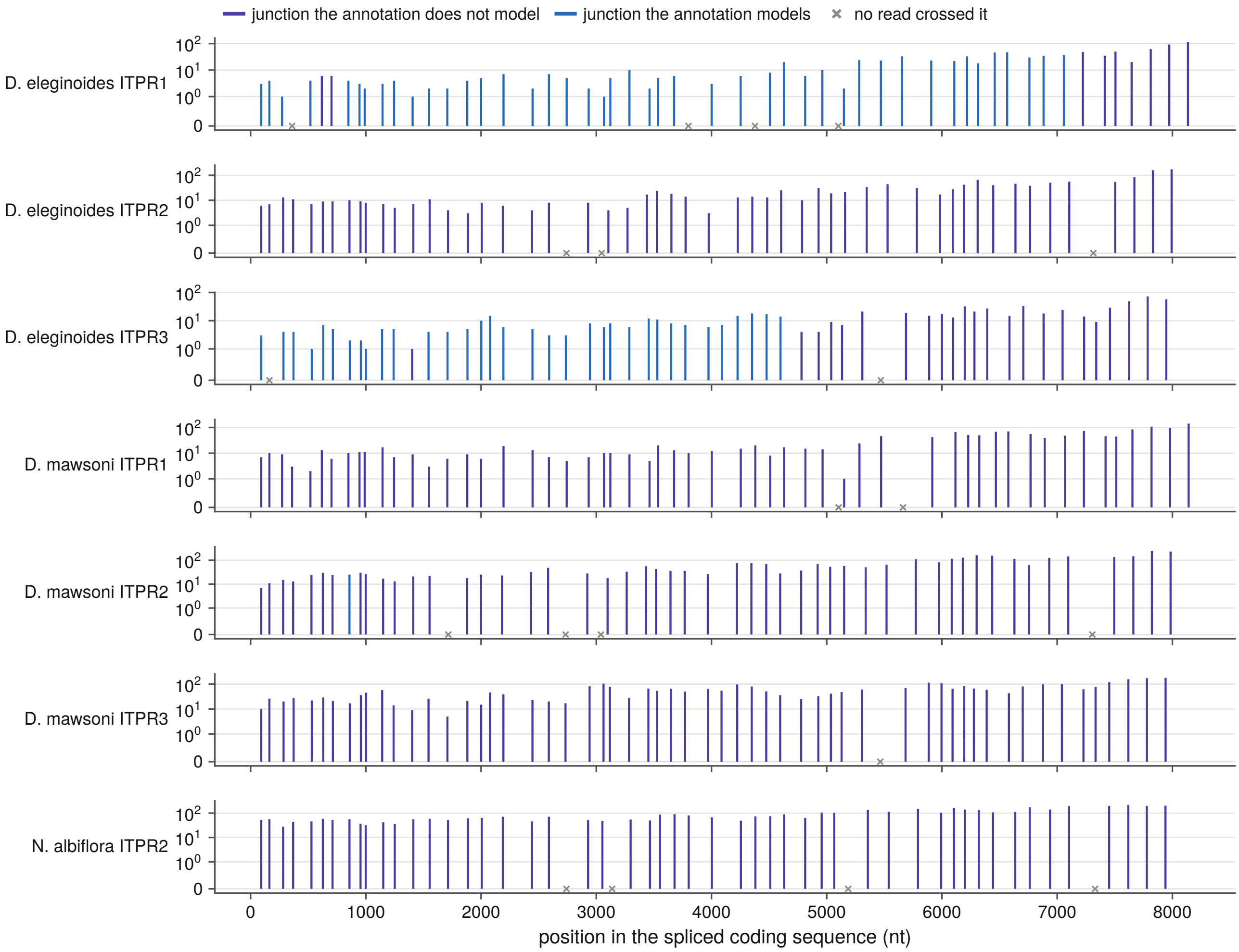
in an annotated CDS
inside a gene, not coding
no annotated gene
annotated model
annotated pseudogene
a Nibea albiflora ITPR2 — 56 of 56 exons have no annotated gene (omission)
no annotated gene model on any exon
SSPN (5 kb)
PLEKHG7 (4 kb)
−10
0
10
20
30
40
50
60
kb along CM024800.1 (locus starts at 12,447,838)
b Dissostichus eleginoides ITPR3 — 23 of 60 exons have no annotated gene (fragmentation)
split
split
KUDE01_018466
KUDE01_017523
KUDE01_018062
KUDE01_018563 (14 kb)
KUDE01_017692 (13 kb)
0
20
40
60
80
kb along CM062267.1 (locus starts at 8,174,094)



a 382 loci the annotation could have got right
eligible loci
10^2
10^1
10^0
failure
0.0
0.2
0.4
0.6
0.8
1.0
annotation loss (1 − best single model's share of the CDS)
b E5: can this annotation build a gene this long?
longest gene the same annotation builds (kb)
10^3
10^2
10^1
passes E5
excluded by E5
locus span (kb)



a Boundaries vs annotated genomes
fraction of reference genomes sharing it
1.0
0.8
0.6
0.4
0.2
0.0
N. albiflora ITPR2
D. eleginoides ITPR3
0
20
40
60
exon boundary (ranked)
b Splice-site dinucleotides
introns
60
50
40
30
20
10
0
ITPR2 canonical
ITPR2 minor
ITPR3 canonical
c Is the reading frame open?
internal stops observed
2.0
1.5
1.0
0.5
0.0
−0.5
case locus
other family locus, same genome
no locus carries a single internal stop
0
5
10
15
20
25
stops expected if neutral



Extended Data Fig. 16 | Two annotation failures validated at the exon and by RNA-seq. (**a**) Exon tracks at true genomic width for the two validated cases, with each exon coloured by what the annotation holds at that interval and the annotated models drawn beneath. *Nibea albiflora ITPR2* has no annotated gene on any of its 56 exons; *Dissostichus eleginoides ITPR3* has 23 of 60 exons with no annotated gene and is called as three separate models. (**b**) The denominator: annotation loss across all 382 eligible loci, and the eligibility rule that does the work — whether the same annotation builds genes that long anywhere else in the same genome. (**c**) The checks on the alignment evidence itself: exon-boundary concordance with independently annotated genomes, splice dinucleotides, and internal stop codons observed against the number expected under neutral drift. (**d**) Every junction of every locus in the expression panel, at its position in the spliced coding sequence, coloured by whether an annotated model spans it and marked where no read crossed it.

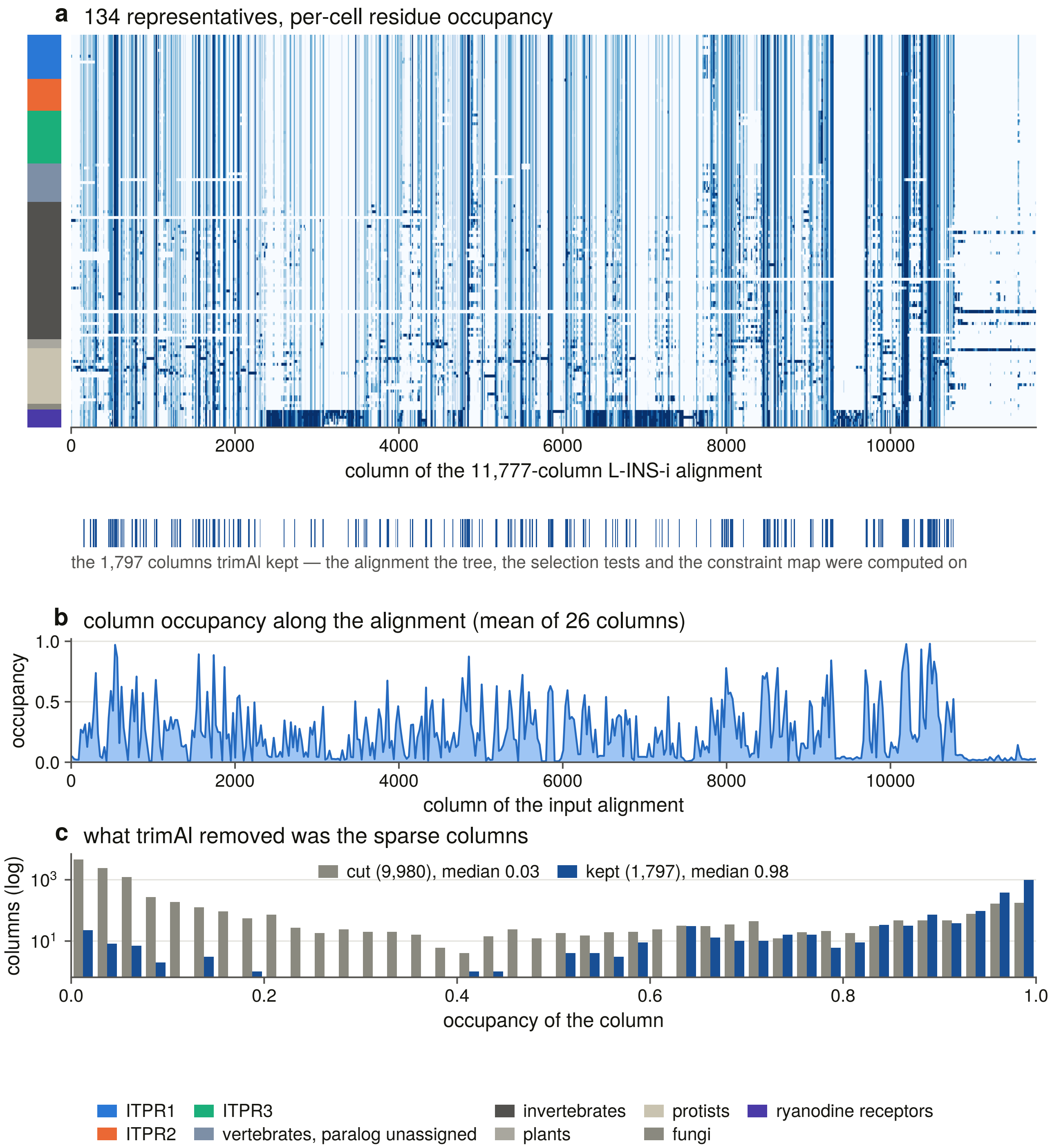
Supplementary information

Supplementary Figures 1–6 show the alignments and structures the main figures rest on: the representative alignment and the columns the tree actually saw; the IP_3-binding core and the pore module at residue resolution across the three paralogues, with each clinically labelled position checked against the residue at its column; the per-paralogue alignments the constraint map is computed on; the trimmed codon alignment behind every ω estimate; the constraint map painted onto the channel; and every labelled variant with its per-element enrichment test. None of them re-aligns or re-renders anything: each is drawn from the same committed file as the main figure it supports, so a supplementary panel that disagreed with its main figure would be a bug rather than a difference of method.

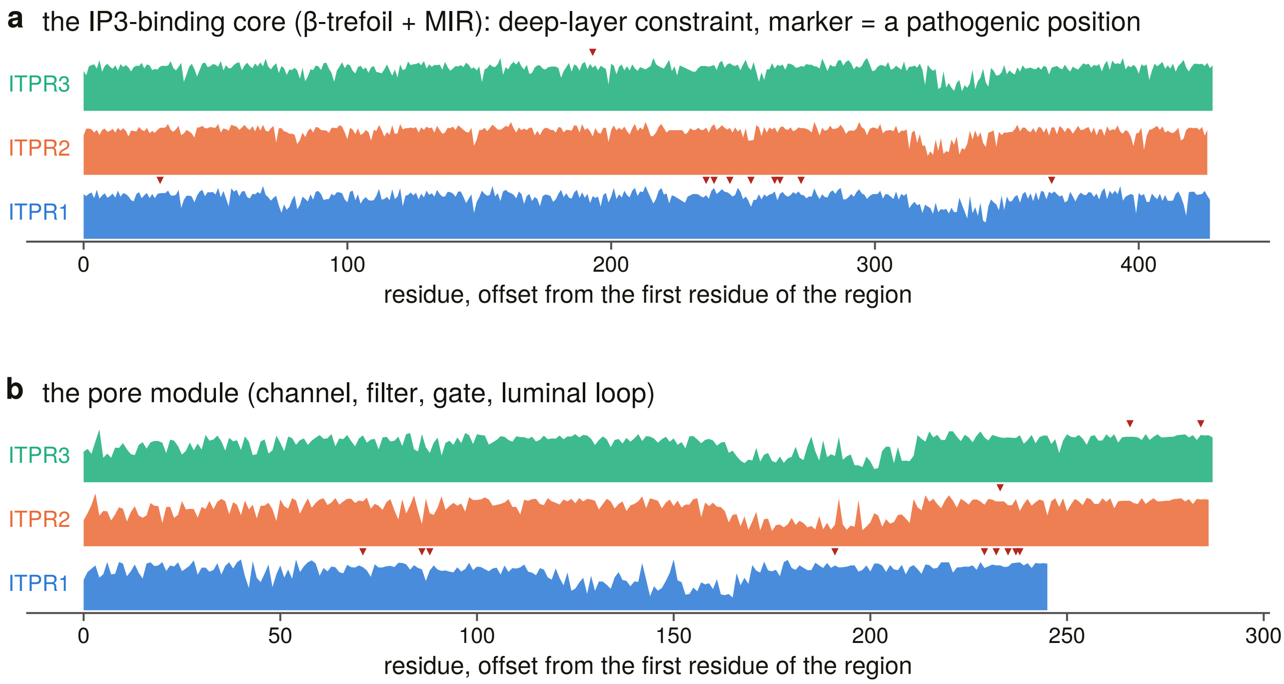
Two joins are checked before any of them is drawn, and both are hard failures. trimAl writes no map between the untrimmed alignment and the trimmed one, so the map committed with the alignment is verified by an exhaustive column walk: all 1,797 trimmed columns against all 134 sequences, because an off-by-one would renumber every residue claim downstream and would have no other symptom. And every clinically labelled position is checked against the residue its own paralogue's per-residue table holds there, and every aligned partner against the residue the *other* paralogue's table holds — 1,780 variants and 2,699 aligned partners.

A structure carries a variant position only if it earns it. Human *ITPR2* (9YKK) and *ITPR3* (8TKG) match the human per-residue table at every residue they share with it; the *ITPR1* cryo-EM reference is a rat structure and the AlphaFold DB model of human *ITPR1* is the 2,695-residue Q14643-4 isoform, so neither carries this gene's numbering and *ITPR1*'s pathogenic positions are not drawn on coordinates that are not theirs. That refusal is the panel, not a caveat to it.

Supplementary figure legends



Supplementary Fig. 1 | The representative alignment, and the columns the tree actually saw. (**a**) The 134 representative proteins over the 11,777 columns MAFFT L-INS-i produced, each cell shaded by the fraction of its column block that is residue rather than gap, rows ordered and side-barred by group. Beneath it, the 1,797 columns trimAl kept — the alignment the tree, the selection tests and the constraint map were all computed on. The map between the two coordinate systems was verified by an exhaustive column walk before this figure was drawn. (**b**) Column occupancy along the input alignment. (**c**) The same occupancy as two distributions, kept against cut: the median kept column is 0.98 occupied and the median cut column 0.03, so what trimAl removed was the sparse columns and not a region.

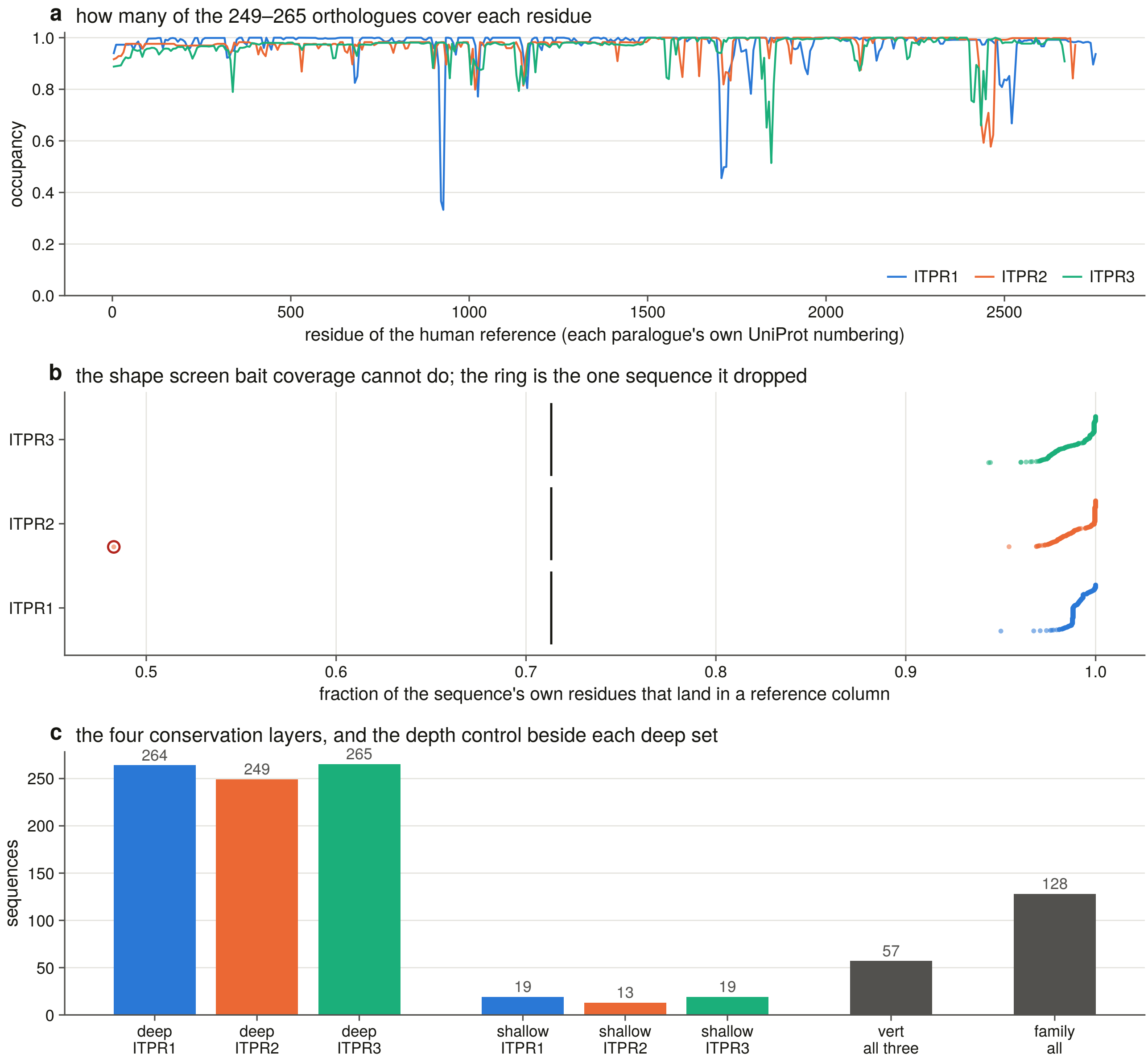


c every pathogenic position in the two regions, and the residue each paralogue carries there

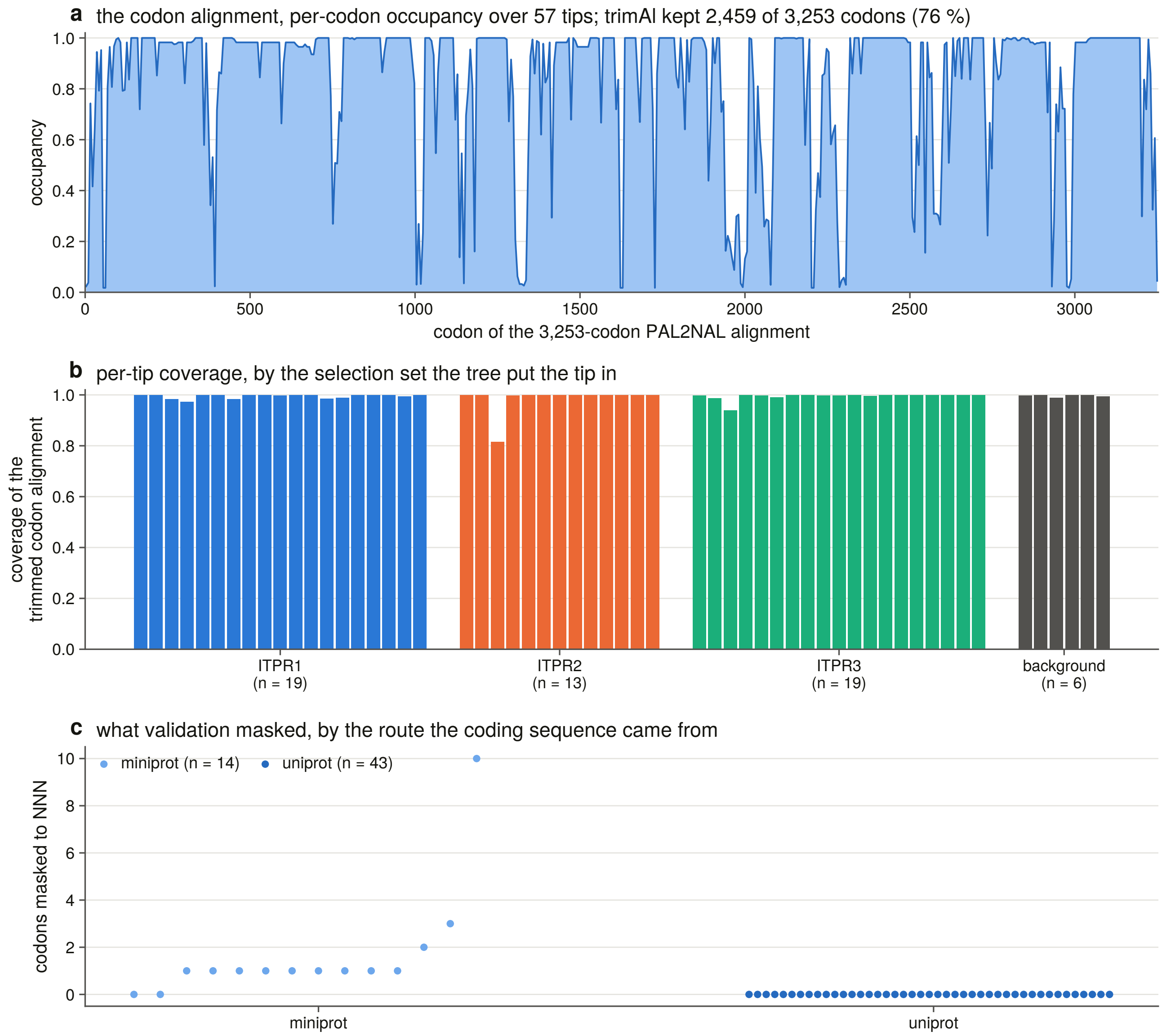
		ITPR1/2/3 1	2	3				ITPR1/2/3 1	2	3	
ITPR1 34	β-trefoil (PF08709)	D	D	D	→N	ITPR1 2449	channel (TM)	I	I	I	→V
ITPR3 196	β-trefoil (PF08709)	A	A	A	→T	ITPR1 2451	channel (TM)	L	L	L	→P
ITPR1 241	MIR	R	R	R	→K	ITPR1 2592	channel (TM)	L	L	L	→P
ITPR1 244	MIR	H	H	H	→P	ITPR1 2600	channel (TM)	T	T	T	→P
ITPR1 250	MIR	F	F	F	→L	ITPR1 2601	channel (TM)	F	F	F	→L
ITPR1 258	MIR	K	K	G	→M	ITPR3 2506	channel (TM)	I	I	I	→N
ITPR1 267	MIR	T	T	T	→R	ITPR3 2524	channel (TM)	R	R	R	→H
ITPR1 269	MIR	R	R	R	→L	ITPR1 2554	filter	G	G	G	→R
ITPR1 277	MIR	S	S	S	→I	ITPR2 2498	filter	G	G	G	→S
ITPR1 372	MIR	P	A	P	→L	ITPR1 2595	gate	G	G	G	→A
ITPR1 2434	channel (TM)	T	T	T	→I	ITPR1 2598	gate	I	I	I	→T

red letter = the other paralogue carries a different residue; every letter checked against that paralogue's own per-residue table

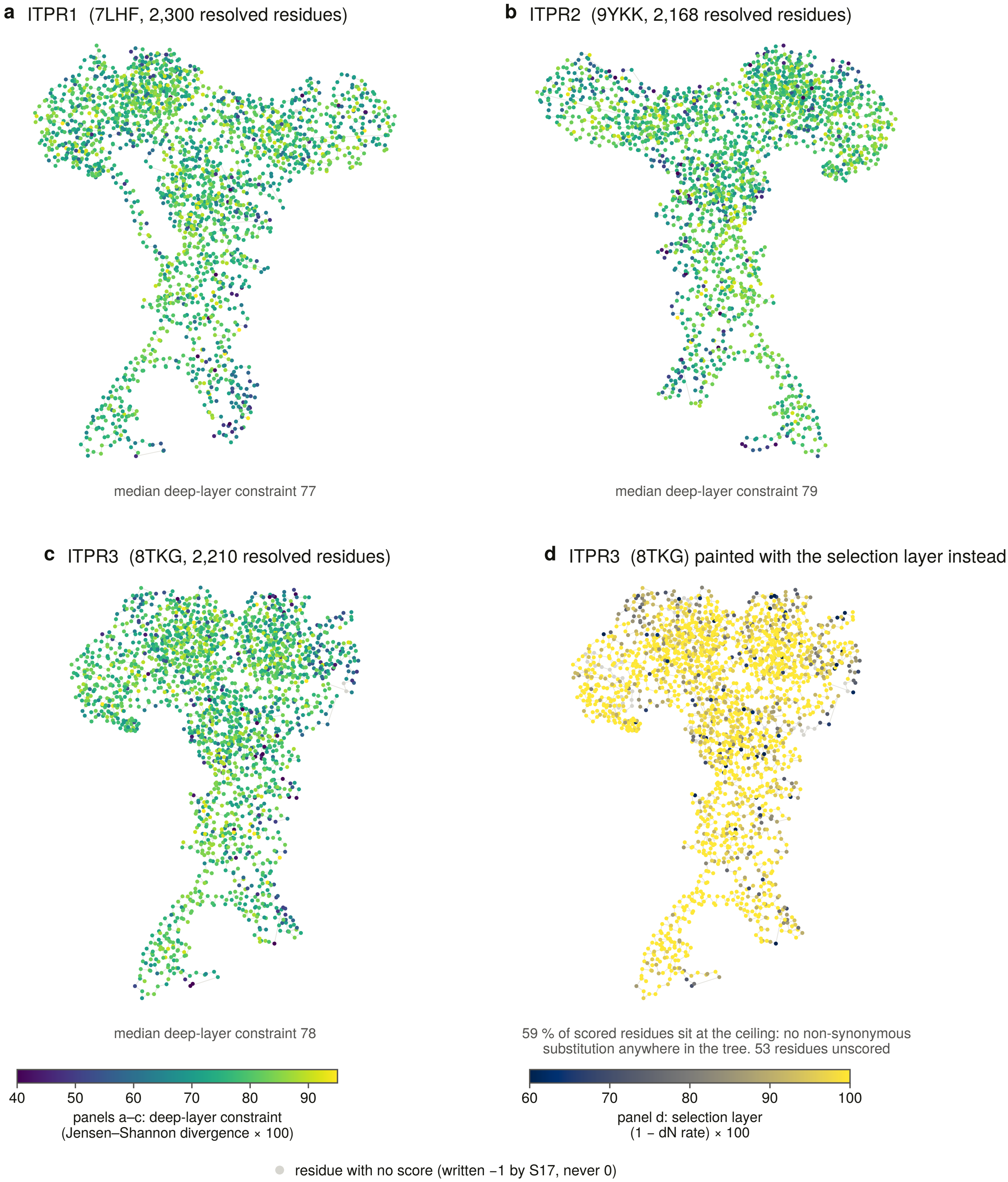
Supplementary Fig. 2 | The ligand core and the pore module at residue resolution across the three paralogues. (**a**) Deep-layer constraint along the β-trefoil and MIR domains of each paralogue in its own numbering, with every pathogenic position marked. (**b**) The same for the channel region — selectivity filter, gate, and the geometrically located luminal loop drawn beside them rather than inside the pore module, since whether it counts as pore is what reverses the comparison in the main text. (**c**) Every pathogenic position in those two regions with the residue each paralogue carries there; a red letter is a paralogue that carries a different residue. Twenty of the 22 positions are the same residue in all three. Every letter was checked against that paralogue's own per-residue table.



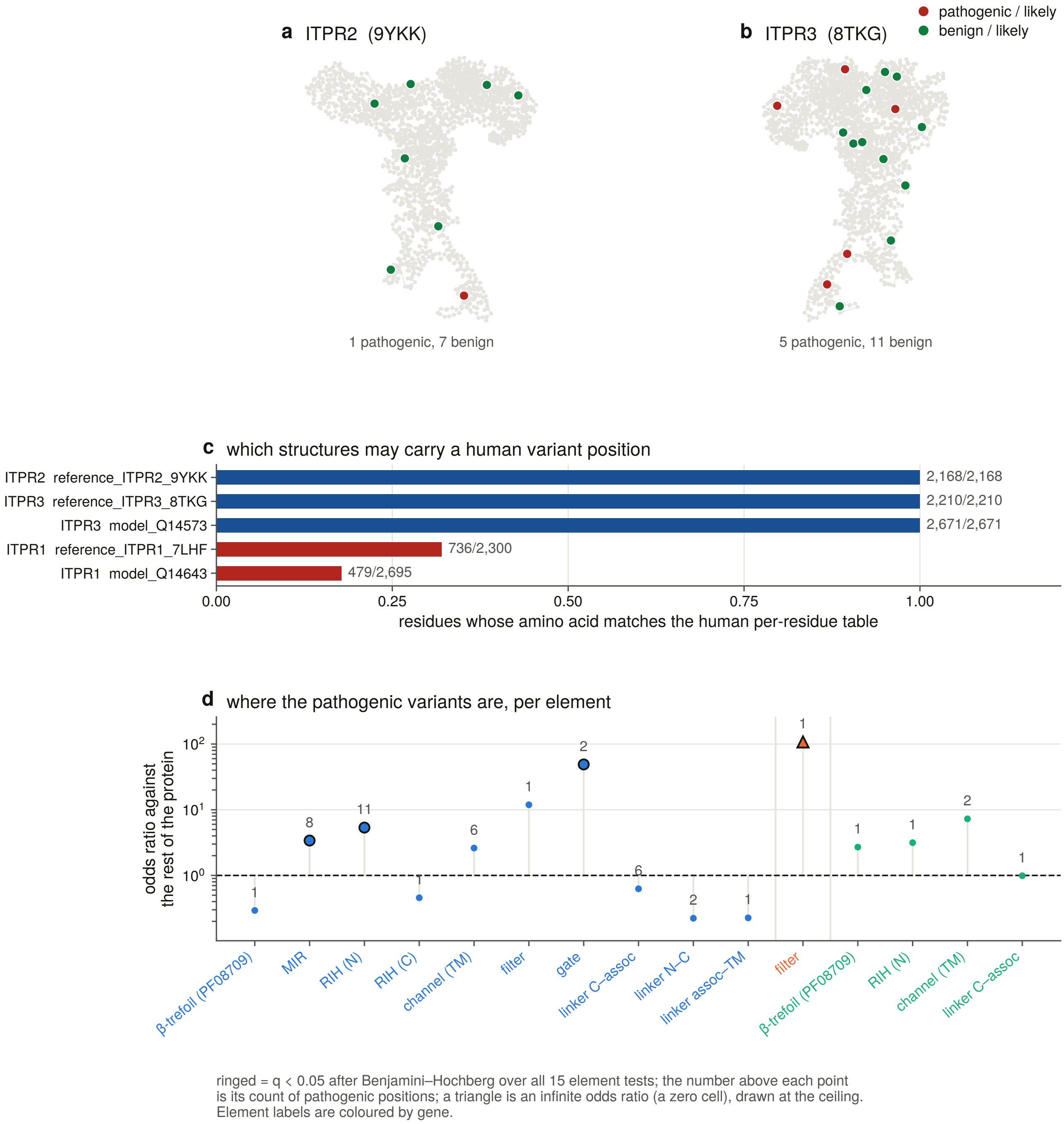
Supplementary Fig. 3 | The within-paralogue alignments the constraint map is computed on. (**a**) How many of each paralogue's 249–265 sweep orthologues cover each residue of the human reference. (**b**) The shape screen that bait coverage cannot do: the fraction of each sequence's own residues that land in a reference column, with the calibrated bar drawn and the one sequence it dropped ringed. (**c**) The four conservation layers, with the shallow msa_v2-only set beside each deep set as the control for what the depth bought.



Supplementary Fig. 4 | The trimmed codon alignment behind every ω estimate. (**a**) Per-codon occupancy of the 3,253-codon PAL2NAL alignment over 57 tips; trimAl kept 2,459 codons. (**b**) Per-tip coverage of the trimmed alignment, grouped by the selection set the tree placed the tip in. (**c**) Codons masked to NNN by validation, by the route the coding sequence came from: the genome gene models carry the masking, the UniProt records almost none.



Supplementary Fig. 5 | The constraint map painted onto the channel. (**a**–**c**) One subunit of each paralogue's cryo-EM reference, projected on its two principal axes and coloured by deep-layer constraint from the file S17 painted. (**d**) The same *ITPR3* structure painted with the selection layer instead; 59 % of its scored residues sit at the ceiling, having no non-synonymous substitution anywhere in the tree. Residues with no score are grey and never the low end of the scale, because the luminal loop is both the least conserved element and the worst resolved.



Supplementary Fig. 6 | Every labelled variant, and the per-element enrichment test. (**a**, **b**) Pathogenic and benign missense positions on the two structures whose numbering the check admits. (**c**) The check: the fraction of each candidate structure's residues whose amino acid matches the human per-residue table. Human *ITPR2* and *ITPR3* match at every residue; the rat *ITPR1* reference and the Q14643-4 isoform AlphaFold DB serves for human *ITPR1* do not, so that gene's pathogenic positions are not placed. (**d**) Pathogenic positions per element as an odds ratio against the rest of the protein, Benjamini–Hochberg corrected across all 15 element tests.

A resource, offered with its limit stated. The per-residue constraint tables carry all four conservation layers, the site-wise selection rate, the structural element and the measured-site flags for every residue of the three human paralogues, and `variants.tsv` joins them onto all 1,753 harvested ClinVar records. `vus_stratification.tsv` places each of the 1,546 uncertain variants against its own gene's labelled distributions. That is a stratification and not a call: it says where a variant sits on an axis that separates the labelled ones, and nothing about the variant's effect. It is offered as evidence to combine with others, not as a classifier to act on.

Supplementary Tables are the committed analysis tables themselves, listed with their sizes and checksums in `manuscript/deposit_manifest.tsv`. The tables a reader is most likely to want are: the genome manifests (the two declared denominators), the census (`census_v6.tsv`, one row per record with both instruments' verdicts), the per-genome ledger, the character matrix, the per-residue constraint tables for the three human paralogues, the correction list, and the per-task statistics files, each of which carries the parameters, the self-test status and the SHA-256 of every table its task wrote.

Data availability

Every committed table, figure, alignment, tree and report in this study is in the deposited archive, listed file by file with its size and SHA-256 in `manuscript/deposit_manifest.tsv`, so any table can be verified against its checksum without re-running an analysis. `manuscript/deposit_notes.md` names each class of bulk data deliberately excluded and gives the command that regenerates it from its public source.

All primary data are public and unmodified: NCBI Datasets genome assemblies and their annotations for the 503 genomes named in the two manifests; UniProt reference proteomes for the 7,691 proteomes swept; InterPro and Pfam signatures; Ensembl Compara paralogy from a pinned dated archive, whose registry is committed beside the map; RCSB PDB entries and AlphaFold DB models for the structure panel; ClinVar missense records; and the SRA runs listed in the expression tables. No new sequence data were generated.

Code availability

The analysis is a single repository of scripted stages: one script prefix per task, each rendering its own report purely from its committed tables, each running a suite of constructed negative controls before it writes anything. `python scripts/s14_assemble.py` rebuilds this manuscript package — figures, claims, stitched text, typeset PDF and deposit manifest — and exits non-zero on a missing figure, a missing section, a cited reference key with no bibliography row, a glyph the document font cannot set, or a load-bearing number that no longer matches its source table.

Author contributions

G.D. designed the study, wrote the analysis code, performed the analyses and wrote the manuscript.

Competing interests

The author declares no competing interests.

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Citations are rendered from `results/s0_baseline/references.tsv`, numbered in order of first appearance; each entry ends with its stable key in that table.

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